GEO-TECHNICAL INVESTIGATION REPORT
(Minor Bridge at Ch. 44+758)

Project:

Geotechnical investigation Report for the Construction of 4 laning Greenfield corridor from Suara (Sasaram) at Ch. 0+000 to Garhani (at SH-12) at Ch 74+432 Package 1 of Patna-Arrah-Sasaram corridor of NH-119A in the State of Bihar on HAM Mode

Project Ref.:25065139

7th February 2026 – R0

Report submitted to:

M/s. NKC Projects Private Limited
Plot No.-872, Phase – V, Udyog Vihar, Gurgaon, Haryana

CELEBRATING 25 YEARS

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BRIEF GEOTECHNICAL INVESTIGATION DATA SUMMARY

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3.	**No. of bore hole and Depth**	3 nos. of Bore hole of 30.0 m depth below EGL
4.	**Seismic Zone**	*Zone III*
5.	**5.1 – Properties of Sub soil strata**	

Ch.	BH No.	Depth below EGL (m)		Sub soil type	γ (t/m³)	Nc Value	C (t/m²)	ϕ (°)
Ch. 44+758	BH-A1	0.0	2.0	Silty Clay of medium plasticity (CI)	1.69	7	4.1	7
		2.0	8.0	Sandy Silt with gravel (ML-CL)	1.70-1.72	14-25	0.4-0.5	28-29
		8.0	12.0	Silty Sand (SM)	-	28-29	0	34
		12.0	16.0	Coarse Sand (SW)	-	30-34	0	34
		16.0	17.0	Sandy Silt with gravel (ML-CL)	1.76	25	0.8	29
		17.0	19.0	Silty Clay of medium plasticity (CI)	-	43	-	-
		19.0	22.0	Sandy Silt (ML-CL)	1.76	27	1.0	28
		22.0	24.0	Silty Clay of medium plasticity (CI)	-	48	-	-
		24.0	28.0	Sandy Silt with gravel (ML-CL)	-	29-30	-	-
		28.0	30.0	Silty Sand (SM)	-	35-35	0	34
	BH-P1	0.0	1.0	Filled up soil	-	-	-	-
		1.0	7.5	Silty Clay of medium plasticity with gravel (CI)	1.70-1.74	13-26	6.4-11.8	8-9
		7.5	11.0	Silty Sand (SM)	-	23-25	0	34
		11.0	15.0	Fine Sand (SP)	-	28-31	0	34
		15.0	16.0	Coarse Sand (SW)	-	34	0	34
		16.0	18.0	Silty Clay of medium plasticity with gravel (CI)	1.76	47	15.7	9
		18.0	21.0	Sandy Silt (ML-CL)	1.78	25-27	0.9	28
		21.0	24.0	Silty Clay of medium plasticity with gravel (CI)	1.78	50-55	19.8	9
		24.0	27.0	Sandy Silt (ML-CL)	1.79	31-32	1.1	29
		27.0	30.0	Fine Sand (SP)	-	36-43	0	34

5.

5.1 – Properties of Sub soil strata

Ch.	BH No.	Depth below EGL (m)		Sub soil type	γ (t/m³)	Nc Value	C (t/m²)	ϕ (°)
Ch. 44+758	BH-A2	0.0	4.5	Silty Clay of medium plasticity with gravel (CI)	1.72	8-20	7.6	6
		4.5	7.0	Sandy Silt with gravel (ML-CL)	-	25-28	-	-
		7.0	8.0	Silty Clay of medium plasticity with gravel (CI)	1.74	39	11.5	8
		8.0	11.0	Silty Sand with gravel (SM)	-	33	0	34
		11.0	16.0	Coarse Sand with gravel (SW)	-	35-39	0	34
		16.0	18.0	Sandy Silt (ML-CL)	1.78	29	0.8	28
		18.0	22.5	Silty Clay of medium plasticity (CI)	1.77-1.79	41-49	16.8-18.9	9-10
		22.5	28.0	Sandy Silt (ML-CL)	1.81	27-31	0.9	29
		28.0	30.0	Silty Sand (SM)	-	36-42	0	34

5.2 – SPT values at different depth

Depth Below EGL (m)	Ch. 44+758					
	BH-A1		BH-P1		BH-A2	
	N_r	N_c	N_r	N_c	N_r	N_c
1.50	7	7	13	13	8	8
3.00	9	14	16	16	20	20
4.50	14	17	21	21	28	25
6.00	18	19	26	26	34	28
7.50	29	25	25	23	39	39
9.00	35	28	28	24	47	33
10.50	40	29	32	25	49	33
12.00	43	30	39	28	55	35
13.50	48	32	46	31	64	38
15.00	55	34	54	34	67	39
16.50	38	25	47	47	47	29
18.00	43	43	42	27	42	42
19.50	44	27	40	25	41	41
21.00	46	27	50	50	49	49
22.50	48	48	55	55	55	30
24.00	53	29	57	31	60	31
25.50	56	30	61	32	53	28
27.00	59	30	74	36	51	27
28.50	71	35	87	41	76	36
30.00	81	38	94	43	94	42

5.

5.3 - Water Table: Water table was encountered at 1.40 m to 1.60 m depth below existing ground surface in different boreholes.

5.4 - Liquefaction Potential: After carrying out assessment of liquefaction potential of subsoil strata as per in **IRC:SP:114-2018 & IS:1893(Part-1):2016** from the SPT data and peak ground acceleration likely to occur at the site during earthquake, it is found that the soil strata up to a depth of 2.00 m below EGL is **susceptible** to liquefaction in BH-A1 only

5.5- Sub Soil strata:

Depth (m)	BH-A1	BH-P1	BH-A2
1.00	SILTY CLAY OF MEDIUM PLASTICITY (CI)	FILLED UP SOIL	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)
2.00		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)	
3.00	SANDY SILT WITH GRAVEL (ML-CL)		
4.00			
5.00			SANDY SILT WITH GRAVEL (ML-CL)
6.00			
7.00			
8.00		SILTY SAND (SM)	SILTY CLAY OF MEDIUM PLASTICITY WITH
9.00	SILTY SAND (SM)		SILTY SAND WITH GRAVEL (SM)
10.00			
11.00			
12.00		FINE SAND (SP)	COARSE SAND WITH GRAVEL (SW)
13.00	COARSE SAND (SW)		
14.00			
15.00			
16.00		COARSE SAND (SW)	
17.00	SANDY SILT WITH GRAVEL	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)	SANDY SILT (ML-CL)
18.00	SILTY CLAY OF MEDIUM PLASTICITY (CI)		
19.00		SANDY SILT (ML-CL)	SILTY CLAY OF MEDIUM PLASTICITY (CI)
20.00	SANDY SILT (ML-CL)		
21.00			
22.00		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)	
23.00	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		SANDY SILT (ML-CL)
24.00			
25.00	SANDY SILT WITH GRAVEL (ML-CL)	SANDY SILT (ML-CL)	
26.00			
27.00			
28.00		FINE SAND (SP)	
29.00	SILTY SAND (SM)		SILTY SAND (SM)
30.00			

6. **Recommendations:**

6.1- Type of Foundation: Open/Raft Foundation

Allowable settlement of 75 mm

BH No.	Founding Level below EGL (m)	Depth of Foundation (m)	Size of Foundation (m)	$q_{net\ safe}$ (t/m²)	Safe Bearing pressure (t/m²)	$q_{allowable}$ (t/m²)
BH-A1	2.00	2.00	8.00 x 8.00	13.26	16.59	13.25
	3.00	3.00	8.00 x 8.00	14.69	17.36	14.50
BH-P1	2.00	2.00	8.00 x 8.00	12.76	30.45	12.75
	3.00	3.00	8.00 x 8.00	13.51	32.08	13.50
BH-A2	2.00	2.00	8.00 x 8.00	12.38	30.35	12.25
	3.00	3.00	8.00 x 8.00	13.47	31.86	13.25

6.2- Type of Foundation: Cast - in - situ bored pile

Pile diameter of 1000mm & 1200 mm Having COL at 2.00 m & 2.30 m depth below EGL

Chainage	BH no.	Pile Dia. (mm)	COL below EGL (m)	Pile Length below COL (m)	Safe Axial Load Carrying Capacity (t)		Safe Lateral Load Carrying Capacity (t)	
					Comp.	Uplift	Free Head	Fix Head
Ch. 44+758	BH-A1	1000	2.00	16.0	170	120	13.0	37.0
				18.0	200	140	13.0	37.0
				19.0	210	150	13.0	37.0
				21.0	240	170	13.0	37.0
		1200	2.30	16.0	220	140	23.0	63.0
				18.0	260	170	23.0	63.0
				19.0	280	190	23.0	63.0
				21.0	310	220	23.0	63.0

The load capacities of piles are based on empirical correlations and it should be confirmed with conducting pile load test as per IS: 2911 (Part IV) on test pes before execution of working piles.

6. **6.2- Type of Foundation: Cast - in - situ bored pile**

Chainage	BH no.	Pile Dia. (mm)	COL below EGL (m)	Pile Length below COL (m)	Safe Axial Load Carrying Capacity (t)		Safe Lateral Load Carrying Capacity (t)	
					Comp.	Uplift	Free Head	Fix Head
Ch. 44+758	BH-P1	1000	2.00	15.0	180	120	13.0	37.0
				17.0	220	140	13.0	37.0
				18.0	230	150	13.0	37.0
				20.0	260	170	13.0	37.0
		1200	2.30	17.0	280	170	23.0	63.0
				18.0	300	190	23.0	63.0
				20.0	330	220	23.0	63.0
				21.0	340	230	23.0	63.0
	BH-A2	1000	2.00	15.0	180	110	15.0	40.0
				17.0	200	140	15.0	40.0
				18.0	210	150	15.0	40.0
				19.0	220	150	15.0	40.0
		1200	2.30	17.0	260	170	24.0	65.0
				18.0	270	180	24.0	65.0
				19.0	280	190	24.0	65.0
				20.0	310	210	24.0	65.0

The load capacities of piles are based on empirical correlations and it should be confirmed with conducting pile load test as per IS: 2911 (Part IV) on test pes before execution of working piles.

1.0 INTRODUCTION

1.1 Project Description

M/s. NKC Projects Private Limited, Plot No.-872, Phase – V, Udyog Vihar, Gurgaon, Haryana has been awarded the work of Consultancy Services for the Construction of 4 laning Greenfield corridor from Suara (Sasaram) at Ch. 0+000 to Garhani (at SH-12) at Ch 74+432 Package 1 of Patna-Arrah-Sasaram corridor of NH-119A in the State of Bihar on HAM Mode.

The project consultant has appointed us to carry out the geotechnical investigation work for the design of suitable foundation system for the construction of the various structures. We have carried out the investigations and testing accordingly and the report for the structure is submitted below.

Type of structure	Chainage
Minor Bridge	Ch. 44+758

1.2 Object of Investigations

To establish the parameters for the foundation design of the structure, various properties and parameters regarding the subsoil at site are required. These parameters are achieved through geo-technical investigations viz. soil profile, engineering properties & physical characteristics of the soil strata, variation in strength of soil strata etc. and can be elaborated as below:

- Sub-surface conditions which will reflect the thickness of the different soil strata
- Depth of ground water table
- Safe bearing capacity of the soil which will need the determination of various engineering properties of the soil strata at different levels
- Depth of the foundations
- Suitable type of foundations
- Requirement of any treatment needed to enhance the engineering properties of the soil beneath the footing

1.3 Scope of Investigations

For achieving the aforesaid objectives, the scope of work, as finalized by the consultant includes:

1. Making **three** bore holes of **30m** depth below existing ground level or **3m** in hard rock after refusal whichever is encountered earlier on the site at each location as per the schedule given below.

2. Conducting Standard Penetration Tests (S. P. T.) at 1.5 m depth interval.
3. Extracting disturbed & undisturbed soil sample at different depth interval.
4. Observing ground water table after a stabilization period of 24 hours.
5. Conducting laboratory tests on disturbed and undisturbed soil samples collected during the subsurface exploration.
6. Compiling and submitting report, containing field and laboratory tests results and suggestion & recommendations regarding type & depth of foundations and allowable load bearing capacity of soil and other desired parameters at various depths.

1.4 Organization of Report

This report has been primarily designed to explain the whole study in a systematic way, keeping in view the various demands of designers as well as the client. Each investigation is backed by its theoretical base and the results obtained either during the field tests or in laboratory are presented herein this report in self-explanatory tabular and/or graphical form. Calculations are shown wherever necessary before incorporating any parameter in the recommendations.

The chapter 'Introduction' describes the details of the project and various contents of this study. 'Site Reconnaissance' provides the general information regarding the site conditions, weather of the region, topography and the geology of the area. Details of various field and laboratory tests are given in the following two chapters. Findings obtained during these tests are summarized in the next chapter.

Foundation Analysis' is an important chapter dealing with all design calculations required for the foundation selection and design. Recommendations are finalized in the concluding chapter.

Various table, figures and graphs are given in the appendices in quite an explanatory mode.

A list of Indian Standard (IS) Codes, which are referred throughout the study, is also attached at the end of this report.

2.0 SITE RECONNAISSANCE

2.1 Location

The site investigated is at different chainage along 4 laning Greenfield corridor from Suara (Sasaram) at Ch. 0+000 to Garhani (at SH-12) at Ch 74+432 Package 1 of Patna-Arrah-Sasaram corridor of NH-119A in the State of Bihar on HAM Mode.

2.2 Physiography, Rainfall and Climate

The proposed 4-lane road project from Sasaram to Arrah passes through the districts of Rohtas and Bhojpur in Bihar. The region lies in the alluvial plains of the Ganga basin with generally flat terrain, fertile soil, and scattered patches of undulating land near the Kaimur plateau. Sasaram, located at the foothills of the Kaimur range, represents a transitional zone from rocky uplands to alluvial plains, while Arrah lies within the central alluvial tract of the state. This physiographic setting provides suitable conditions for agriculture and transport development.

The road alignment falls under the middle Ganga plain region which receives rainfall mainly from the southwest monsoon. The average annual rainfall in the districts of Rohtas and Bhojpur ranges between 900 mm to 1,200 mm, with more than 80% concentrated in the monsoon months of June to September. Pre-monsoon showers and occasional winter rainfall also contribute to the water availability, which supports both agriculture and groundwater recharge along the project corridor.

The climate of the Sasaram–Arrah region is predominantly tropical with marked seasonal variations. Summers are hot and dry, with maximum temperatures often crossing 40°C during May and June. The monsoon season brings high humidity and heavy rainfall, moderating the heat. Winters, from November to February, are cool and pleasant with minimum temperatures sometimes dropping to around 6°C–8°C. Overall, the climate is characterized by three distinct seasons—summer, monsoon, and winter—typical of the middle Ganga plain in Bihar.

2.3 General Geology

The Sasaram–Arrah project area covers parts of Rohtas and Bhojpur districts in Bihar, lying within the Middle Ganga Plain. Geologically, the region is dominated by thick alluvial deposits of Quaternary age, consisting mainly of sand, silt, clay, and occasional kankar (calcareous concretions). These alluvial formations are the result of sedimentation by the Ganga and its tributaries over time. Towards Sasaram, the influence of the Kaimur plateau becomes noticeable, where older Vindhyan rocks comprising sandstones, shales, and limestones occur in the upland areas. However, along most of the road alignment, the subsurface strata are represented by unconsolidated alluvium with high fertility and groundwater potential.

2.4 Seismicity

Region along the surveyed site lies partly in the seismic zone III (Moderate damage risk zone). This means it increases the vulnerability of the section towards earthquake to a great deal. The damage caused by earthquake can be tremendous with huge loss of lives, infrastructure and property. Suitable seismic coefficient may be adopted in the design of structure commensurate to the Indian Standard seismic zoning of the country **IRC:SP:114-2018 & IS:1893(Part-1):2016**.

3.0 FIELD INVESTIGATION

3.1 Introduction

For achieving various soil parameters, Field Investigations are carried out at site. Field Investigation comprises site reconnaissance, detailed exploration including extensive boring program and conducting specified field tests viz. Standard Penetration Test.

3.2 Subsurface Exploration

Subsurface Exploration was carried out at locations specified by the consultant as per the following schedule:

Chainage	Bore holes	Depth (m)	Water Table below EGL (m)	Date of start	Date of Finish
Ch. 44+758	**BH-A1**	30.0	1.40	10-12-2025	11-12-2025
	BH-P1	30.0	1.50	07-12-2025	09-12-2025
	BH-A2	30.0	1.60	04-12-2025	06-12-2025

The depth of these bore- holes was taken from the existing ground surface. Disturbed and undisturbed samples were collected from these bore- holes at various depths.

3.3 Standard Penetration Test

Standard Penetration Test conducted by means of the split spoon sampler furnishes data about resistance of the soils to penetration, which can be used to evaluate standard strength data, such as N values (number of blows per 30 cm of penetration using standard split spoon) of the soil.

Standard Penetration Tests were conducted in the boreholes at 1.5 m interval as per the provisions of IS 2131:1981. The tests were conducted by means of the split spoon sampler conforming to IS 9640:1980.

N values have been also utilized to establish the final shear parameters.

3.4 Ground Water Conditions

Water table was encountered at 1.40m to 1.60m depth below existing ground surface in different boreholes.

4.0 LABORATORY INVESTIGATIONS

The laboratory tests to determine the physical properties, the engineering properties and the engineering characteristics of the soil were conducted in accordance with IS 2720. The tests performed are as follows.

4.1 Bulk Density and Natural Moisture Content

Undisturbed samples were collected from the boreholes in thin wall steel sample tubes by taking the dimensions and weight of these sample tubes, the bulk density of the soil is determined. Moisture content of the soil has been calculated by Oven Drying Method.

4.2 Grain Size Analysis

The grain size analysis has been carried out utilizing both sieve and hydrometer analysis. The sieve analysis was carried out by wet sieving method in which the material was first washed through a 4.75 mm test sieve nested in a 75 µm test sieve. The soils retained in the sieves were then dried in an oven. The dried soils were then sieved by dry sieving by passing the soils through a series of square mesh sieves, which become progressively finer down to 75 µm mesh. Each fraction thus collected was then weighed and the percentage retained on each sieve was calculated by dividing individual weights by the total sample weight.

The soils passing through 75 μm mesh was analyzed by sedimentation using hydrometer method which involves measuring the rate of settlement of fine particles suspended in a solution. Utilizing the principle of Stokes' law, particle size can be directly related to its rate of settlement in a fluid such as water. From this process, the particle diameter and percentage finer is calculated.

4.3 Atterberg Limits

Atterberg Limits in the form of liquid limit, plastic limit and shrinkage limit are determined for the soil to establish its consistency. In the case of cohesionless soil, plastic limit is first determined and if it cannot be determined the soil sample is reported to be non-plastic.

4.4 Specific Gravity

The sample is dried overnight in an oven at 110° C, cooled in desiccators, grind and sieved through 425µ IS Sieve. About 10gm of sieved sample is taken in a specific gravity bottle and sufficient distilled water is added to just cover the soil and left it for soaking for 10-15 minutes after which it is shaken well and more distilled water added to fill the bottle about half. It is then placed in a sand bath to de-air. After air is totally removed, it is cooled and fills completely with water.

Various weights, e.g. , weight of empty bottle, weight of bottle filled with water, weight of bottle filled with water and sample, etc are taken from which specific gravity is calculated.

4.5 Direct Shear Test

Direct Shear Test is a strength test, which is performed on the soil sample to determine the value of angle of internal friction. The direct shear test is generally conducted on cohesionless soil as consolidated drained (CD) test.

4.6 Triaxial Shear Test

Triaxial Shear Test is a strength test, which is performed on the soil sample to determine the value of cohesion and angle of internal friction. In the present case, test samples were prepared from undisturbed samples and were tested in the Triaxial Apparatus with un-drained / drained condition and un-consolidated / consolidated condition simulating the actual site conditions.

UU testis performed as a set of three single stage tests. The test is performed on clayey soils. Three specimens were taken from a single undisturbed sample. The soil specimens were trimmed and cut until the length to diameter ratio is approximately two. The specimens were then weighed, measured and placed in a tri-axial cell and were sheared under un-drained conditions at a constant cell pressure and strain rate. Axial load and displacement were recorded at regular intervals until a maximum deviator stress, or 20% of strain, is reached. Cell pressures of 1.0, 2.0 and 3.0 kg/cm^2 have been used for three specimens.

CU & CD test were performed as a set of three single stage tests. Three specimens were taken from a single undisturbed sample wherever possible, otherwise remolding of samples were done under SMC. The soil specimens were trimmed and cut until the length to diameter ratio is approximately two. The specimens were then weighed, measured and placed in a tri-axial cell and left for saturation. After the saturation is completed the sample is left for consolidation under the desired cell pressure and then were sheared under un-drained (for CU test)/drained (for CD test) conditions at a strain rate depending on type of test. During shearing, axial load and displacement along with pore water pressure in CU Test and burette reading in CD Test were recorded at regular intervals until a maximum deviator stress, or 20% of strain, is reached. Cell pressures of 0.5, 1.0, 1.5 kg/cm^2 for sample upto 15m depth and 1.0, 2.0 and 3.0 kg/cm^2 for sample above15m depth have been used for three specimens.

4.7 Consolidation Test

Consolidation Test is conducted on clayey soil with an apparatus known as consolidometer. The test is performed on the soil sample to determine the values of void ratio, coefficient of volume change and coefficient of consolidation.

Summary of Laboratory Tests results for all boreholes is shown in tabular form and the same is presented in the annexure of this report.

5.0 FINDINGS OF INVESTIGATIONS

Based on various field and laboratory tests, following soil parameters for the design of the suitable foundation system have been taken for various sites:

1A-Properties of Subsoil strata:

Ch.	BH No.	Depth below EGL (m)		Sub soil type	γ (t/m³)	Nc Value	C (t/m²)	ϕ (°)
Ch. 44+758	BH-A1	0.0	2.0	Silty Clay of medium plasticity (CI)	1.69	7	4.1	7
		2.0	8.0	Sandy Silt with gravel (ML-CL)	1.70-1.72	14-25	0.4-0.5	28-29
		8.0	12.0	Silty Sand (SM)	-	28-29	0	34
		12.0	16.0	Coarse Sand (SW)	-	30-34	0	34
		16.0	17.0	Sandy Silt with gravel (ML-CL)	1.76	25	0.8	29
		17.0	19.0	Silty Clay of medium plasticity (CI)	-	43	-	-
		19.0	22.0	Sandy Silt (ML-CL)	1.76	27	1.0	28
		22.0	24.0	Silty Clay of medium plasticity (CI)	-	48	-	-
		24.0	28.0	Sandy Silt with gravel (ML-CL)	-	29-30	-	-
		28.0	30.0	Silty Sand (SM)	-	35-35	0	34
	BH-P1	0.0	1.0	Filled up soil	-	-	-	-
		1.0	7.5	Silty Clay of medium plasticity with gravel (CI)	1.70-1.74	13-26	6.4-11.8	8-9
		7.5	11.0	Silty Sand (SM)	-	23-25	0	34
		11.0	15.0	Fine Sand (SP)	-	28-31	0	34
		15.0	16.0	Coarse Sand (SW)	-	34	0	34
		16.0	18.0	Silty Clay of medium plasticity with gravel (CI)	1.76	47	15.7	9
		18.0	21.0	Sandy Silt (ML-CL)	1.78	25-27	0.9	28
		21.0	24.0	Silty Clay of medium plasticity with gravel (CI)	1.78	50-55	19.8	9
		24.0	27.0	Sandy Silt (ML-CL)	1.79	31-32	1.1	29
		27.0	30.0	Fine Sand (SP)	-	36-43	0	34

Ch.	BH No.	Depth below EGL (m)		Sub soil type	γ (t/m³)	Nc Value	C (t/m²)	ϕ (°)
Ch. 44+758	BH-A2	0.0	4.5	Silty Clay of medium plasticity with gravel (CI)	1.72	8-20	7.6	6
		4.5	7.0	Sandy Silt with gravel (ML-CL)	-	25-28	-	-
		7.0	8.0	Silty Clay of medium plasticity with gravel (CI)	1.74	39	11.5	8
		8.0	11.0	Silty Sand with gravel (SM)	-	33	0	34
		11.0	16.0	Coarse Sand with gravel (SW)	-	35-39	0	34
		16.0	18.0	Sandy Silt (ML-CL)	1.78	29	0.8	28
		18.0	22.5	Silty Clay of medium plasticity (CI)	1.77-1.79	41-49	16.8-18.9	9-10
		22.5	28.0	Sandy Silt (ML-CL)	1.81	27-31	0.9	29
		28.0	30.0	Silty Sand (SM)	-	36-42	0	34

1B-<u>SPT values at different depth</u>

Depth Below EGL (m)	Ch. 44+758					
	BH-A1		BH-P1		BH-A2	
	N_r	N_c	N_r	N_c	N_r	N_c
1.50	7	7	13	13	8	8
3.00	9	14	16	16	20	20
4.50	14	17	21	21	28	25
6.00	18	19	26	26	34	28
7.50	29	25	25	23	39	39
9.00	35	28	28	24	47	33
10.50	40	29	32	25	49	33
12.00	43	30	39	28	55	35
13.50	48	32	46	31	64	38
15.00	55	34	54	34	67	39
16.50	38	25	47	47	47	29
18.00	43	43	42	27	42	42
19.50	44	27	40	25	41	41
21.00	46	27	50	50	49	49
22.50	48	48	55	55	55	30
24.00	53	29	57	31	60	31
25.50	56	30	61	32	53	28
27.00	59	30	74	36	51	27
28.50	71	35	87	41	76	36
30.00	81	38	94	43	94	42

6.0 FOUNDATION ANALYSIS

6.1 Foundation Type

A Foundation is required for distributing the loads of the superstructure on a larger ground area. The dead and live load of the proposed structure are to be transferred to the underlying supporting soil through suitable foundation.

Foundation may be broadly classified into two categories:
i) Shallow Foundation and ii) Deep Foundation.

6.2 Shallow Foundation

Shallow foundation transmits the loads to the strata at shallow depth. A shallow foundation is one the width of which is greater than its depth. Shallow Foundations are located just below the lowest part of the wall or a column, which they support.

6.3 Deep Foundation

When the soil at or near the ground surface is not capable of supporting a structure, deep foundations are required to transfer the loads to deeper strata. Deep foundations are, therefore, used when surface soil is unsuitable for shallow foundation, and a firm stratum is so deep that it cannot be reached economically by shallow foundations. The most common types of deep foundations are piles, wells and caissons.

6.4 Assessment of Liquefaction (As per IRC:SP:114-2018 & IS:1893 (Part-1):2016)

Liquefaction is the sudden loss of shear strength of the loose fine-grained sands due to earthquake-induced vibration under saturated conditions. Liquefaction generally takes place in loose fine-grained sands (fines < 10 %, D_{60}, 0.20 mm to 1.0 mm and C_u between 2 to 5) with N value less than 15. In case of soil strata having N>15, liquefaction of soil will not take place normally.

Assessment of liquefaction potential of foundation strata is carried out using the procedure as given in **IRC:SP:114-2018 & IS:1893(Part-1):2016** from the SPT data and peak ground acceleration likely to occur at the site..

In this method, cyclic shear stress likely to be induced in the foundation strata by Design Basis Earthquake (DBE) is first evaluated. Next threshold cyclic shear stress, which is good enough to cause liquefaction, is determined from SPT data and the empirical relations. Finally, comparison of these two stresses is used in the estimation of liquefaction susceptibility of the foundation strata

Cyclic Stress Ratio (CSR)

The equivalent average of shear stress τ_{av} likely to be induced in the foundation material due to an earthquake is calculated by using the equation

τ_{av}	=	$0.65 * \gamma * h * (a_{max} / g)* r_d$	
τ_{av}	=	equivalent average of shear stress to be induced by DBE	
γ	=	Unit weight of foundation material	
h	=	depth at which cyclic shear stress is calculated	
a_{max}	=	maximum surface acceleration taken as 0.16.g	
r_d	=	Stress reduction factor	
	=	1.0 – 0.00765 * h	if h < 9.15 m
	=	1.174 – 0.0267 * h	if h = 9.15 m to 23 m
	=	0.744 – 0.008 * h	if h = 23.0 m to 30.0 m
	=	0.50	if h > 30.0 m

If the equivalent average of shear stress τ_{av} is normalized with the initial effective overburden pressure (σ_o), the term is called seismic demand of soil layer or cyclic stress ratio (CSR).

$$CSR = 0.65 * (\sigma_o / \sigma_o')* (a_{max} / g)* r_d$$

Cyclic Resistance Ratio (CRR)

It expresses capacity of soil to resist liquefaction. CRR is determined using correlation between corrected blow count $(N_1)_{60}$ and CRR for earthquake of magnitude 7.5. $(N_1)_{60}$ is the SPT blow count corrected to an effective overburden pressure of 100 kPa and to hammer energy efficiency of 60 %.

The corrected blow count C_{60} is determined as follows.

$$C_{60} = C_{HT}\, C_{HW}\, C_{SS}\, C_{RL}\, C_{BD}$$

Where,

C_{HT} & C_{HW} = Correction factors for non-standard weight or height of fall

C_{HT} =0.75

C_{HW} =HW/48387

Where H = height of fall (mm), W = hammer weight (kg)

C_{BD} =Correction factor for borehole dia= 1.05 for150 mm dia borehole

C_{RL} = Correction factor for rod length =0.75for less than 3.0m
=0.80 for 3.0m to 4.0 m
=0.85 for 4.0 m to 6.0 m
=0.95 for 6.0 m to 10.0 m
=1.0 for 10.0 m to 30.0 m

C_{SS} = Correction factor for standard sampler =1.0

Correction factor for effective overburden pressure (C_N) is given by the following relation.

$$C_N = (100 / \sigma_o')^{0.50} <= 1.70$$

The value of SPT blow count for soil with fines content (FC) can be adjusted to the equivalent clean sand value of $(N_1)_{60CS}$ as follows:

$(N_1)_{60CS} = \alpha + \beta\,(N_1)_{60}$

where α and β can be determined as follows.

α = 0.0 and β = 1.0 for FC <= 5.0 %

$\alpha = \exp[(1.76 - (190/FC^2)]$ & $\beta = [0.99 + (FC^{1.5}/1000)]$ for 5.0 %<FC<35.0 %

α = 5.0 and β = 1.20 for FC >= 35.0 %

$CRR_{M=7.5}$ is given by the following equation.

$CRR_{M=7.5}=[1/\ (34-(N_1)_{60CS})] + [(N_1)_{60CS}/135]+[\ 50/\{10*(N_1)_{60CS} + 45\}^2]-[1/200]$

Hence the CRR for a particular earthquake magnitude is determined as

$CRR = CRR_{M=7.5} * MSF * K_\sigma * K_\alpha$

The MSF value for earthquake having magnitude of 6.5 is 1.44.

The value of K_σ and K_α is determined as given in code.
The factor of safety against liquefaction, FS_L, is given as

$FS_L = CRR/CSR$

The value of CSR and CRR are computed at different depth and depth susceptible to liquefaction is determined.

Liquefaction is probable when FS_L is less than 1.0.

Computation at different depth for each borehole of each site is given in details of Bore Hole and the depth of liquefiable strata.

Structure and Chainage	Bore holes	Liquefiable depth (m)
Minor Bridge at Ch. 44+758	BH-A1	Up to 2.00
	BH-P1	Nil
	BH-A2	Nil

6.5 Selection of Foundation

Selection of suitable type of foundation for the proposed structure depends upon the (i) intensity & type of loading to be transferred from the superstructure and (ii) the properties & behavior of sub soil.

In the present case, the proposed structures, which is likely to transfer normal load on the subsoil. For the prevailing soil conditions and type of structures, Either Raft foundation or cast in situ bored pile shall be adopted.

6.6 Bearing Capacity for Shallow Foundations

Bearing capacity of soil for foundation has been calculated in accordance with IS: 6403 –1981. Here Allowable bearing pressure has been evaluated by:
(a) Shear Failure criterion and (b) Settlement criterion taking SPT values & consolidation parameters. the minimum of the two has been recommended as Allowable bearing capacity.

(a) Shear Failure criterion

$$q_a = \frac{1}{F}\left[c\, N_c S_c d_c i_c + \gamma\, D_f (Nq-1) Sq\, dq\, iq + 0.5\gamma B N_\gamma S_\gamma d_\gamma i_\gamma W'\right]$$

Where,	q_a	= Allowable Bearing Capacity
	F	= Factor of safety, taken equal to 2.5
	c	= Cohesion
	γ	= Unit weight of soil
	W’	= Water table correction factor
	D_f	= Depth of foundation
	B	= Width of foundation
	N_c, Nq, N_γ	= Bearing Capacity factors
	S_c, Sq, S_γ	= Shape Factors
	d_c, dq, d_γ	= Depth Factor
	i_c, iq, i_γ	= Inclination Factor

For various values of D_f & B, calculations are done and the values for net safe bearing capacity have been obtained.

(b) Settlement criterion taking SPT values

As per IS: 8009-1976 Part – I, the settlement for different width has been computed. For the allowable total settlement of 75 mm for Raft Foundation (as per IS: 1904 and IS: 8009).

6.7 Suitability of Deep Pile Foundations

Piles transmit foundation loads through soil strata of low bearing capacity to deeper soil having a higher bearing capacity value. Piles carry loads as a combination of side friction and point bearing resistance.

Piles are suitable due to the following specific advantages over spread footings / raft foundation:

- Completely non-displacement.
- Carry the heavy superstructure loads into or through a soil stratum. Both vertical and lateral loads may be involved.
- Controls settlements when spread footing/raft foundation is on a marginal soil.
- Can resist uplift, or overturning.

- Applicable for a wide variety of soil conditions.
- Safe against liquefaction in case of earthquake.

6.8 Load Carrying Capacity of Piles (As per IS:2911 (Part1/Sec.2): 2010)

The ultimate pile capacity can be determined as follows:

Ultimate Pile Capacity (Qu) $= Q_s + Q_p$

Where, Q_s = Skin friction resistance &Q_p = Point bearing resistance

$Q_s = \Sigma K_i P_{Di} \tan\delta . A_{si}$ (for Granular Soils, C = 0)

$= \alpha i\, C_u A_s$ (for Cohesive soils, $\phi = 0$)

Where, (As per IS: 2911 Part 1/ Sec2):

K = Coefficient of earth pressure

P_{Di} = effective overburden pressure for the i^{th} layer (t/m^2) at mid depth

δ = angle of wall friction between pile and soil, in degrees.

A_{si} = Surface area of pile stem in m^2 in the i^{th} layer, where i varies from 1 to n

α = reduction factor

C = Average cohesion throughout the length of pile (t/m2)

$Qp = A_p (P_d N_q)$ for non-cohesive soils; Nq shall be taken from Fig.1, IS: 2911 Part1/Sec2)

$= Ap\, N_c\, C_p$ for cohesive soils; N_c usually taken as 9,

C_p = average cohesion at pile tip

In working out pile capacities using static formula, for piles longer than 15 times pile diameter, maximum effective overburden at the pile tip should be correspond to pile length equal to 15 diameters.

As the soils are c - ϕ type, skin friction shall be determined using both the relations for cohesive and non-cohesive soils.

Adhesion factor shall be taken from IS 2911 (Part I/ sec 2) – 2010 based on SPT value. A factor of safety of 2.5 has been used to determine safe vertical load capacity of pile.

6.9 Pile Capacity

As per the requirement of the structure, the load carrying capacity of cast in situ bored piles of 1000 mm and 1200 mm diameter with different effective length has been worked out.

Determination of safe axial load carrying capacity (Both in compression and uplift) and lateral load carrying capacity of cast in situ bored pile has been made using different calculation criterion and the recommended values and the detailed calculations have been given in the annexure.

7.0 RECOMMENDATIONS

In view of these findings & results obtained during the field and laboratory investigations and the analysis carried out thereafter, following general recommendation is being made for the foundation design of the proposed structure of Patna-Arrah-Sasaram corridor of NH-119A in the State of Bihar on HAM Mode.

1. The type of foundation depends upon the configuration of loading & loading intensity as well as characteristics & behavior of subsoil. Considering the type of loading and based on various findings of the subsoil, in the present case, the type of foundation can be adopted as Raft foundation / Cast in situ bored pile foundation.

2. Recommended Bearing Capacity shall be adopted corresponding to various parameters as following:

BH No.	Founding Level below EGL (m)	Depth of Foundation (m)	Size of Foundation (m)	$q_{net\ safe}$ (t/m^2)	Safe Bearing pressure (t/m^2)	$q_{allowable}$ (t/m^2)
BH-A1	2.00	2.00	8.00 x 8.00	13.26	16.59	13.25
	3.00	3.00	8.00 x 8.00	14.69	17.36	14.50
BH-P1	2.00	2.00	8.00 x 8.00	12.76	30.45	12.75
	3.00	3.00	8.00 x 8.00	13.51	32.08	13.50
BH-A2	2.00	2.00	8.00 x 8.00	12.38	30.35	12.25
	3.00	3.00	8.00 x 8.00	13.47	31.86	13.25

3. Recommended safe axial load carrying capacity (both in compression and uplift) and lateral load carrying capacity of individual pile shall be adopted corresponding to various parameters as following:

Chainage	BH no.	Pile Dia. (mm)	COL below EGL (m)	Pile Length below COL (m)	Safe Axial Load Carrying Capacity (t)		Safe Lateral Load Carrying Capacity (t)	
					Comp.	Uplift	Free Head	Fix Head
Ch. 44+758	BH-A1	1000	2.00	16.0	170	120	13.0	37.0
				18.0	200	140	13.0	37.0
				19.0	210	150	13.0	37.0
				21.0	240	170	13.0	37.0
		1200	2.30	16.0	220	140	23.0	63.0
				18.0	260	170	23.0	63.0
				19.0	280	190	23.0	63.0
				21.0	310	220	23.0	63.0

Chainage	BH no.	Pile Dia. (mm)	COL below EGL (m)	Pile Length below COL (m)	Safe Axial Load Carrying Capacity (t)		Safe Lateral Load Carrying Capacity (t)	
					Comp.	Uplift	Free Head	Fix Head
Ch. 44+758	BH-P1	1000	2.00	15.0	180	120	13.0	37.0
				17.0	220	140	13.0	37.0
				18.0	230	150	13.0	37.0
				20.0	260	170	13.0	37.0
		1200	2.30	17.0	280	170	23.0	63.0
				18.0	300	190	23.0	63.0
				20.0	330	220	23.0	63.0
				21.0	340	230	23.0	63.0
	BH-A2	1000	2.00	15.0	180	110	15.0	40.0
				17.0	200	140	15.0	40.0
				18.0	210	150	15.0	40.0
				19.0	220	150	15.0	40.0
		1200	2.30	17.0	260	170	24.0	65.0
				18.0	270	180	24.0	65.0
				19.0	280	190	24.0	65.0
				20.0	310	210	24.0	65.0

4. The load capacities of piles are based on empirical correlations and it should be confirmed with conducting pile load test as per IS: 2911 (Part IV) on test piles before execution of working piles.

5. If the design parameters corresponding to any other foundation type, size or depth are required, the matter may be please referred back to us.

6. Our analysis and recommendations are based only on the data collected from different bore-hole at each site. Subsoil strata observed in these bore-holes at each site has been assumed as the representative of the site.

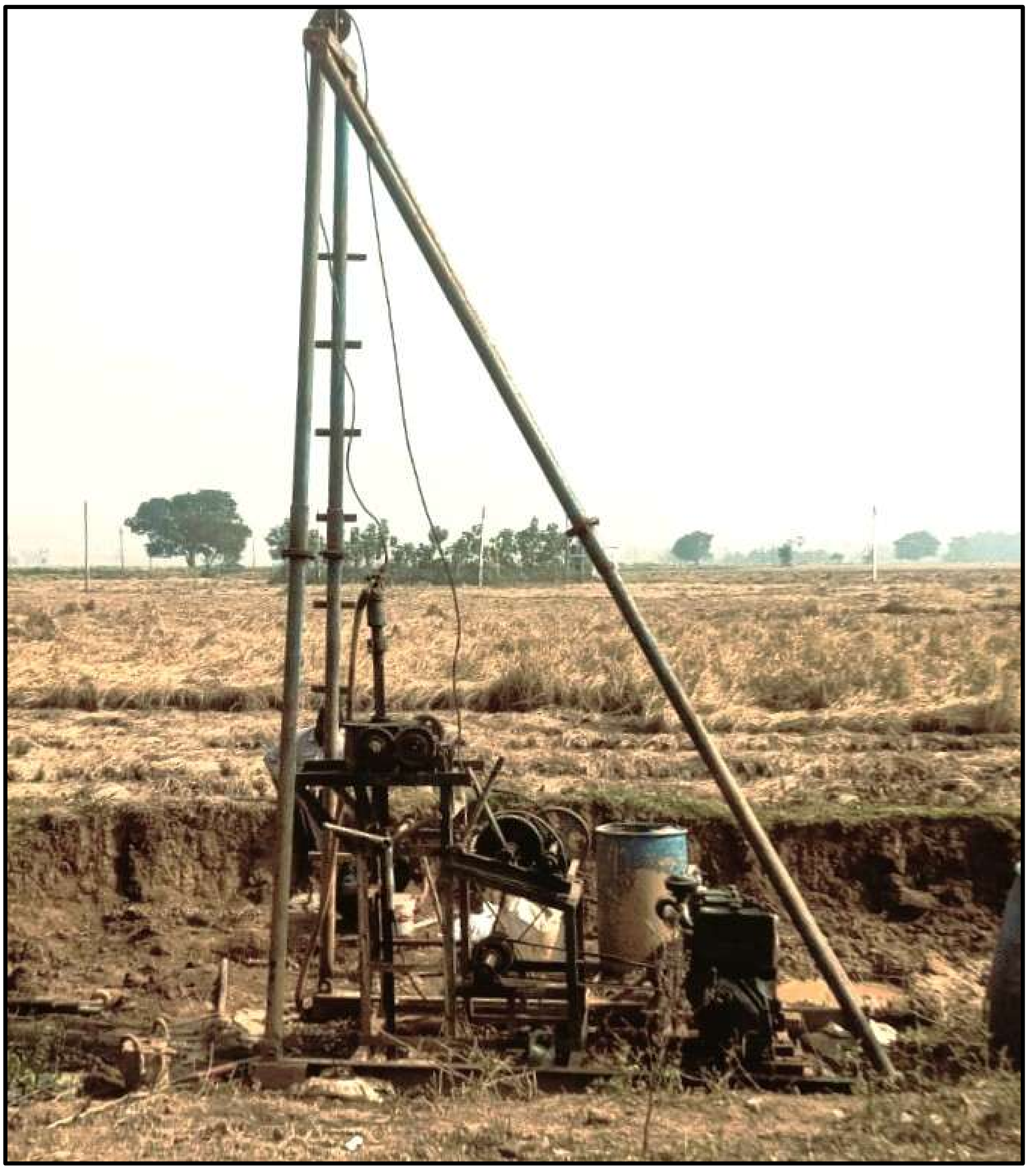
For **Explore Engineering Consultants Pvt. Ltd.**

Explore Engineering Consultants Pvt. Ltd.

(Dr. A. K. SINGH) (Dr. A. P. SINGH)

Date: 07/02/2026
Place: Noida

ANNEXURES



Minor Bridge at Design Chainage: 44+758

Details of Bore Holes

Details of BH - A1

FIELD BORELOG

Table No.- 1

METHOD OF BORING	: Shell & Auger Method	BORE HOLE NO.	: BH- A1
CASING TYPE	: SX-6	LOCATION	: Ch. 44+758
WATER TABLE	: 1.4m	DATE START	: 10-12-2025
DEPTH OF BORING	: 30.0m b.g.l	DATE COMPLETION	: 11-12-2025

DEPTH(m)	DISCRIPTION OF SOIL STRATA	SOIL CLASS.	LEGEND	STRATA THICK. (m)	SAMPLES DETAILS			SPT BLOWS COUNTS				REMARKS
					TYPE	NO.	TEST DEPTH (m)	0-15cm	15-30cm	30-45cm	"N"	
1.0	SILTY CLAY OF MEDIUM PLASTICITY	CI		2.00	DS	1	0.00-0.50	COLLECTED				
2.0					UDS	1	1.00-1.30	COLLECTED				
					SPT	1	1.50-1.95	3	3	4	7	
3.0	SANDY SILT WITH GRAVEL	ML-CL		6.00								
4.0					SPT	2	3.00-3.45	4	4	5	9	
5.0					UDS	2	4.00-4.30	COLLECTED				
					SPT	3	4.50-4.95	6	7	7	14	
6.0												
7.0					SPT	4	6.00-6.45	6	8	10	18	
8.0					UDS	3	7.00-7.30	COLLECTED				
					SPT	5	7.50-7.95	9	14	15	29	
9.0	SILTY SAND	SM		4.00								
10.0					SPT	6	9.00-9.45	14	16	19	35	
11.0					UDS	4	10.00-10.30	SLIPPED				
					SPT	7	10.50-10.95	12	18	22	40	
12.0												
13.0	COARSE SAND	SW		4.00	SPT	8	12.00-12.45	16	19	24	43	
14.0					UDS	5	13.00-13.30	SLIPPED				
					SPT	9	13.50-13.95	17	22	26	48	
15.0												
16.0					SPT	10	15.00-15.45	16	26	29	55	
17.0	SANDY SILT WITH GRAVEL	ML-CL		1.00	UDS	6	16.00-16.30	COLLECTED				
					SPT	11	16.50-16.95	13	15	23	38	
18.0	SILTY CLAY OF MEDIUM PLASTICITY	CI		2.00								
19.0					SPT	12	18.00-18.45	12	17	26	43	
20.0	SANDY SILT	ML-CL		3.00	UDS	7	19.00-19.30	COLLECTED				
					SPT	13	19.50-19.95	14	20	24	44	
21.0												
22.0					SPT	14	21.00-21.45	14	21	25	46	
23.0	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		2.00	UDS	8	22.00-22.30	SLIPPED				
					SPT	15	22.50-22.95	17	22	26	48	
24.0												
25.0	SANDY SILT WITH GRAVEL	ML-CL		4.00	SPT	16	24.00-24.45	16	25	28	53	
26.0					UDS	9	25.00-25.30	SLIPPED				
					SPT	17	25.50-25.95	18	26	30	56	
27.0												
28.0					SPT	18	27.00-27.45	19	28	31	59	
29.0	SILTY SAND	SM		2.00	UDS	10	28.00-28.30	SLIPPED				
					SPT	19	28.50-28.95	24	34	37	71	
30.0					SPT	20	29.55-30.00	27	39	42	81	

ABBREVIATION:- SPT - Standard Penetration Test UDS- Undisturbed Sample

LABORATORY TEST RESULTS OF BH-A1

Table No.-2

DEPTH BELOW GL	TYPE / SAMPLE NO.	DEPTH OF SAMPLE	SPT VALUE 'N' OBSERVED	SPT VALUE 'N' CORRECTED	DESCRIPTION OF STRATA	LEGEND	GRAIN SIZE ANALYSIS GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)	WEIGHTED MEAN DIA. (mm)	SILT FACTOR	ATTERBERG LIMIT LIQUID LIMIT (%)	PLASTIC LIMIT (%)	PLASTICITY INDEX	BULK DENSITY in t/m^3	DRY DENSITY in t/m3	MOISTURE CONTENT (%)	SPECIFIC GRAVITY	SHEAR PARAMETER COHESION 'C' (t/m^2)	ANGLE of FRICTION (Φ)	VOID RATIO, e0	COMPRESSION INDEX, Cc	Coefficient of volume compressibility (mv) (10-3 m2 / t)
	DS	0.00-0.50	COLLECTED		SILTY CLAY OF MEDIUM PLASTICITY (CI)		0	6	77	17	0.06	-												
1.0																								
	UDS	1.00-1.30	COLLECTED				0	9	75	16	0.03	2.87	40	22	18	1.69	1.52	11.53	2.68	4.10	7	0.769	0.226	0.94
2.0	SPT	1.50-1.95	7	7			0	7	75	18	0.05	-												
3.0					SANDY SILT WITH GRAVEL (ML-CL)																			
	SPT	3.00-3.45	9	14			5	5	84	6	0.41	1.12												
4.0																								
	UDS	4.00-4.30	COLLECTED				4	8	80	8	0.40	1.11	27	20	7	1.70	1.52	11.72	2.67	0.40	28	0.755	-	-
5.0	SPT	4.50-4.95	14	17			3	7	81	9	0.28	0.93												
6.0																								
	SPT	6.00-6.45	18	19			0	23	71	6	0.09	0.53												
7.0																								
	UDS	7.00-7.30	COLLECTED				2	16	75	7	0.25	0.88	27	21	6	1.72	1.53	12.31	2.67	0.50	29	0.743	-	-
8.0	SPT	7.50-7.95	29	25			4	23	67	6	0.42	1.14												
9.0					SILTY SAND (SM)																			
	SPT	9.00-9.45	35	28			0	75	25	0	0.32	0.99	NON PLASTIC								34			
10.0																								
	UDS	10.00-10.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
11.0	SPT	10.50-10.95	40	29			0	72	28	0	0.22	0.83									34			
12.0																								
	SPT	12.00-12.45	43	30	COARSE SAND (SW)		0	89	11	0	0.94	1.71	NON PLASTIC								34			
13.0																								
	UDS	13.00-13.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
14.0	SPT	13.50-13.95	48	32			0	90	10	0	0.96	1.73									34			
15.0																								
	SPT	15.00-15.45	55	34			0	88	12	0	0.86	1.64									34			
16.0																								
	UDS	16.00-16.30	COLLECTED		SANDY SILT WITH GRAVEL (ML-CL)		2	5	85	8	0.19	0.77	28	21	7	1.76	1.55	13.52	2.67	0.80	29	0.722	-	-
17.0	SPT	16.50-16.95	38	25			4	3	84	9	0.30	0.96												
18.0					SILTY CLAY OF MEDIUM PLASTICITY (CI)																			
	SPT	18.00-18.45	43	43			2	25	58	15	0.28	-												
19.0																								
	UDS	19.00-19.30	COLLECTED		SANDY SILT (ML-CL)		3	11	77	9	0.29	0.95	27	20	7	1.76	1.54	13.93	2.67	1.00	28	0.728	-	-
20.0	SPT	19.50-19.95	44	27			1	26	67	6	0.22	0.83												
21.0																								
	SPT	21.00-21.45	46	27			0	14	78	8	0.04	0.35												
22.0																								
	UDS	22.00-22.30	SLIPPED		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
23.0	SPT	22.50-22.95	48	48			10	11	63	16	0.94	-												
24.0																								
	SPT	24.00-24.45	53	29	SANDY SILT WITH GRAVEL (ML-CL)		6	14	73	7	0.61	1.37												
25.0																								
	UDS	25.00-25.30	SLIPPED				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
26.0	SPT	25.50-25.95	56	30			0	3	89	8	0.00	0.12												
27.0																								
	SPT	27.00-27.45	59	30			0	3	90	7	0.00	0.12												
28.0																								
	UDS	28.00-28.30	SLIPPED		SILTY SAND (SM)		-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
29.0	SPT	28.50-28.95	71	35			0	68	32	0	0.25	0.88	NON PLASTIC								34			
30.0	SPT	29.55-30.00	81	38			0	66	34	0	0.23	0.84									34			

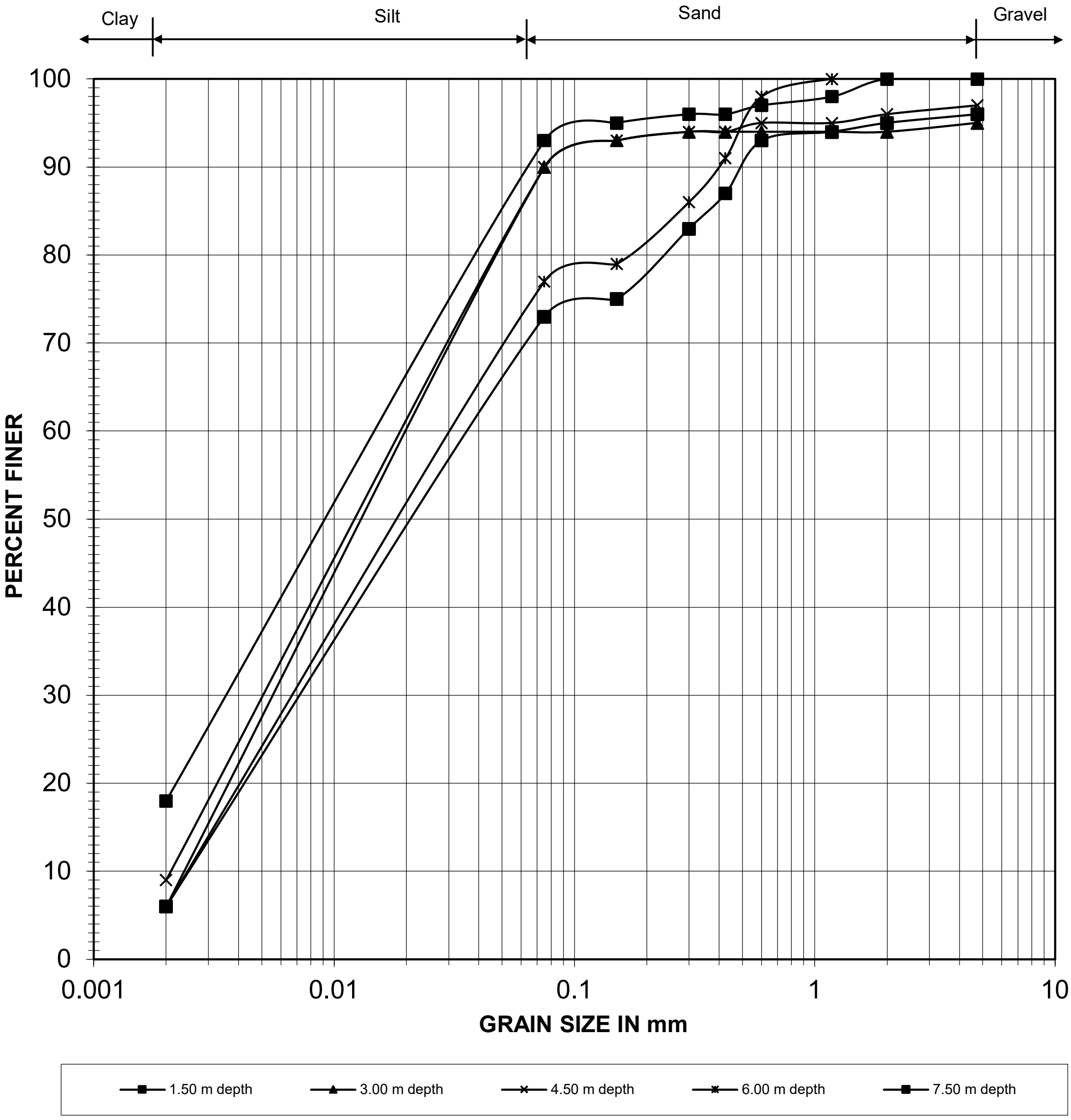
Computation of Liquefaction Potential as per IS-1893 (PART-1): 2016 — Table No.-3

Borehole No.	BH-A1
Borehole dia.	150 mm
Design Water Table (m)	1.40

Computation Depth (h)	Bulk/Saturated density (t/m3)	Fine Content (%)	PGA a_{max}/g	Earthquake magnitude (Mw)	Stress reduction coefficient (rd)	Total overburden (σo)	Pore Water Pressure (u_o)	Eff. overburden (σ_o)'	Cyclic Stress ratio (CSR)	SPT Value (N)	C_N	C_{SS}	C_{RL}	C_{BD}	C_{60}	NC_{60}	SPT Corrected $(N1)_{60}$	α	β	$(N1)_{60cs}$	$CRR_{M=7.5}$	Relative Density, Dr%	f	K_σ	K_α	MSF	CRR	FS_L	REMARK
1.50	1.95	93	0.16	6.5	0.989	0.19	0.10	0.09	0.211	7	1.70	1.0	0.75	1.05	0.58	4.05	6.9	5.00	1.20	13	0.14	NA	NA	1.00	1.0	1.44	0.21	0.98	L
3.00	1.95	90	0.16	6.5	0.977	3.12	1.60	1.52	0.209	9	1.70	1.0	0.80	1.05	0.62	5.56	9.4	5.00	1.20	16	0.17	NA	NA	1.00	1.0	1.44	0.25	1.20	N L
4.50	1.95	90	0.16	6.5	0.966	6.05	3.10	2.95	0.206	14	1.70	1.0	0.85	1.05	0.66	9.18	15.6	5.00	1.20	24	0.27	NA	NA	1.00	1.0	1.44	0.39	1.88	N L
6.00	1.95	77	0.16	6.5	0.954	8.98	4.60	4.38	0.204	18	1.51	1.0	0.95	1.05	0.73	13.20	20.0	5.00	1.20	29	0.41	NA	NA	1.00	1.0	1.44	0.59	2.89	N L
7.50	1.96	73	0.16	6.5	0.943	11.91	6.10	5.81	0.201	29	1.31	1.0	0.95	1.05	0.73	21.26	27.9	5.00	1.20	38	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
9.00	1.96	25	0.16	6.5	0.931	14.85	7.60	7.25	0.198	35	1.17	1.0	0.95	1.05	0.73	25.66	30.1	4.29	1.12	38	NA	65.14	0.67	1.00	1.0	1.44	NA	>1.0	N L
10.50	1.96	28	0.16	6.5	0.894	17.79	9.10	8.69	0.190	40	1.07	1.0	1.00	1.05	0.77	30.87	33.1	4.56	1.14	42	NA	68.12	0.66	1.00	1.0	1.44	NA	>1.0	N L
12.00	1.96	11	0.16	6.5	0.854	20.72	10.60	10.12	0.182	43	0.99	1.0	1.00	1.05	0.77	33.19	33.0	1.21	1.03	35	NA	67.98	0.66	1.00	1.0	1.44	NA	>1.0	N L
13.50	1.96	10	0.16	6.5	0.814	23.66	12.10	11.56	0.173	48	0.93	1.0	1.00	1.05	0.77	37.04	34.5	0.87	1.02	36	NA	69.46	0.65	0.95	1.0	1.44	NA	>1.0	N L
15.00	1.96	12	0.16	6.5	0.774	26.60	13.60	13.00	0.165	55	0.88	1.0	1.00	1.05	0.77	42.45	37.2	1.55	1.03	40	NA	72.23	0.64	0.91	1.0	1.44	NA	>1.0	N L
16.50	1.97	93	0.16	6.5	0.733	29.55	15.10	14.45	0.156	38	0.83	1.0	1.00	1.05	0.77	29.33	24.4	5.00	1.20	34	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
18.00	1.97	73	0.16	6.5	0.693	32.51	16.60	15.91	0.147	43	0.79	1.0	1.00	1.05	0.77	33.19	26.3	5.00	1.20	37	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
19.50	1.97	73	0.16	6.5	0.653	35.45	18.10	17.35	0.139	44	0.76	1.0	1.00	1.05	0.77	33.96	25.8	5.00	1.20	36	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
21.00	1.97	86	0.16	6.5	0.613	38.40	19.60	18.80	0.130	46	0.73	1.0	1.00	1.05	0.77	35.50	25.9	5.00	1.20	36	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
22.50	1.97	79	0.16	6.5	0.573	41.35	21.10	20.25	0.122	48	0.70	1.0	1.00	1.05	0.77	37.04	26.0	5.00	1.20	36	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
24.00	1.97	80	0.16	6.5	0.552	44.30	22.60	21.70	0.117	53	0.68	1.0	1.00	1.05	0.77	40.90	27.8	5.00	1.20	38	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
25.50	1.97	97	0.16	6.5	0.540	47.25	24.10	23.15	0.115	56	0.66	1.0	1.00	1.05	0.77	43.22	28.4	5.00	1.20	39	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
27.00	1.97	97	0.16	6.5	0.528	50.20	25.60	24.60	0.112	59	0.64	1.0	1.00	1.05	0.77	45.53	29.0	5.00	1.20	40	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
28.50	1.97	32	0.16	6.5	0.516	53.15	27.10	26.05	0.109	71	0.62	1.0	1.00	1.05	0.77	54.79	33.9	4.83	1.17	45	NA	68.95	0.66	0.72	1.0	1.44	NA	>1.0	N L
29.55	1.97	34	0.16	6.5	0.508	55.22	28.15	27.07	0.108	81	0.61	1.0	1.00	1.05	0.77	62.51	38.0	4.93	1.19	50	NA	73.00	0.64	0.70	1.0	1.44	NA	>1.0	N L

Note:			
C_{HT} =	Correction factor for non standard weight = 0.75 for drop hammer with rope and pulley		
C_{HW} =	Correction factor for non standard height of fall =HW/48387		
Kσ	Correction for high overburden stress (for effective oberburden pressure>10 T/m^2)		
Kα	Correction for static shear stress is required only for sloping ground		
L	Liquefiable	NL	Non-Liquefiable

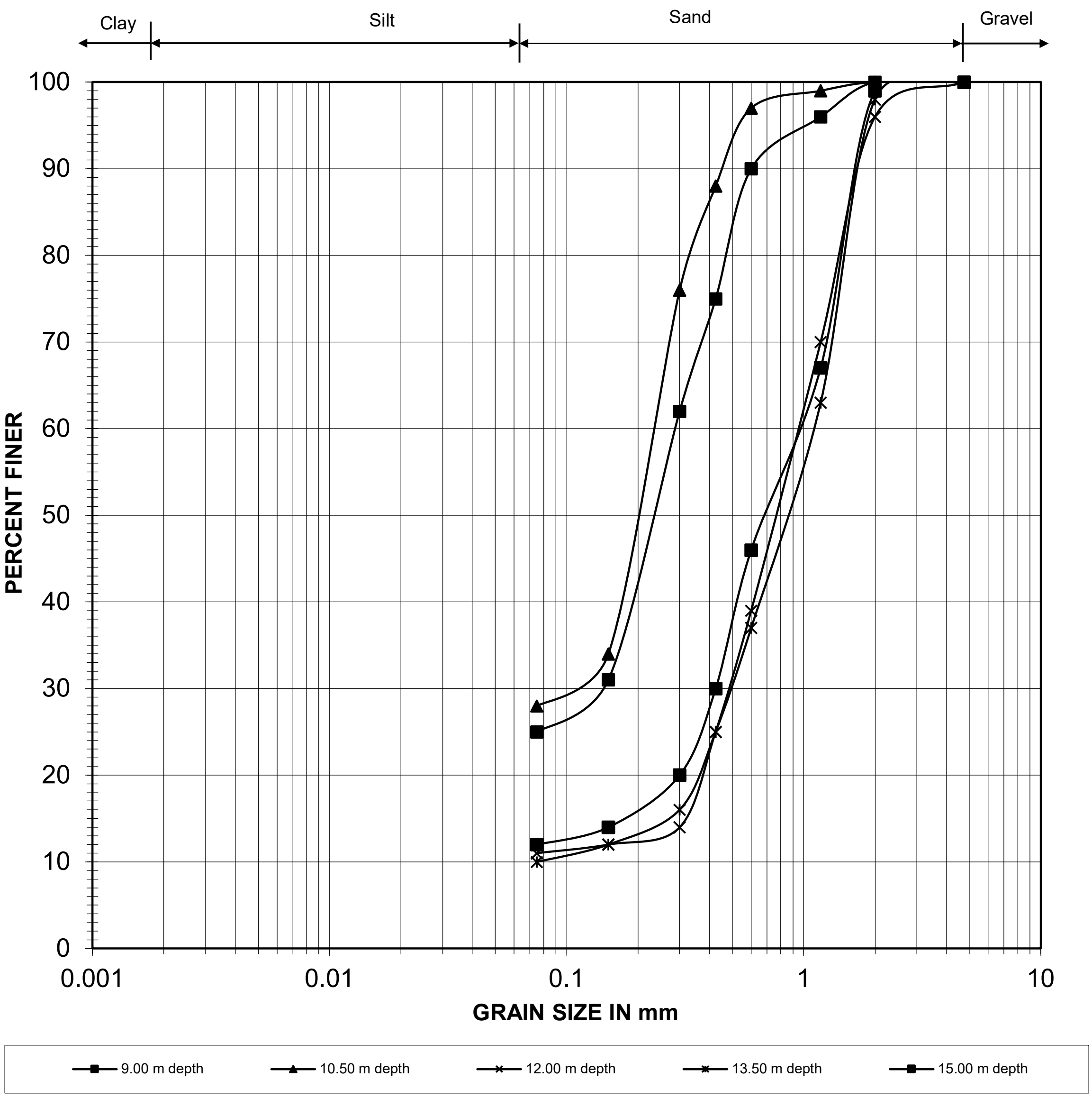
PARTICLE SIZE DISTRIBUTION CURVE for BH-A1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
1.50 m depth	0	7	75	18
3.00 m depth	5	5	84	6
4.50 m depth	3	7	81	9
6.00 m depth	0	23	71	6
7.50 m depth	4	23	67	6

Fig. - 1

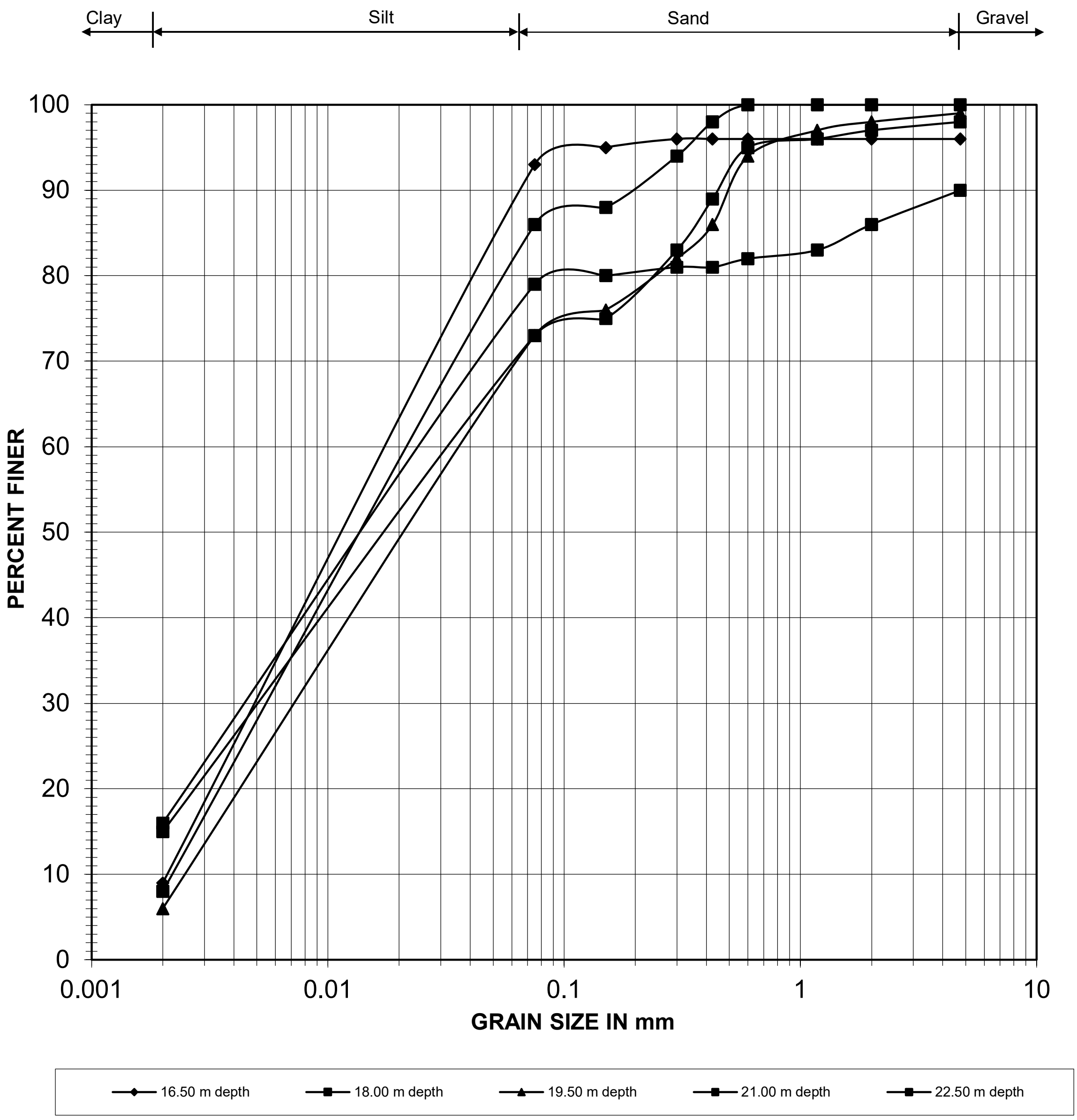
PARTICLE SIZE DISTRIBUTION CURVE for BH-A1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
9.00 m depth	0	75	25	0
10.50 m depth	0	72	28	0
12.00 m depth	0	89	11	0
13.50 m depth	0	90	10	0
15.00 m depth	0	88	12	0

Fig. - 2

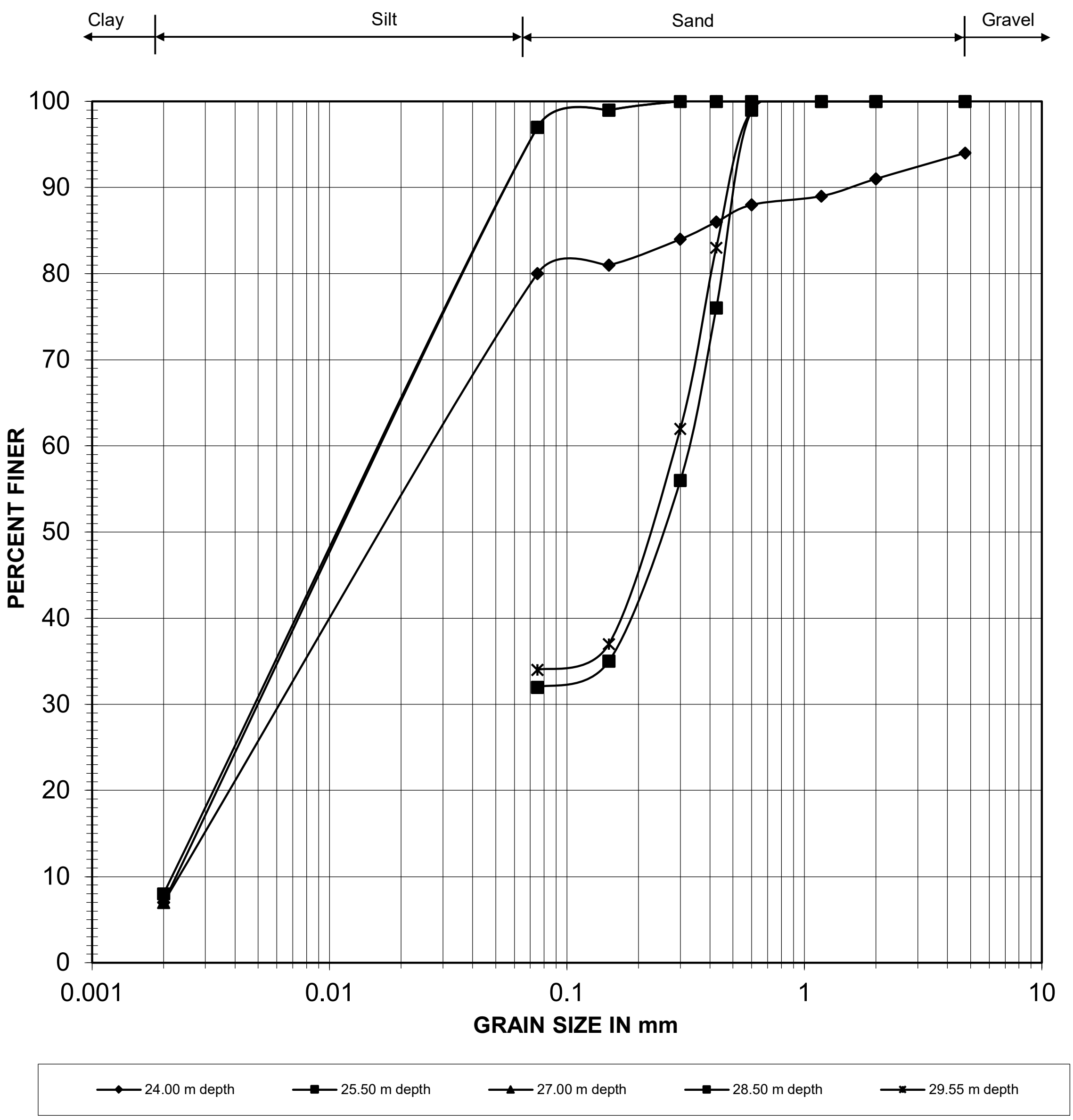
PARTICLE SIZE DISTRIBUTION CURVE for BH-A1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
16.50 m depth	4	3	84	9
18.00 m depth	2	25	58	15
19.50 m depth	1	26	67	6
21.00 m depth	0	14	78	8
22.50 m depth	10	11	63	16

Fig. - 3

PARTICLE SIZE DISTRIBUTION CURVE for BH-A1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
24.00 m depth	6	14	73	7
25.50 m depth	0	3	89	8
27.00 m depth	0	3	90	7
28.50 m depth	0	68	32	0
29.55 m depth	0	66	34	0

Fig. - 4

VARIATION OF SPT 'N' WITH DEPTH for BH-A1

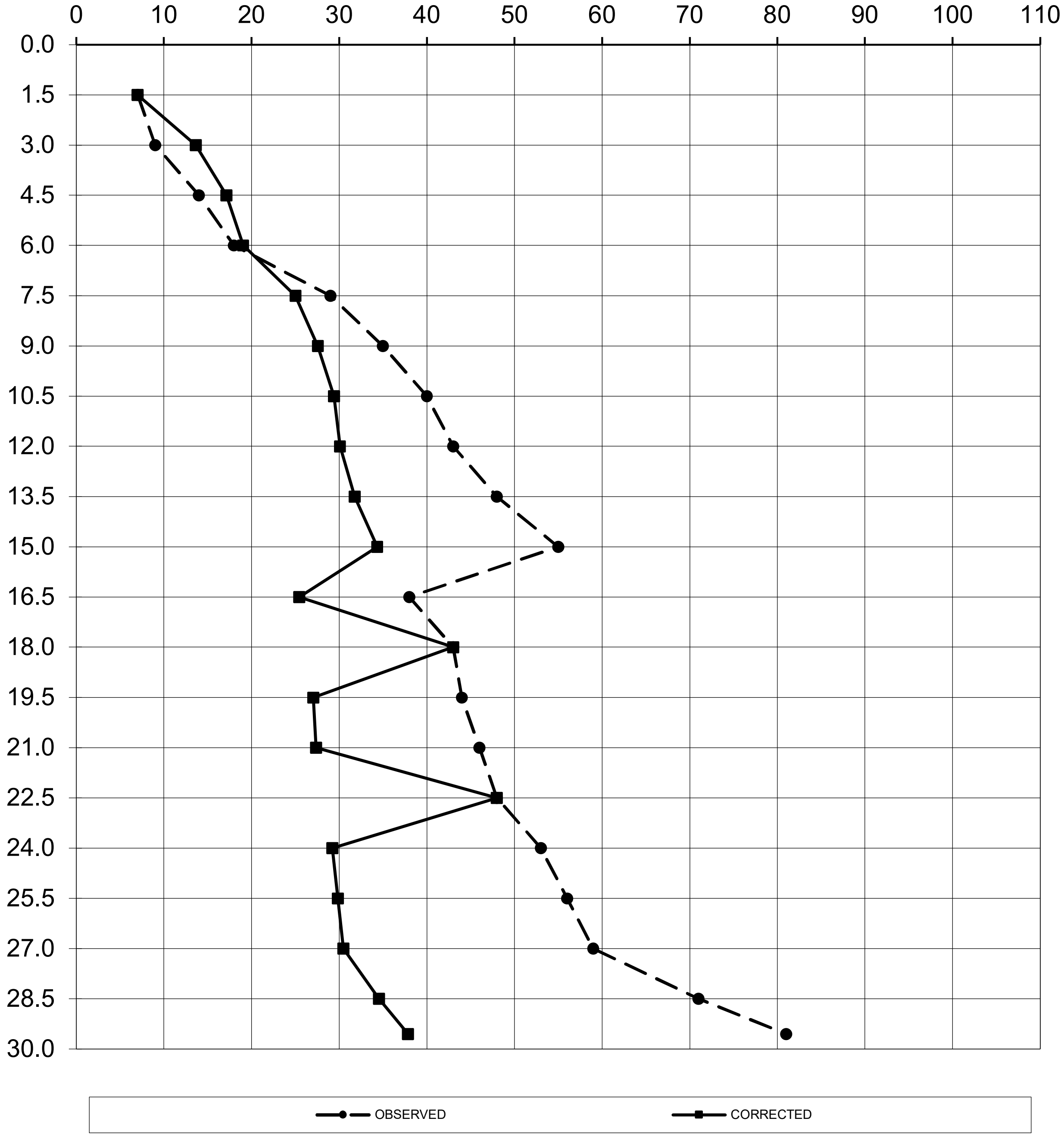


Fig. - 5

Details of BH - P1

EXPLORE

(Project Ref.-25065139)

FIELD BORELOG

Table No.- 4

METHOD OF BORING	: Shell & Auger Method	BORE HOLE NO.	: BH- P1
CASING TYPE	: SX-6	LOCATION	: Ch. 44+758
WATER TABLE	: 1.5m	DATE START	: 07-12-2025
DEPTH OF BORING	: 30.0m b.g.l	DATE COMPLETION	: 09-12-2025

DEPTH(m)	DISCRIPTION OF SOIL STRATA	SOIL CLASS.	LEGEND	STRATA THICK. (m)	SAMPLES DETAILS			SPT BLOWS COUNTS				REMARKS
					TYPE	NO.	TEST DEPTH (m)	0-15cm	15-30cm	30-45cm	"N"	
1.0	FILLED UP SOIL	-		1.00	DS	1	0.00-0.50	-				
2.0	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		6.50	UDS	1	1.00-1.30	COLLECTED				
					SPT	1	1.50-1.95	4	6	7	13	
3.0												
4.0					SPT	2	3.00-3.45	4	8	8	16	
5.0					UDS	2	4.00-4.30	COLLECTED				
					SPT	3	4.50-4.95	6	10	11	21	
6.0												
7.0					SPT	4	6.00-6.45	9	12	14	26	
8.0					UDS	3	7.00-7.30	COLLECTED				
	SILTY SAND	SM		3.50	SPT	5	7.50-7.95	8	10	15	25	
9.0												
10.0					SPT	6	9.00-9.45	10	12	16	28	
11.0					UDS	4	10.00-10.30	SLIPPED				
					SPT	7	10.50-10.95	12	14	18	32	
12.0	FINE SAND	SP		4.00								
13.0					SPT	8	12.00-12.45	13	17	22	39	
14.0					UDS	5	13.00-13.30	SLIPPED				
					SPT	9	13.50-13.95	15	19	27	46	
15.0												
16.0	COARSE SAND	SW		1.00	SPT	10	15.00-15.45	19	24	30	54	
17.0	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		2.00	UDS	6	16.00-16.30	COLLECTED				
					SPT	11	16.50-16.95	16	22	25	47	
18.0												
19.0	SANDY SILT	ML-CL		3.00	SPT	12	18.00-18.45	13	19	23	42	
20.0					UDS	7	19.00-19.30	COLLECTED				
					SPT	13	19.50-19.95	16	18	22	40	
21.0												
22.0	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		3.00	SPT	14	21.00-21.45	14	24	26	50	
23.0					UDS	8	22.00-22.30	COLLECTED				
					SPT	15	22.50-22.95	17	27	28	55	
24.0												
25.0	SANDY SILT	ML-CL		3.00	SPT	16	24.00-24.45	20	26	31	57	
26.0					UDS	9	25.00-25.30	COLLECTED				
					SPT	17	25.50-25.95	22	27	34	61	
27.0												
28.0	FINE SAND	SP		3.00	SPT	18	27.00-27.45	26	36	38	74	
29.0					UDS	10	28.00-28.30	SLIPPED				
					SPT	19	28.50-28.95	30	42	45	87	
30.0					SPT	20	29.55-30.00	34	46	48	94	

ABBREVIATION:- SPT - Standard Penetration Test UDS- Undisturbed Sample

LABORATORY TEST RESULTS OF BH-P1

Table No.-5

DEPTH BELOW GL	TYPE / SAMPLE NO.	DEPTH OF SAMPLE	SPT VALUE 'N'		DESCRIPTION OF STRATA	LEGEND	GRAIN SIZE ANALYSIS						ATTERBERG LIMIT			BULK DENSITY in t/m^3	DRY DENSITY in t/m3	MOISTURE CONTENT (%)	SPECIFIC GRAVITY	SHEAR PARAMETER		VOID RATIO, e0	COMPRESSION INDEX, Cc	Coefficient of volume compressibility (mv) (10-3 m2 / t)
			OBSERVED	CORRECTED			GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)	WEIGHTED MEAN DIA. (mm)	SILT FACTOR	LIQUID LIMIT (%)	PLASTIC LIMIT (%)	PLASTICITY INDEX					COHESION 'C' (t/m^2)	ANGLE of FRICTION (φ)			
1.0	DS	0.00-0.50	-		FILLED UP SOIL		-	-	-	-	-	-												
2.0	UDS	1.00-1.30	COLLECTED		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		6	9	67	18	0.50	3.15	39	22	17	1.70	1.53	11.39	2.68	6.40	8	0.756	-	-
	SPT	1.50-1.95	13	13			9	18	57	16	0.81	-												
3.0																								
4.0	SPT	3.00-3.45	16	16			7	11	65	17	0.58	-												
5.0	UDS	4.00-4.30	COLLECTED				4	7	67	22	0.39	2.85	40	24	16	1.71	1.53	11.81	2.68	8.10	8	0.752	0.217	0.79
	SPT	4.50-4.95	21	21			6	7	68	19	0.50	-												
6.0																								
7.0	SPT	6.00-6.45	26	26			17	7	58	18	-	-												
8.0	UDS	7.00-7.30	COLLECTED				3	15	62	20	0.34	3.13	40	22	18	1.74	1.55	12.42	2.68	11.80	9	0.732	-	-
	SPT	7.50-7.95	25	23	SILTY SAND (SM)		0	77	23	0	0.29	0.95	NON PLASTIC								34			
9.0																								
10.0	SPT	9.00-9.45	28	24			0	74	26	0	0.27	0.92									34			
11.0	UDS	10.00-10.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
	SPT	10.50-10.95	32	25			1	81	18	0	0.34	1.03									34			
12.0					FINE SAND (SP)								NON PLASTIC											
13.0	SPT	12.00-12.45	39	28			0	91	9	0	0.51	1.25									34			
14.0	UDS	13.00-13.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
	SPT	13.50-13.95	46	31			0	94	6	0	0.53	1.28									34			
15.0																								
16.0	SPT	15.00-15.45	54	34	COARSE SAND (SW)		0	89	11	0	0.43	1.16									34			
17.0	UDS	16.00-16.30	COLLECTED		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		7	11	65	17	0.62	3.94	38	20	18	1.76	1.55	13.69	2.68	15.70	9	0.731	0.206	0.64
	SPT	16.50-16.95	47	47			8	18	59	15	0.78	-												
18.0																								
19.0	SPT	18.00-18.45	42	27	SANDY SILT (ML-CL)		0	25	67	8	0.11	0.59												
20.0	UDS	19.00-19.30	COLLECTED				0	25	66	9	0.07	0.48	27	20	7	1.78	1.56	13.96	2.67	0.90	28	0.709	-	-
	SPT	19.50-19.95	40	25			0	36	57	7	0.12	0.61												
21.0																								
22.0	SPT	21.00-21.45	50	50	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		3	12	69	16	0.40	-												
23.0	UDS	22.00-22.30	COLLECTED				4	9	69	18	0.40	4.21	39	21	18	1.78	1.55	14.51	2.67	19.80	9	0.718	0.199	0.58
	SPT	22.50-22.95	55	55			6	18	59	17	0.63	-												
24.0																								
25.0	SPT	24.00-24.45	57	31	SANDY SILT (ML-CL)		0	38	56	6	0.16	0.71												
26.0	UDS	25.00-25.30	COLLECTED				2	30	60	8	0.30	0.97	27	21	6	1.79	1.56	14.88	2.67	1.10	29	0.714	-	-
	SPT	25.50-25.95	61	32			2	13	76	9	0.29	0.96												
27.0																								
28.0	SPT	27.00-27.45	74	36	FINE SAND (SP)		0	91	9	0	0.52	1.27	NON PLASTIC								34			
29.0	UDS	28.00-28.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
	SPT	28.50-28.95	87	41			0	89	11	0	0.47	1.20									34			
30.0	SPT	29.55-30.00	94	43			0	85	15	0	0.46	1.19									34			

EXPLORE

(Project Ref.-25065139)

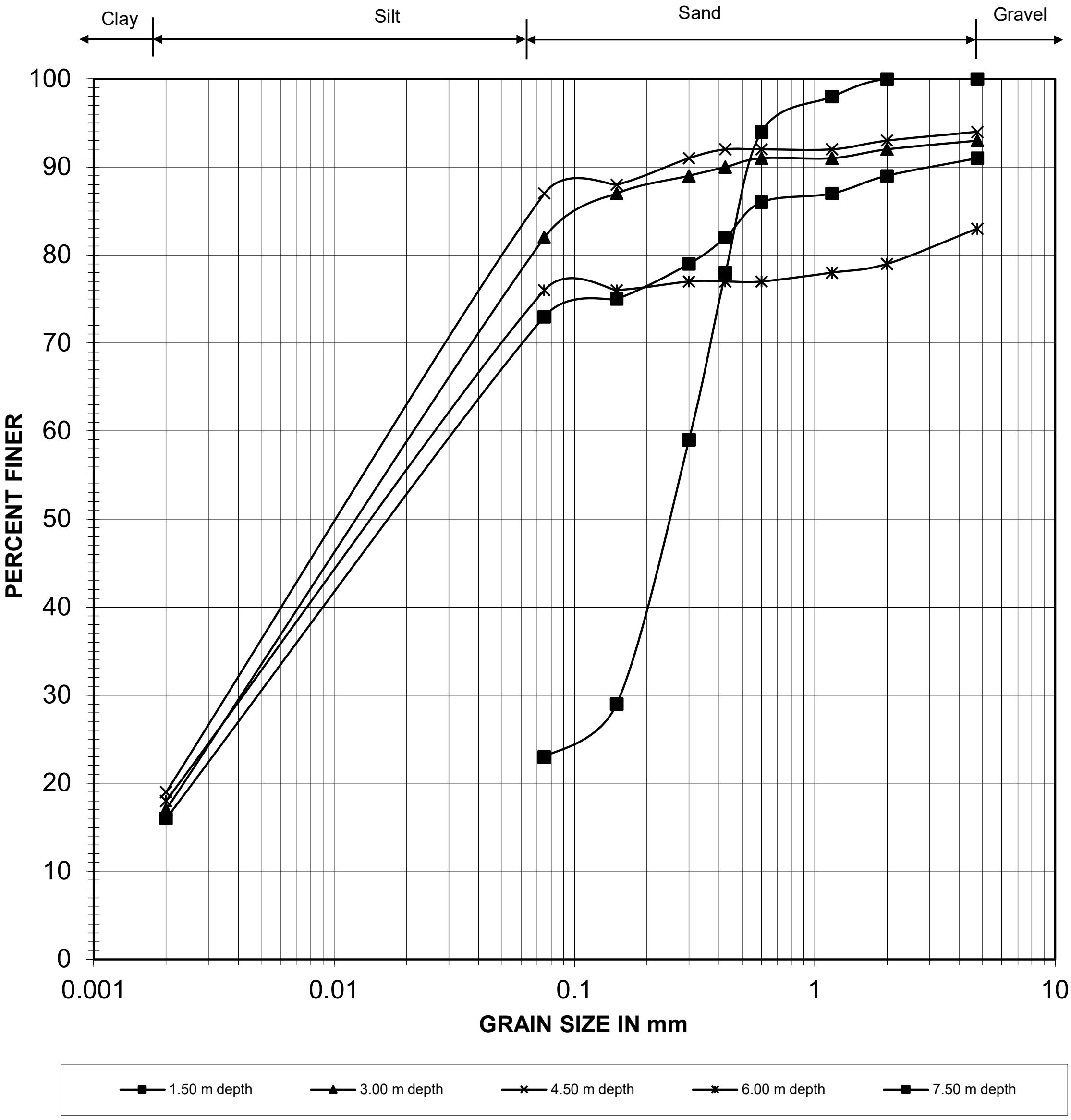
Computation of Liquefaction Potential as per IS-1893 (PART-1): 2016 — Table No.-6

Borehole No.	BH-P1
Borehole dia.	150 mm
Design Water Table (m)	1.50

Computation Depth (h)	Bulk/Saturated density (t/m3)	Fine Content (%)	PGA a_{max}/g	Earthquake magnitude (Mw)	Stress reduction coefficient (rd)	Total overburden (σo)	Pore Water Pressure (u_o)	Eff. overburden (σ_o)'	Cyclic Stress ratio (CSR)	SPT Value (N)	C_N	C_{SS}	C_{RL}	C_{BD}	C_{60}	NC_{60}	SPT Corrected $(N1)_{60}$	α	β	$(N1)_{60cs}$	$CRR_{M=7.5}$	Relative Density, Dr%	f	K_σ	K_α	MSF	CRR	FS_L	REMARK
1.50	1.70	73	0.16	6.5	0.989	2.55	-	2.55	0.103	13	1.70	1.0	0.75	1.05	0.58	7.52	12.8	5.00	1.20	20	0.22	NA	NA	1.00	1.0	1.44	0.32	3.08	N L
3.00	1.96	82	0.16	6.5	0.977	5.49	1.50	3.99	0.140	16	1.58	1.0	0.80	1.05	0.62	9.88	15.6	5.00	1.20	24	0.27	NA	NA	1.00	1.0	1.44	0.39	2.78	N L
4.50	1.96	87	0.16	6.5	0.966	8.42	3.00	5.42	0.156	21	1.36	1.0	0.85	1.05	0.66	13.78	18.7	5.00	1.20	27	0.35	NA	NA	1.00	1.0	1.44	0.51	3.25	N L
6.00	1.96	76	0.16	6.5	0.954	11.36	4.50	6.86	0.164	26	1.21	1.0	0.95	1.05	0.73	19.06	23.0	5.00	1.20	33	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
7.50	1.97	23	0.16	6.5	0.943	14.32	6.00	8.32	0.169	25	1.10	1.0	0.95	1.05	0.73	18.33	20.1	4.06	1.10	26	0.32	42.72	0.79	1.00	1.0	1.44	0.46	2.71	N L
9.00	1.97	26	0.16	6.5	0.931	17.27	7.50	9.77	0.171	28	1.01	1.0	0.95	1.05	0.73	20.53	20.8	4.39	1.12	28	0.36	44.23	0.78	1.00	1.0	1.44	0.52	3.03	N L
10.50	1.97	18	0.16	6.5	0.894	20.23	9.00	11.23	0.167	32	0.94	1.0	1.00	1.05	0.77	24.70	23.3	3.23	1.07	28	0.37	49.94	0.75	0.97	1.0	1.44	0.52	3.12	N L
12.00	1.97	9	0.16	6.5	0.854	23.18	10.50	12.68	0.162	39	0.89	1.0	1.00	1.05	0.77	30.10	26.7	0.56	1.02	28	0.36	57.63	0.71	0.93	1.0	1.44	0.49	2.99	N L
13.50	1.97	6	0.16	6.5	0.814	26.14	12.00	14.14	0.156	46	0.84	1.0	1.00	1.05	0.77	35.50	29.9	0.03	1.00	30	NA	64.68	0.68	0.89	1.0	1.44	NA	>1.0	N L
15.00	1.97	11	0.16	6.5	0.774	29.09	13.50	15.59	0.150	54	0.80	1.0	1.00	1.05	0.77	41.67	33.4	1.21	1.03	35	NA	68.37	0.66	0.86	1.0	1.44	NA	>1.0	N L
16.50	1.97	74	0.16	6.5	0.733	32.05	15.00	17.05	0.143	47	0.77	1.0	1.00	1.05	0.77	36.27	27.8	5.00	1.20	38	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
18.00	1.97	75	0.16	6.5	0.693	35.00	16.50	18.50	0.136	42	0.74	1.0	1.00	1.05	0.77	32.41	23.8	5.00	1.20	34	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
19.50	1.98	64	0.16	6.5	0.653	37.97	18.00	19.97	0.129	40	0.71	1.0	1.00	1.05	0.77	30.87	21.8	5.00	1.20	31	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
21.00	1.98	85	0.16	6.5	0.613	40.94	19.50	21.44	0.122	50	0.68	1.0	1.00	1.05	0.77	38.59	26.4	5.00	1.20	37	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
22.50	1.97	76	0.16	6.5	0.573	43.89	21.00	22.89	0.114	55	0.66	1.0	1.00	1.05	0.77	42.45	28.1	5.00	1.20	39	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
24.00	1.97	62	0.16	6.5	0.552	46.85	22.50	24.35	0.110	57	0.64	1.0	1.00	1.05	0.77	43.99	28.2	5.00	1.20	39	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
25.50	1.97	85	0.16	6.5	0.540	49.81	24.00	25.81	0.108	61	0.62	1.0	1.00	1.05	0.77	47.08	29.3	5.00	1.20	40	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
27.00	1.97	9	0.16	6.5	0.528	52.78	25.50	27.28	0.106	74	0.61	1.0	1.00	1.05	0.77	57.11	34.6	0.56	1.02	36	NA	69.58	0.65	0.71	1.0	1.44	NA	>1.0	N L
28.50	1.97	11	0.16	6.5	0.516	55.74	27.00	28.74	0.104	87	0.59	1.0	1.00	1.05	0.77	67.14	39.6	1.21	1.03	42	NA	74.61	0.63	0.67	1.0	1.44	NA	>1.0	N L
29.55	1.97	15	0.16	6.5	0.508	57.81	28.05	29.76	0.103	94	0.58	1.0	1.00	1.05	0.77	72.54	42.1	2.50	1.05	47	NA	77.05	0.61	0.66	1.0	1.44	NA	>1.0	N L

Note:			
C_{HT} =	Correction factor for non standard weight = 0.75 for drop hammer with rope and pulley		
C_{HW} =	Correction factor for non standard height of fall =HW/48387		
Kσ	Correction for high overburden stress (for effective oberburden pressure>10 T/m^2)		
Kα	Correction for static shear stress is required only for sloping ground		
L	Liquefiable	NL	Non-Liquefiable

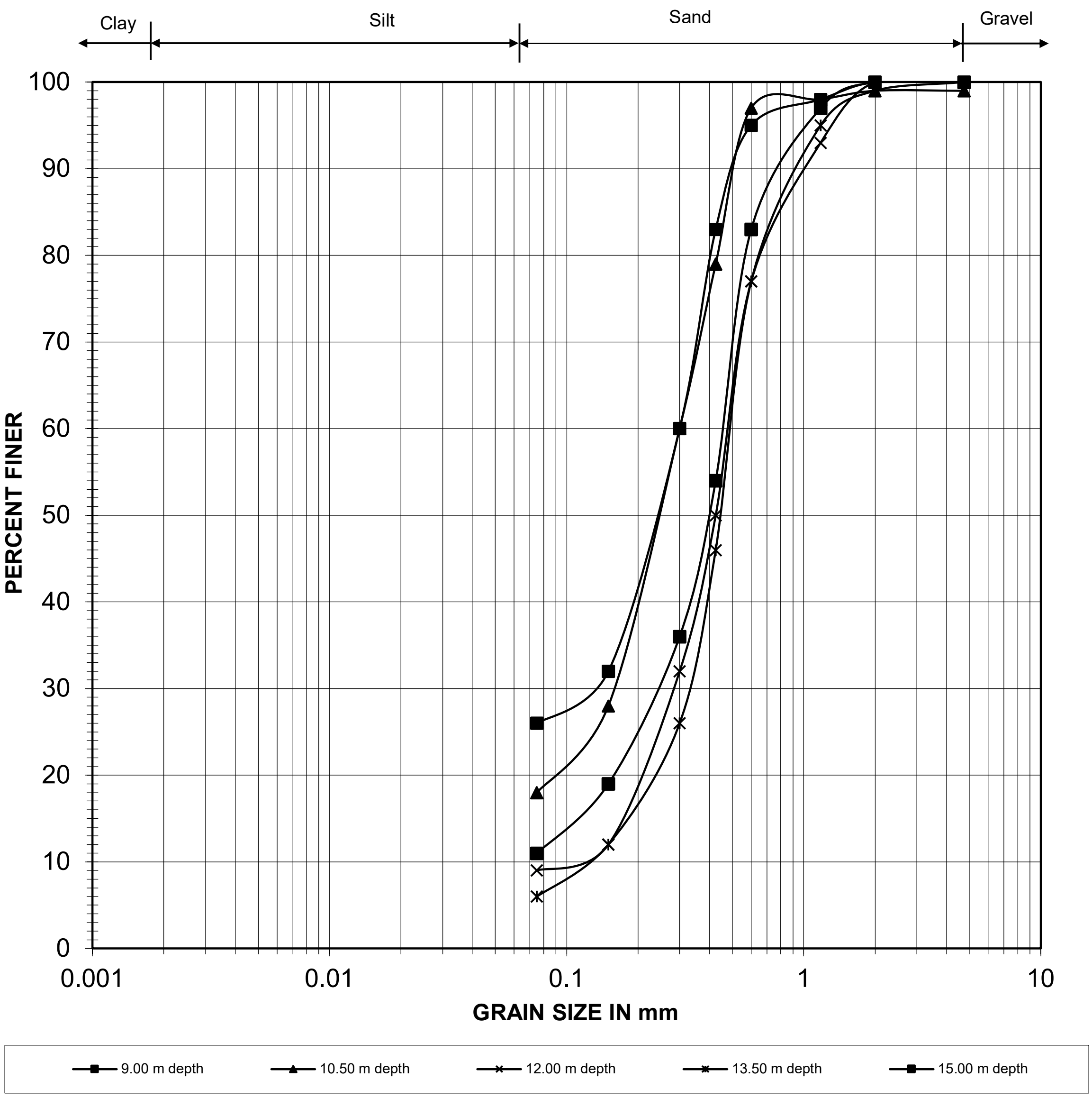
PARTICLE SIZE DISTRIBUTION CURVE for BH-P1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
1.50 m depth	9	18	57	16
3.00 m depth	7	11	65	17
4.50 m depth	6	7	68	19
6.00 m depth	17	7	58	18
7.50 m depth	0	77	23	0

Fig. - 6

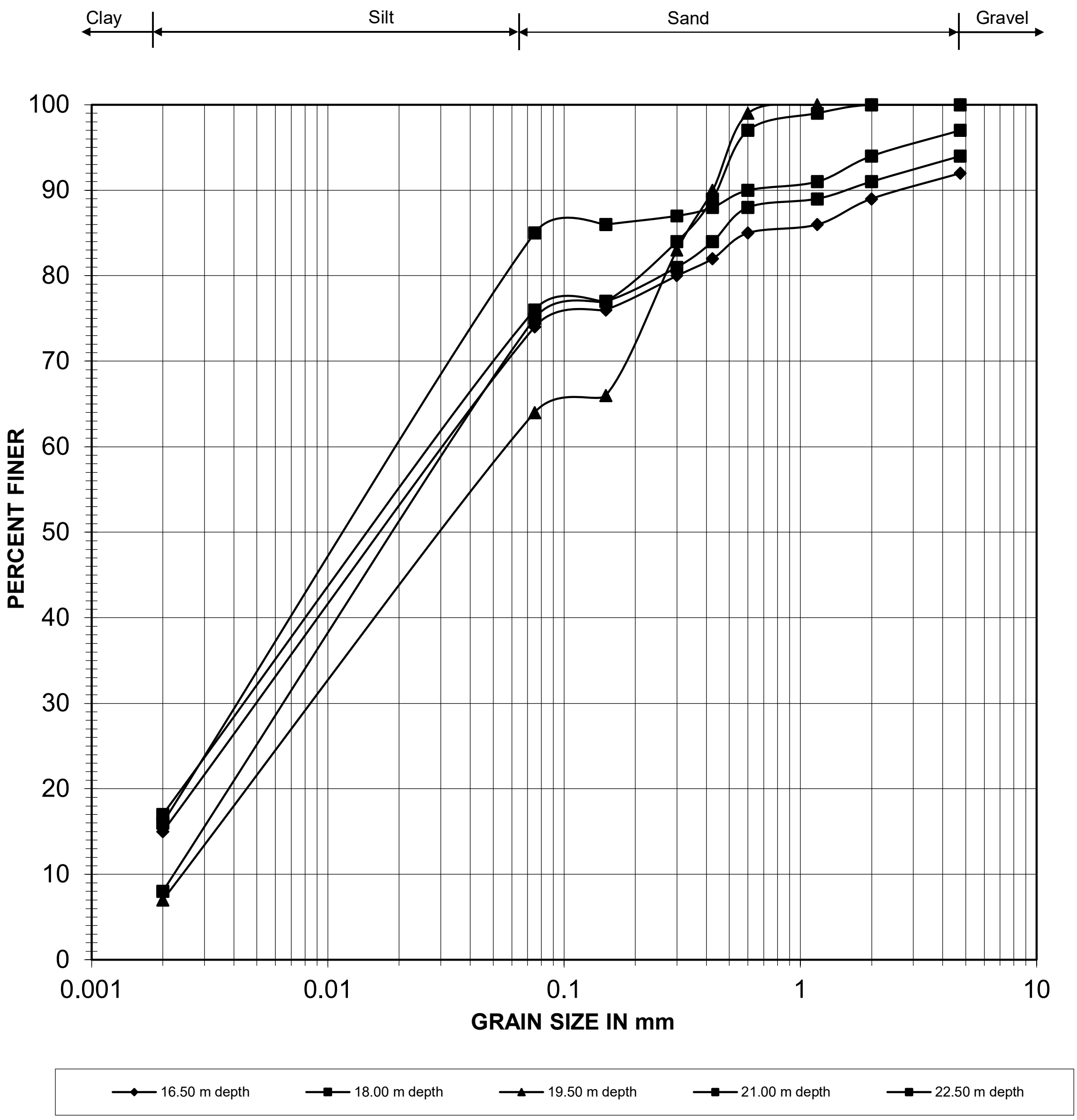
PARTICLE SIZE DISTRIBUTION CURVE for BH-P1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
9.00 m depth	0	74	26	0
10.50 m depth	1	81	18	0
12.00 m depth	0	91	9	0
13.50 m depth	0	94	6	0
15.00 m depth	0	89	11	0

Fig. - 7

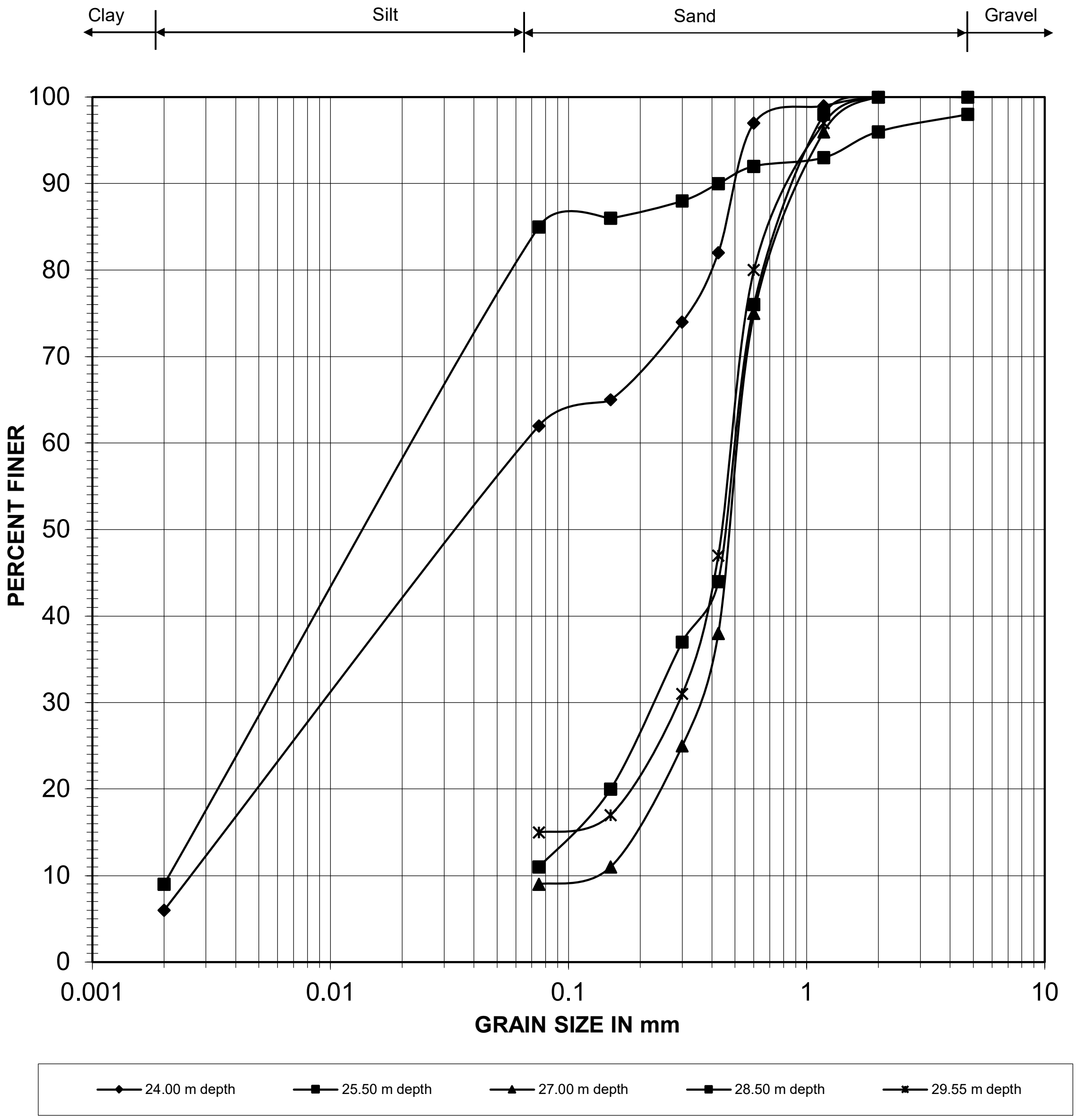
PARTICLE SIZE DISTRIBUTION CURVE for BH-P1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
16.50 m depth	8	18	59	15
18.00 m depth	0	25	67	8
19.50 m depth	0	36	57	7
21.00 m depth	3	12	69	16
22.50 m depth	6	18	59	17

Fig. - 8

PARTICLE SIZE DISTRIBUTION CURVE for BH-P1



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
24.00 m depth	0	38	56	6
25.50 m depth	2	13	76	9
27.00 m depth	0	91	9	0
28.50 m depth	0	89	11	0
29.55 m depth	0	85	15	0

Fig. - 9

VARIATION OF SPT 'N' WITH DEPTH for BH-P1

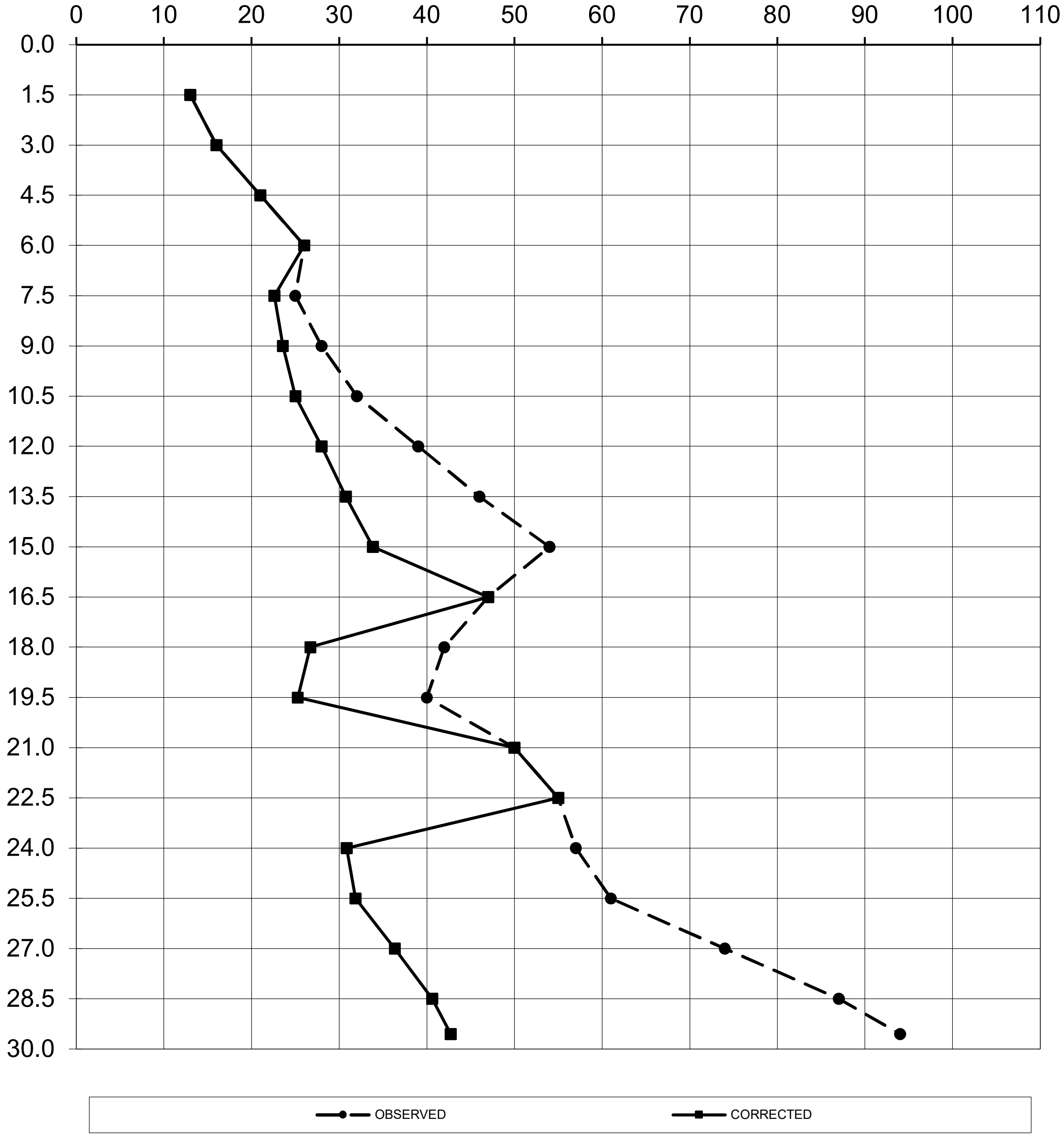


Fig. - 10

Details of BH - A2

EXPLORE

(Project Ref.-25065139)

FIELD BORELOG

Table No.- 7

METHOD OF BORING	: Shell & Auger Method	BORE HOLE NO.	: BH- A2
CASING TYPE	: SX-6	LOCATION	: Ch. 44+758
WATER TABLE	: 1.6m	DATE START	: 04-12-2025
DEPTH OF BORING	: 30.0m b.g.l	DATE COMPLETION	: 06-12-2025

DEPTH(m)	DISCRIPTION OF SOIL STRATA	SOIL CLASS.	LEGEND	STRATA THICK. (m)	SAMPLES DETAILS TYPE	NO.	TEST DEPTH (m)	SPT BLOWS COUNTS 0-15cm	15-30cm	30-45cm	"N"	REMARKS
	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		4.50	DS	1	0.00-0.50	COLLECTED				
1.0												
					UDS	1	1.00-1.30	SLIPPED				
2.0					SPT	1	1.50-1.95	3	4	4	8	
3.0												
					SPT	2	3.00-3.45	7	9	11	20	
4.0												
					UDS	2	4.00-4.30	COLLECTED				
5.0	SANDY SILT WITH GRAVEL	ML-CL		2.50	SPT	3	4.50-4.95	10	13	15	28	
6.0												
					SPT	4	6.00-6.45	12	16	18	34	
7.0												
	SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL	CI		1.00	UDS	3	7.00-7.30	COLLECTED				
8.0					SPT	5	7.50-7.95	16	18	21	39	
9.0	SILTY SAND WITH GRAVEL	SM		3.00								
10.0					SPT	6	9.00-9.45	19	22	25	47	
					UDS	4	10.00-10.30	SLIPPED				
11.0					SPT	7	10.50-10.95	21	23	26	49	
12.0	COARSE SAND WITH GRAVEL	SW		5.00								
13.0					SPT	8	12.00-12.45	20	27	28	55	
					UDS	5	13.00-13.30	SLIPPED				
14.0					SPT	9	13.50-13.95	24	30	34	64	
15.0												
16.0					SPT	10	15.00-15.45	26	31	36	67	
	SANDY SILT	ML-CL		2.00	UDS	6	16.00-16.30	COLLECTED				
17.0					SPT	11	16.50-16.95	13	19	28	47	
18.0												
19.0	SILTY CLAY OF MEDIUM PLASTICITY	CI		4.50	SPT	12	18.00-18.45	11	18	24	42	
					UDS	7	19.00-19.30	COLLECTED				
20.0					SPT	13	19.50-19.95	14	19	22	41	
21.0												
22.0					SPT	14	21.00-21.45	13	22	27	49	
					UDS	8	22.00-22.30	COLLECTED				
23.0	SANDY SILT	ML-CL		5.50	SPT	15	22.50-22.95	16	25	30	55	
24.0												
25.0					SPT	16	24.00-24.45	19	27	33	60	
					UDS	9	25.00-25.30	COLLECTED				
26.0					SPT	17	25.50-25.95	18	25	28	53	
27.0												
28.0					SPT	18	27.00-27.45	16	24	27	51	
	SILTY SAND	SM		2.00	UDS	10	28.00-28.30	SLIPPED				
29.0					SPT	19	28.50-28.95	30	36	40	76	
30.0					SPT	20	29.55-30.00	37	46	48	94	

ABBREVIATION:- SPT - Standard Penetration Test UDS- Undisturbed Sample

EXPLORE

(Project Ref.-25065139)

LABORATORY TEST RESULTS OF BH-A2

Table No.-8

DEPTH BELOW GL	TYPE / SAMPLE NO.	DEPTH OF SAMPLE	SPT VALUE 'N' OBSERVED	SPT VALUE 'N' CORRECTED	DESCRIPTION OF STRATA	LEGEND	GRAIN SIZE ANALYSIS GRAVEL (%)	SAND (%)	SILT (%)	CLAY (%)	WEIGHTED MEAN DIA. (mm)	SILT FACTOR	ATTERBERG LIMIT LIQUID LIMIT (%)	PLASTIC LIMIT (%)	PLASTICITY INDEX	BULK DENSITY in t/m³	DRY DENSITY in t/m3	MOISTURE CONTENT (%)	SPECIFIC GRAVITY	SHEAR PARAMETER COHESION 'C' (t/m²)	ANGLE of FRICTION (φ)	VOID RATIO, e0	COMPRESSION INDEX, Cc	Coefficient of volume compressibility (mv) (10-3 m2 / t)
	DS	0.00-0.50	COLLECTED		SILTY CLAY OF MEDIUM PLASTICITY WITH GRAVEL (CI)		0	8	76	16	0.04	-												
1.0																								
	UDS	1.00-1.30	SLIPPED				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.0	SPT	1.50-1.95	8	8			10	8	67	15	0.84	-												
3.0																								
	SPT	3.00-3.45	20	20			13	8	60	19	1.03	-												
4.0																								
	UDS	4.00-4.30	COLLECTED				12	11	60	17	1.00	2.81	39	21	18	1.72	1.54	11.62	2.68	7.60	6	0.739	0.210	0.89
5.0	SPT	4.50-4.95	28	25	SANDY SILT WITH GRAVEL (ML-CL)		5	9	78	8	0.49	1.23												
6.0																								
	SPT	6.00-6.45	34	28			5	7	78	10	0.43	1.15												
7.0																								
	UDS	7.00-7.30	COLLECTED		SILTY CLAY OF MEDIUM PLASTICITY WITH		4	6	72	18	0.39	3.12	38	21	17	1.74	1.55	12.14	2.68	11.60	8	0.727	-	-
8.0	SPT	7.50-7.95	39	39			5	26	53	16	0.57	-												
9.0																								
	SPT	9.00-9.45	47	33	SILTY SAND WITH GRAVEL (SM)		0	54	46	0	0.32	0.99	NON PLASTIC								34			
10.0																								
	UDS	10.00-10.30	SLIPPED				-	-	-	-	-	-				-	-	-	-	-	-	-	-	-
11.0	SPT	10.50-10.95	49	33			8	52	40	0	0.96	1.73									34			
12.0																								
	SPT	12.00-12.45	55	35	COARSE SAND WITH GRAVEL (SW)		21	70	9	0	2.78	2.93									34			
13.0																								
	UDS	13.00-13.30	SLIPPED				-	-	-	-	-	-	NON PLASTIC			-	-	-	-	-	-	-	-	-
14.0	SPT	13.50-13.95	64	38			2	90	8	0	1.17	1.90									34			
15.0																								
	SPT	15.00-15.45	67	39			2	91	7	0	1.33	2.03									34			
16.0																								
	UDS	16.00-16.30	COLLECTED		SANDY SILT (ML-CL)		1	11	78	10	0.14	0.67	27	20	7	1.78	1.57	13.37	2.67	0.80	28	0.701	-	-
17.0	SPT	16.50-16.95	47	29			1	7	83	9	0.10	0.56												
18.0																								
	SPT	18.00-18.45	42	42	SILTY CLAY OF MEDIUM PLASTICITY (CI)		2	9	71	18	0.24	-												
19.0																								
	UDS	19.00-19.30	COLLECTED				3	10	71	16	0.32	4.02	39	21	18	1.77	1.55	13.93	2.67	16.80	9	0.719	0.199	0.57
20.0	SPT	19.50-19.95	41	41			1	4	76	19	0.11	-												
21.0																								
	SPT	21.00-21.45	49	49			0	5	75	20	0.01	-												
22.0																								
	UDS	22.00-22.30	COLLECTED				0	5	78	17	0.01	4.16	36	20	16	1.79	1.56	14.52	2.68	18.90	10	0.715	-	-
23.0	SPT	22.50-22.95	55	30	SANDY SILT (ML-CL)		1	7	82	10	0.09	0.52												
24.0																								
	SPT	24.00-24.45	60	31			0	4	87	9	0.01	0.13												
25.0																								
	UDS	25.00-25.30	COLLECTED				1	29	60	10	0.23	0.84	27	21	6	1.81	1.58	14.89	2.66	0.90	29	0.688	-	-
26.0	SPT	25.50-25.95	53	28			0	39	53	8	0.20	0.79												
27.0																								
	SPT	27.00-27.45	51	27			0	40	54	6	0.14	0.67												
28.0																								
	UDS	28.00-28.30	SLIPPED		SILTY SAND (SM)		-	-	-	-	-	-	NON PLASTIC			-	-	-	-	-	-	-	-	-
29.0	SPT	28.50-28.95	76	36			0	67	33	0	0.26	0.90									34			
30.0	SPT	29.55-30.00	94	42			0	80	20	0	0.32	0.99									34			

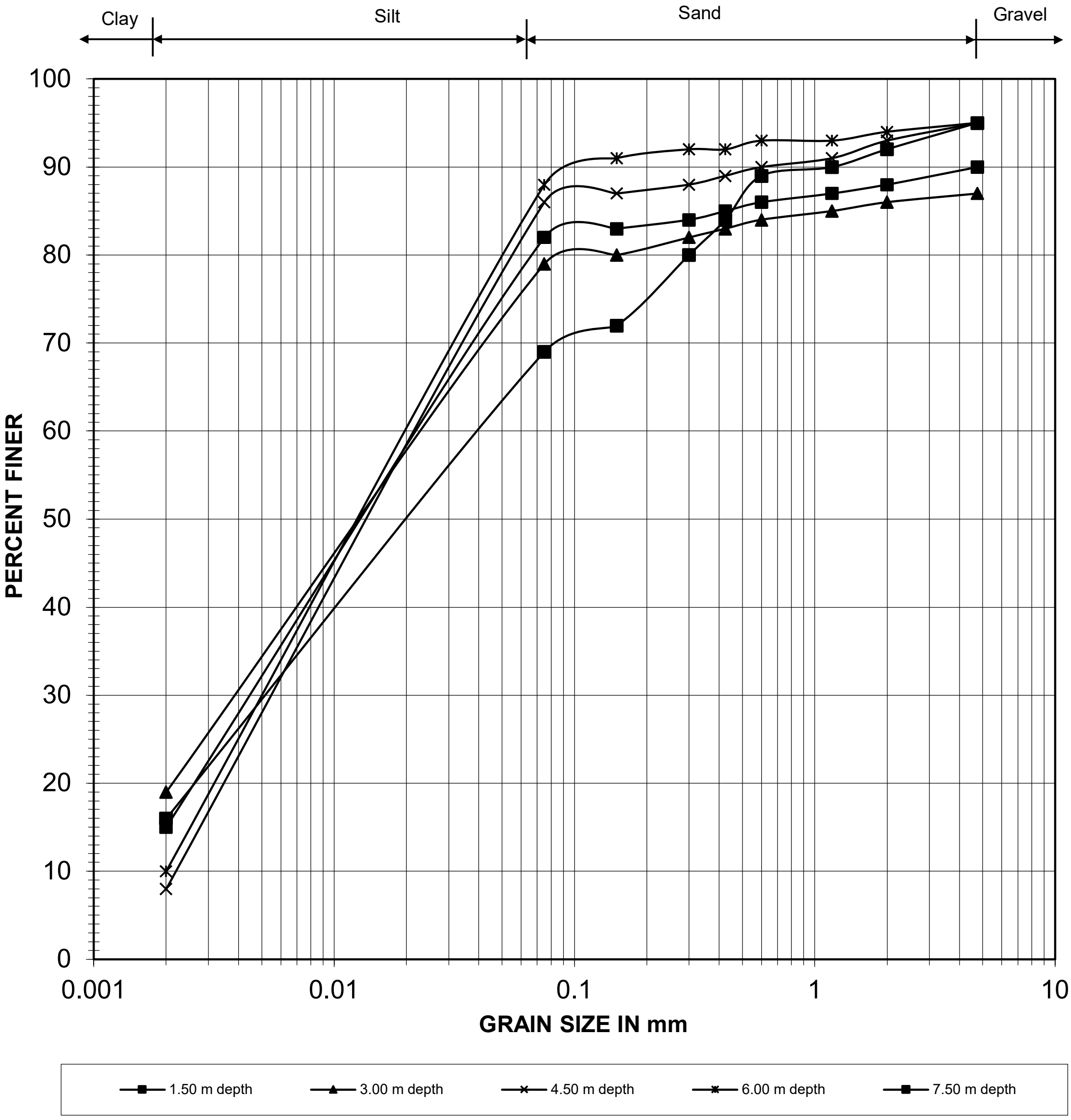
Computation of Liquefaction Potential as per IS-1893 (PART-1): 2016 | Table No.-9

Borehole No.	BH-A2
Borehole dia.	150 mm
Design Water Table (m)	1.60

Computation Depth (h)	Bulk/Saturated density (t/m3)	Fine Content (%)	PGA a_{max}/g	Earthquake magnitude (Mw)	Stress reduction coefficient (rd)	Total overburden (σo)	Pore Water Pressure (u_o)	Eff. overburden (σ_o)'	Cyclic Stress ratio (CSR)	SPT Value (N)	C_N	C_{SS}	C_{RL}	C_{BD}	C_{60}	NC_{60}	SPT Corrected $(N1)_{60}$	α	β	$(N1)_{60cs}$	$CRR_{M=7.5}$	Relative Density, Dr%	f	K_σ	K_α	MSF	CRR	FS_L	REMARK
1.50	1.72	82	0.16	6.5	0.989	2.58	-	2.58	0.103	8	1.70	1.0	0.75	1.05	0.58	4.63	7.9	5.00	1.20	14	0.15	NA	NA	1.00	1.0	1.44	0.22	2.17	N L
3.00	1.97	79	0.16	6.5	0.977	5.50	1.40	4.10	0.136	20	1.56	1.0	0.80	1.05	0.62	12.35	19.3	5.00	1.20	28	0.37	NA	NA	1.00	1.0	1.44	0.54	3.96	N L
4.50	1.97	86	0.16	6.5	0.966	8.45	2.90	5.55	0.153	28	1.34	1.0	0.85	1.05	0.66	18.37	24.6	5.00	1.20	35	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
6.00	1.97	88	0.16	6.5	0.954	11.40	4.40	7.00	0.162	34	1.20	1.0	0.95	1.05	0.73	24.93	29.8	5.00	1.20	41	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
7.50	1.97	69	0.16	6.5	0.943	14.36	5.90	8.46	0.166	39	1.09	1.0	0.95	1.05	0.73	28.59	31.1	5.00	1.20	42	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
9.00	1.97	46	0.16	6.5	0.931	17.32	7.40	9.92	0.169	47	1.00	1.0	0.95	1.05	0.73	34.46	34.6	5.00	1.20	47	NA	69.60	0.65	1.00	1.0	1.44	NA	>1.0	N L
10.50	1.97	40	0.16	6.5	0.894	20.28	8.90	11.38	0.166	49	0.94	1.0	1.00	1.05	0.77	37.82	35.4	5.00	1.20	48	NA	70.45	0.65	0.96	1.0	1.44	NA	>1.0	N L
12.00	1.97	9	0.16	6.5	0.854	23.24	10.40	12.84	0.161	55	0.88	1.0	1.00	1.05	0.77	42.45	37.5	0.56	1.02	39	NA	72.46	0.64	0.91	1.0	1.44	NA	>1.0	N L
13.50	1.97	8	0.16	6.5	0.814	26.20	11.90	14.30	0.155	64	0.84	1.0	1.00	1.05	0.77	49.39	41.3	0.30	1.01	42	NA	76.31	0.62	0.87	1.0	1.44	NA	>1.0	N L
15.00	1.97	7	0.16	6.5	0.774	29.16	13.40	15.76	0.149	67	0.80	1.0	1.00	1.05	0.77	51.71	41.2	0.12	1.01	42	NA	76.19	0.62	0.84	1.0	1.44	NA	>1.0	N L
16.50	1.98	92	0.16	6.5	0.733	32.13	14.90	17.23	0.142	47	0.76	1.0	1.00	1.05	0.77	36.27	27.6	5.00	1.20	38	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
18.00	1.98	89	0.16	6.5	0.693	35.10	16.40	18.70	0.135	42	0.73	1.0	1.00	1.05	0.77	32.41	23.7	5.00	1.20	33	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
19.50	1.97	95	0.16	6.5	0.653	38.06	17.90	20.16	0.128	41	0.70	1.0	1.00	1.05	0.77	31.64	22.3	5.00	1.20	32	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
21.00	1.97	95	0.16	6.5	0.613	41.02	19.40	21.62	0.121	49	0.68	1.0	1.00	1.05	0.77	37.82	25.7	5.00	1.20	36	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
22.50	1.98	92	0.16	6.5	0.573	43.99	20.90	23.09	0.114	55	0.66	1.0	1.00	1.05	0.77	42.45	27.9	5.00	1.20	39	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
24.00	1.98	96	0.16	6.5	0.552	46.96	22.40	24.56	0.110	60	0.64	1.0	1.00	1.05	0.77	46.31	29.5	5.00	1.20	40	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
25.50	1.98	61	0.16	6.5	0.540	49.93	23.90	26.03	0.108	53	0.62	1.0	1.00	1.05	0.77	40.90	25.4	5.00	1.20	35	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
27.00	1.98	60	0.16	6.5	0.528	52.91	25.40	27.51	0.106	51	0.60	1.0	1.00	1.05	0.77	39.36	23.7	5.00	1.20	33	NA	NA	NA	1.00	1.0	1.44	NA	>1.0	N L
28.50	1.98	33	0.16	6.5	0.516	55.88	26.90	28.98	0.103	76	0.59	1.0	1.00	1.05	0.77	58.65	34.5	4.88	1.18	46	NA	69.45	0.65	0.69	1.0	1.44	NA	>1.0	N L
29.55	1.98	20	0.16	6.5	0.508	57.96	27.95	30.01	0.102	94	0.58	1.0	1.00	1.05	0.77	72.54	41.9	3.61	1.08	49	NA	76.87	0.62	0.66	1.0	1.44	NA	>1.0	N L

Note:			
C_{HT} =	Correction factor for non standard weight = 0.75 for drop hammer with rope and pulley		
C_{HW} =	Correction factor for non standard height of fall =HW/48387		
Kσ	Correction for high overburden stress (for effective oberburden pressure>10 T/m^2)		
Kα	Correction for static shear stress is required only for sloping ground		
L	Liquefiable	NL	Non-Liquefiable

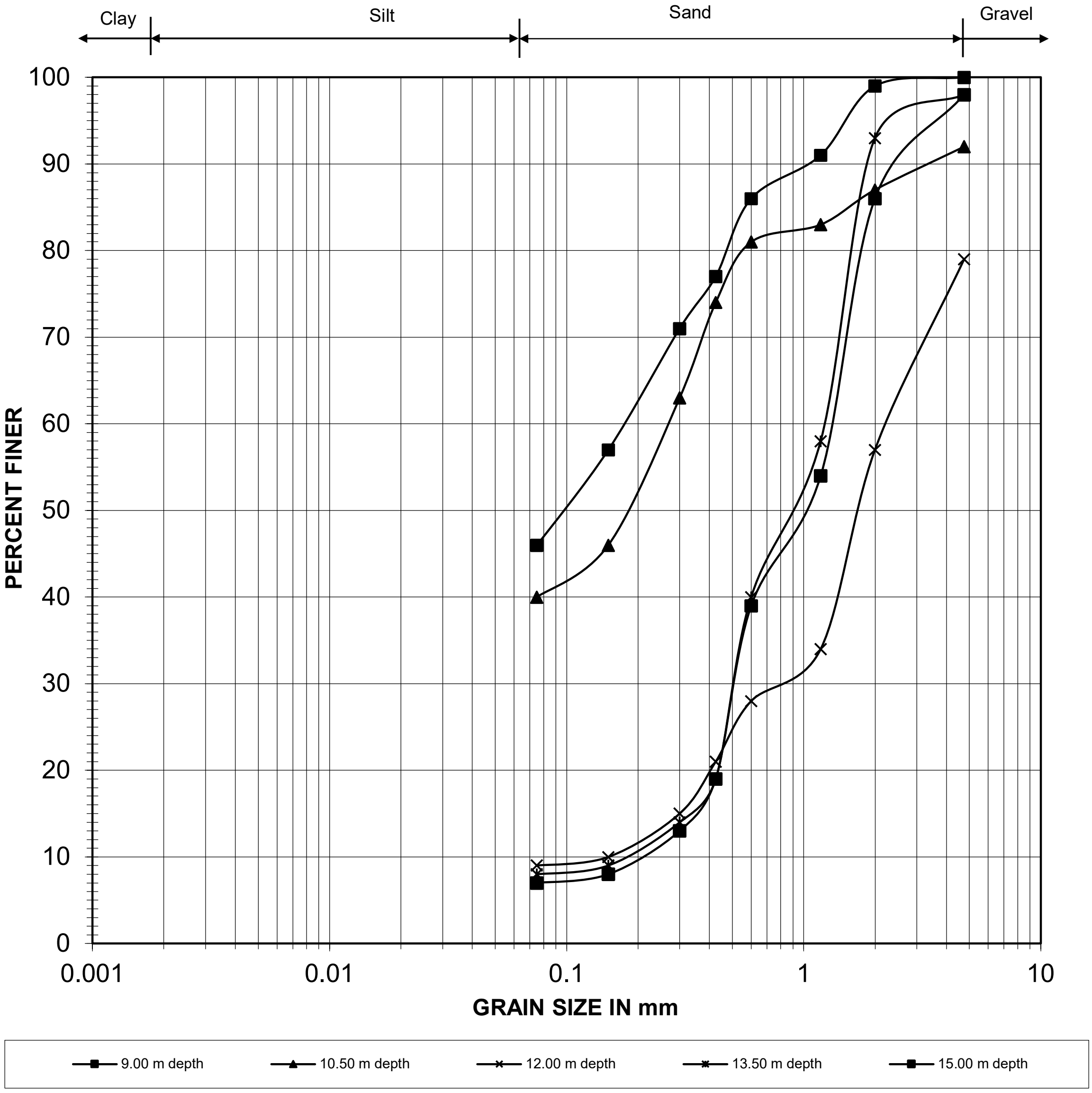
PARTICLE SIZE DISTRIBUTION CURVE for BH-A2



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
1.50 m depth	10	8	67	15
3.00 m depth	13	8	60	19
4.50 m depth	5	9	78	8
6.00 m depth	5	7	78	10
7.50 m depth	5	26	53	16

Fig. - 11

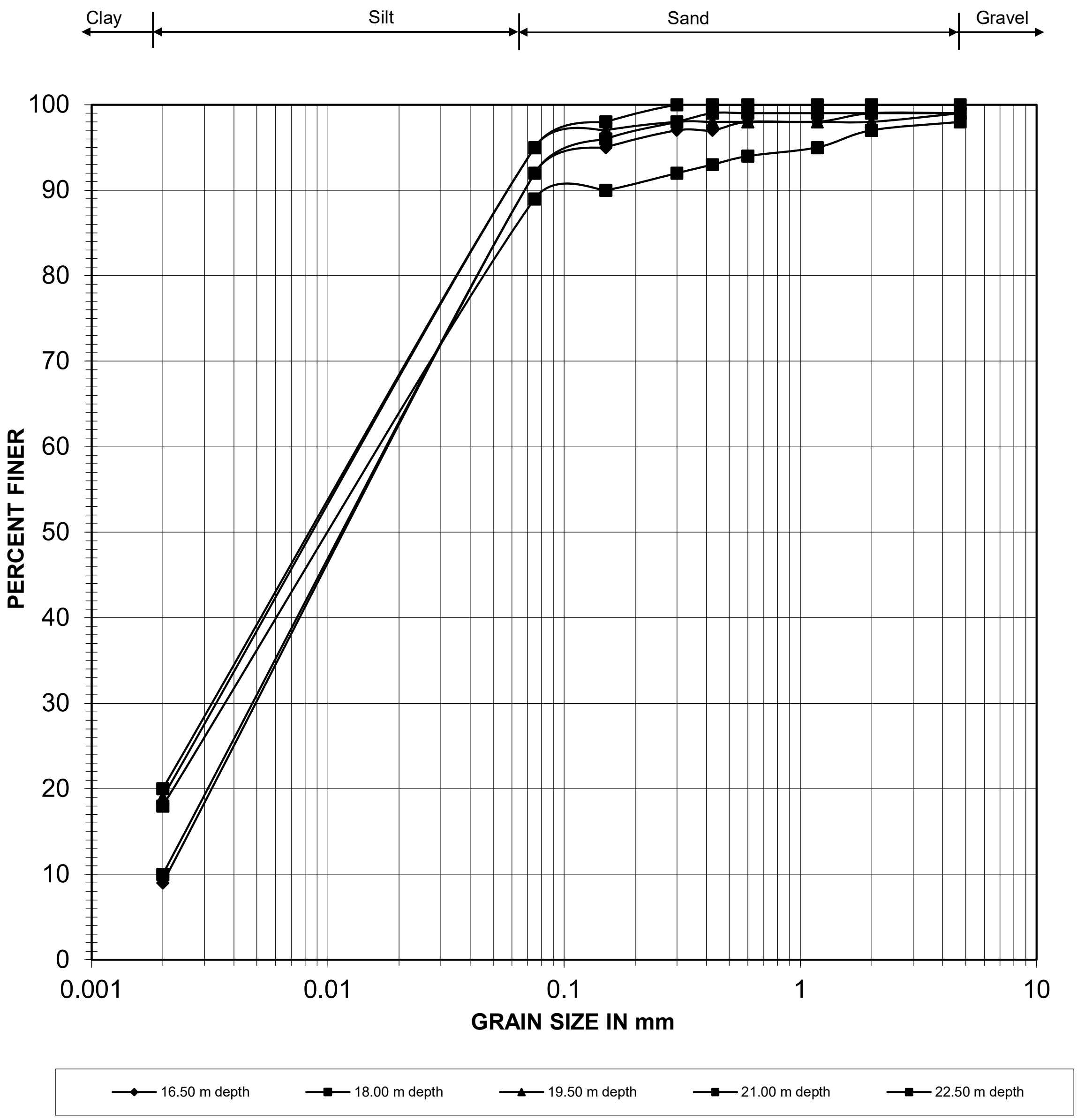
PARTICLE SIZE DISTRIBUTION CURVE for BH-A2



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
9.00 m depth	0	54	46	0
10.50 m depth	8	52	40	0
12.00 m depth	21	70	9	0
13.50 m depth	2	90	8	0
15.00 m depth	2	91	7	0

Fig. - 12

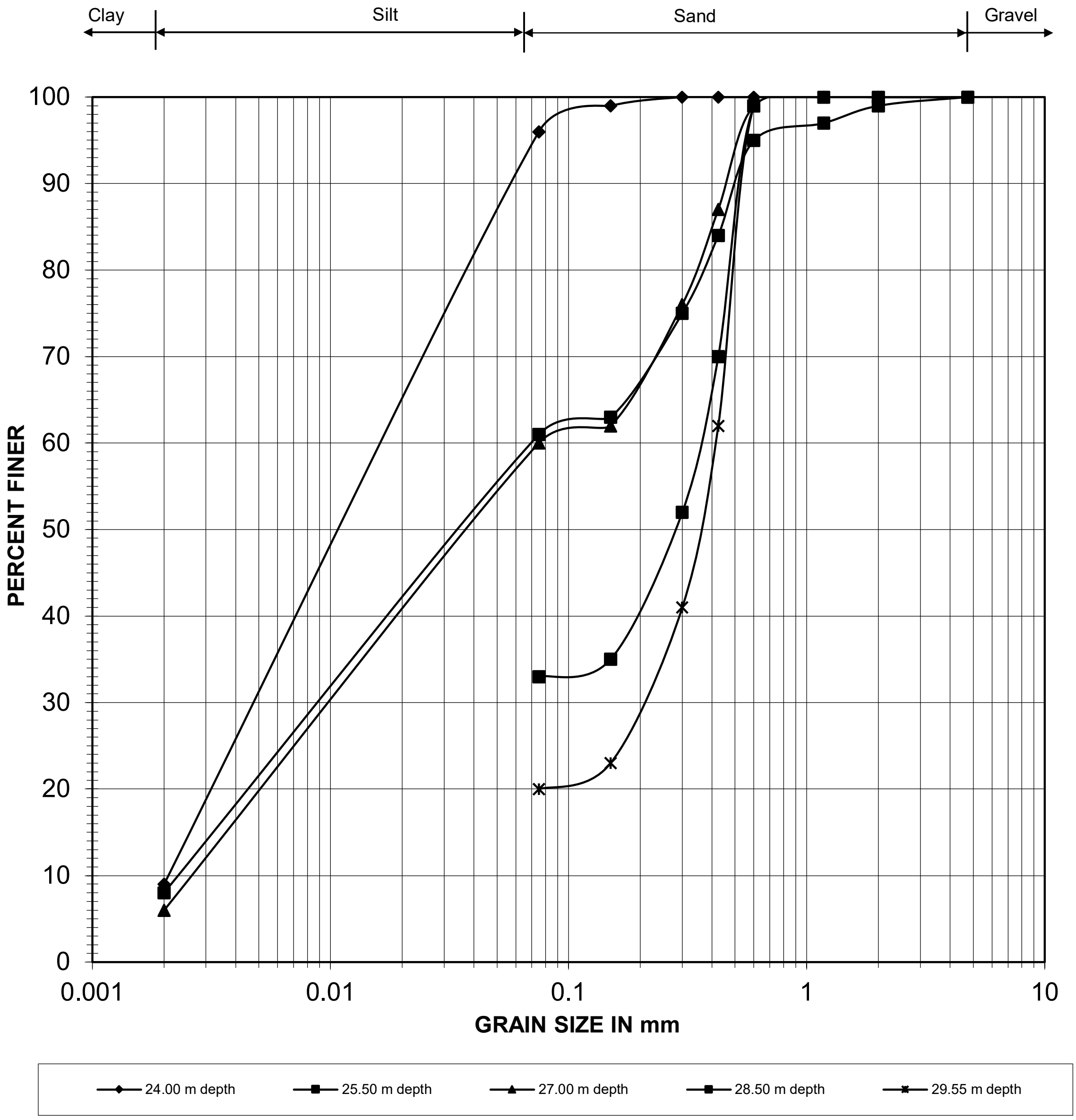
PARTICLE SIZE DISTRIBUTION CURVE for BH-A2



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
16.50 m depth	1	7	83	9
18.00 m depth	2	9	71	18
19.50 m depth	1	4	76	19
21.00 m depth	0	5	75	20
22.50 m depth	1	7	82	10

Fig. - 13

PARTICLE SIZE DISTRIBUTION CURVE for BH-A2



Depth	Gravel (%)	Sand (%)	Silt (%)	Clay (%)
24.00 m depth	0	4	87	9
25.50 m depth	0	39	53	8
27.00 m depth	0	40	54	6
28.50 m depth	0	67	33	0
29.55 m depth	0	80	20	0

Fig. - 14

VARIATION OF SPT 'N' WITH DEPTH for BH-A2

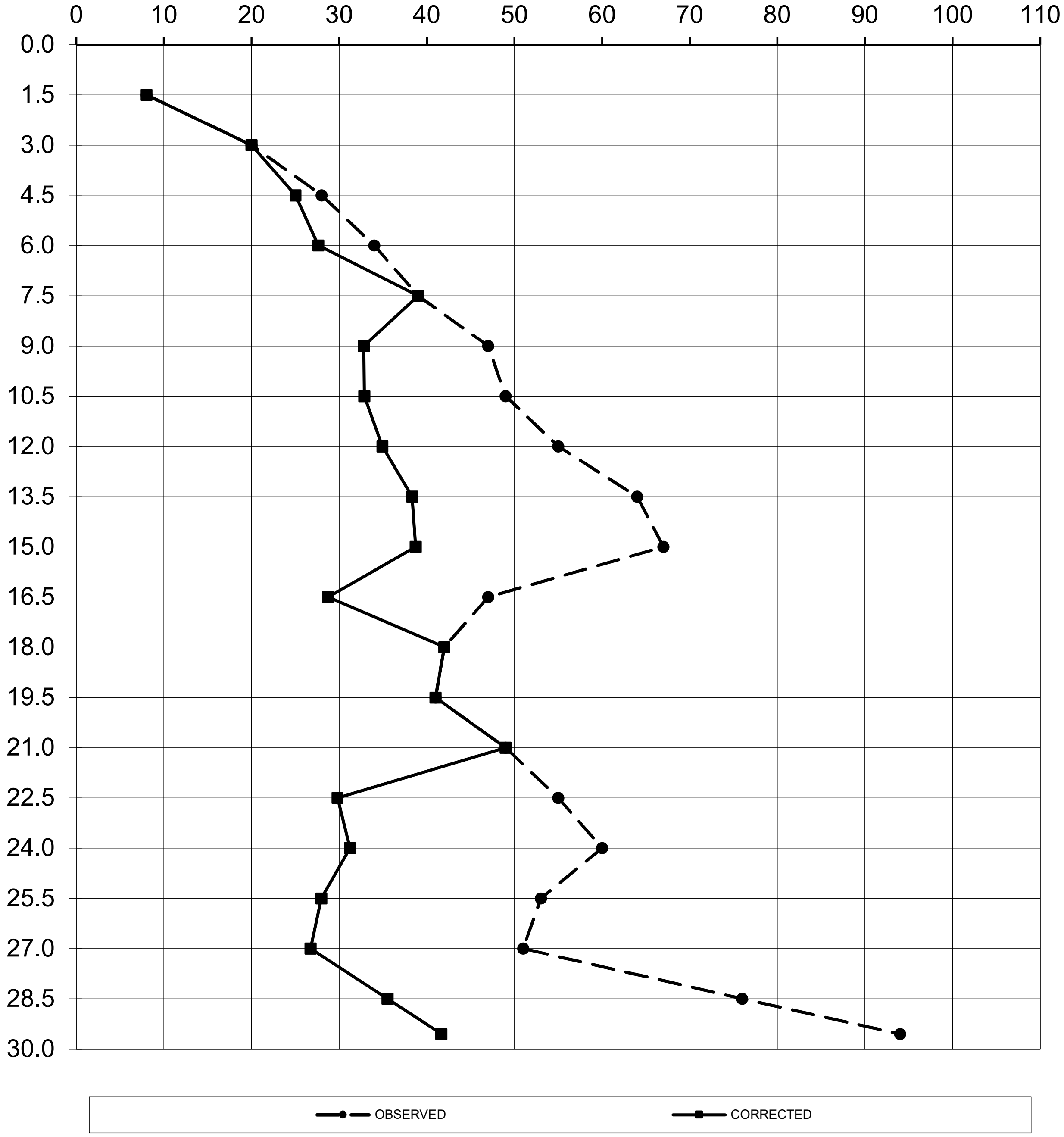


Fig. - 15

Annexure- A
Bearing Capacity Calculations

ANNEXURE-A
Bearing Capacity Calculations
(As per IS: 6403-1981)

I. Shear Failure Criterion: for BH-A1

Founding level below EGL (m)	=	**2.00**	**3.00**
Average Bulk density [t/m^3]	=	1.70	1.70
Average Dry density [t/m^3]	=	1.52	1.52
C [t/m^2]	=	0.35	0.25
ϕ[Degree]	=	29°	29°

ϕ = 29° $\Rightarrow N_c$ = 15.10 N_q = 6.58 N_γ = 5.61

BH No.	Founding Level Below EGL (m)	Depth of Foundation (m)	Size of Foundation (m)	S_c	S_q	S_γ	d_c	$d_q = d_\gamma$	q net safe (t/m^2)
BH-A1	2.00	2.00	8.00 x 8.00	1.30	1.20	0.80	1.085	1.042	13.26
	3.00	3.00	8.00 x 8.00	1.30	1.20	0.80	1.127	1.064	14.69

The values of bearing capacity factors N_c, N_q and N_γ have been arrived at from table 1 of IS: 6403-1981.

The depth factors d_c, d_q and d_γ have been calculated as per clause 5.1.2.2 of IS: 6403-1981.

The Shape factors S_c, S_q, S_γ have taken from clause 5.1.2.1 of IS: 6403-1981.

II. Settlement Criterion:

As per IS: 8009-1976 Part-I- Clause-9.1.4, the settlement for different width has been computed. For the allowable total settlement of 50mm for isolated footing and 75mm for raft foundation (as per IS:1904) the safe bearing pressure is computed and tabulated as following:

BH No.	Founding Level Below EGL (m)	Size of Foundation (m)	N	Settlement in mm for a pressure of 10 t/m^2	Safe Bearing Pressure (t/m^2)
BH-A1	2.00	8.00 x 8.00	18	30.14	16.59
	3.00	8.00 x 8.00		28.80	17.36

Hence the values of safe bearing capacity as chosen the minimum value (rounded off) from above two criterions are as following:

BH No.	Founding Level below EGL (m)	Depth of Foundation (m)	Size of Foundation (m)	$q_{net\ safe}$ (t/m²)	Safe Bearing pressure (t/m²)	$q_{allowable}$ (t/m²)
BH-A1	2.00	2.00	8.00 x 8.00	13.26	16.59	13.25
	3.00	3.00	8.00 x 8.00	14.69	17.36	14.50

ANNEXURE-A
Bearing Capacity Calculation (BH - P1)

Foundation Size:	8.00 x 8.00	m
Depth of Foundation (m):	2.00	m
Founding level below EGL (m):	2.00	m

GL Design GWT

8.00

Silty Clay

2.00 m

γ'= 0.70 t/m^3

γ= 1.70 t/m^3

Layer - I

SILTY CLAY

Z_1

N_{av}=	21	
ϕ_{av}=	8	degree
c_{av}=	8.1	t/m^2
γ'=	1.00	t/m^3
γ=	1.71	t/m^3

7.50 m

SILTY / FINE SAND

Layer - II

......Z_2

N_{av}=	26	
ϕ_{av}=	31	degree
c_{av}=	0	t/m^2
γ'=	1.00	t/m^3
γ=	1.74	t/m^3

14.00 m

Safe Bearing Capacity from Shear Failure

Design ϕ= 12 degree

Design C= 4.0 t/m^2

Q(safe)= $(2/3cN_cs_cd_ci_c+(\gamma*D)(N_q-1)s_qd_qi_q+0.5B\gamma N_\gamma s_\gamma d_\gamma i_\gamma W/FS$

FS= 2.5 w= 0.5

N_c= 7.55 N_q= 2.07 N_γ= 0.87

S_c= 1.30 S_q= 1.20 S_γ= 0.80

dc= 1+0.2*(D/B)*tan(45+φ/2)= 1.058

dq=dγ= 1+0.1*(D/B)*tan(45+φ/2) 1.029

ic=iq= (1-α/90)^2= 1.00 ig= (1-α/φ)^2= 1.00 α=0

Q_{safe-I}= 12.76 t/m^2

Design Bearing Capacity= 12.76 t/m^2

Settlement Calculation

Settlement for Layer - I

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= 0.00079 μ_g= 0.7

1st layer I = 0.85

δ (mm)= [2.303*(H/C)*$\log_{10}$$\{(p_o+\Delta p)/p_o\}$]*$d_f$*Rigidity Factor (0.8) for sand

C=1.5*(C_{kd}/p_o)= 198.95 C_{kd}/N= 30 t/m^2

p_o= 4.75 p = 12.76 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.83

δ_1 (mm)= 21.72

Settlement for Layer-II

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= - μ_g= 0.7

2nd layer I = 0.292

δ (mm)= [2.303*(H/C)*$\log_{10}$((p$_o$+Δp)/p$_o$)]*d_f*Rigidity Factor (0.8) for sand IS:8009 (Part I)

C=1.5*(C_{kd}/p_o)= 108.84 C_{kd}/N= 30 t/m^2

p_o= 10.75 p= 12.76 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.68

δ_2 (mm)= 9.71

Total settlement =	**31.43**	**mm**
Safe bearing pressure for 75 mm settlement =	**30.45**	**t/m^2**
Allowable bearing capacity for 50 mm settlement =	**12.75**	**t/m^2**

ANNEXURE-A
Bearing Capacity Calculation (BH - P1)

Foundation Size:	8.00 x 8.00	m
Depth of Foundation (m):	3.00	m
Founding level below EGL (m):	3.00	m

GL Design GWT

8.00

Silty Clay

3.00 m

γ'= 0.70 t/m^3

γ= 1.70 t/m^3

Layer - I Z_1

	SILTY	CLAY	
	N_{av}=	21	
	ϕ_{av}=	8	degree
	c_{av}=	8.1	t/m^2
	γ'=	1.00	t/m^3
7.50 m	γ=	1.71	t/m^3

Layer - II ……Z_2

	SILTY / FINE	SAND	
	N_{av}=	26	
	ϕ_{av}=	31	degree
	c_{av}=	0	t/m^2
	γ'=	1.00	t/m^3
15.00 m	γ=	1.74	t/m^3

Safe Bearing Capacity from Shear Failure

Design ϕ= 14 degree

Design C= 3.5 t/m^2

Q(safe)= $(2/3cN_cs_cd_ci_c+(\gamma*D)(N_{q-}1)s_qd_qi_q+0.5B\gamma N_\gamma s_\gamma d_\gamma i_{\gamma W}/FS$

FS=	2.5			w=	0.5
N_c=	8.10	N_q=	2.35	N_γ=	1.12
S_c=	1.30	S_q=	1.20	S_γ=	0.80

dc= 1+0.2*(D/B)*tan(45+φ/2)= 1.086

dq=dγ= 1+0.1*(D/B)*tan(45+φ/2) 1.043

ic=iq= (1-α/90)^2= 1.00 ig= (1-α/φ)^2= 1.00 α=0

Q_{safe-I}= 13.51 t/m^2

Design Bearing Capacity= 13.51 t/m^2

Settlement Calculation

Settlement for Layer - I

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= 0.00079 μ_g= 0.7

1st layer I = 0.88

δ (mm)= [2.303*(H/C)*$log_{10(}$(p_o+Δp)/$p_{o)}$]*d_f*Rigidity Factor (0.8) for sand

C=1.5*(C_{kd}/p_o)= 180.00 C_{kd}/N= 30 t/m^2

p_o= 5.25 p = 13.51 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.81

δ_1 (mm)= 19.21

Settlement for Layer-II

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= - μ_g= 0.7

2nd layer I = 0.312

δ (mm)= [2.303*(H/C)*$\log_{10}$((p$_o$+Δp)/p$_o$)]*d_f*Rigidity Factor (0.8) for sand IS:8009 (Part I)

C=1.5*(C_{kd}/p_o)= 104.00 C_{kd}/N= 30 t/m^2

p_o= 11.25 p= 13.51 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.67

δ_2 (mm)= 12.37

Total settlement =	**31.58**	**mm**
Safe bearing pressure for 75 mm settlement =	**32.08**	**t/m^2**
Allowable bearing capacity for 50 mm settlement =	**13.50**	**t/m^2**

ANNEXURE-A
Bearing Capacity Calculation (BH - A2)

Foundation Size:	8.00 x 8.00	m
Depth of Foundation (m):	2.00	m
Founding level below EGL (m):	2.00	m

GL Design GWT

8.00

Silty Clay

2.00 m

γ'= 0.70 t/m^3

γ= 1.70 t/m^3

Layer - I — SILTY CLAY — Z_1

N_{av}=	28	
ϕ_{av}=	6	degree
c_{av}=	7.6	t/m^2
γ'=	1.00	t/m^3
γ=	1.72	t/m^3

8.00 m

Layer - II — SILTY / COARSE SAND — ……Z_2

N_{av}=	35	
ϕ_{av}=	31	degree
c_{av}=	0	t/m^2
γ'=	1.00	t/m^3
γ=	1.75	t/m^3

14.00 m

Safe Bearing Capacity from Shear Failure

Design ϕ= 9 degree

Design C= 4.5 t/m^2

Q(safe)= $(2/3cN_cs_cd_ci_c+(\gamma*D)(N_q-1)s_qd_qi_q+0.5B\gamma N_\gamma s_\gamma d_\gamma i_\gamma W)/FS$

FS= 2.5 w= 0.5

N_c=	6.82	N_q=	1.72	N_γ=	0.57
S_c=	1.30	S_q=	1.20	S_γ=	0.80

dc= 1+0.2*(D/B)*tan(45+φ/2)= 1.056

dq=dγ= 1+0.1*(D/B)*tan(45+φ/2) 1.028

ic=iq= (1-α/90)^2= 1.00 ig= (1-α/φ)^2= 1.00 α=0

Q_{safe-I}= 12.38 t/m^2

Design Bearing Capacity= 12.38 t/m^2

Settlement Calculation

Settlement for Layer - I

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= 0.00089 μ_g= 0.7

1st layer I = 0.80

δ (mm)= [2.303*(H/C)*$\log_{10}$((p$_o$+Δp)/p$_o$)]*d_f*Rigidity Factor (0.8) for sand

C=1.5*(C_{kd}/p_o)= 252.00 C_{kd}/N= 30 t/m^2

p_o= 5 p = 12.38 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.81

δ_1 (mm)= 24.19

Settlement for Layer-II

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= - μ_g= 0.7

2nd layer I = 0.288

δ (mm)= [2.303*(H/C)*$\log_{10}$(p_o+Δp)/p_o]*d_f*Rigidity Factor (0.8) for sand IS:8009 (Part I)

C=1.5*(C_{kd}/p_o)= 143.18 C_{kd}/N= 30 t/m^2

p_o= 11 p= 12.38 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.68

δ_2 (mm)= 6.39

Total settlement =	**30.58**	**mm**
Safe bearing pressure for 75 mm settlement =	**30.35**	**t/m^2**
Allowable bearing capacity for 50 mm settlement =	**12.25**	**t/m^2**

ANNEXURE-A
Bearing Capacity Calculation (BH - A2)

Foundation Size:	8.00 x 8.00	m
Depth of Foundation (m):	3.00	m
Founding level below EGL (m):	3.00	m

GL Design GWT

8.00

Silty Clay

3.00 m

γ'= 0.70 t/m^3

γ= 1.70 t/m^3

Z_1

Layer - I

	SILTY	CLAY	
	N_{av}=	28	
	ϕ_{av}=	6	degree
	c_{av}=	7.6	t/m^2
	γ'=	1.00	t/m^3
8.00 m	γ=	1.72	t/m^3

Layer - II

......Z_2

	SILTY / COARSE	SAND	
	N_{av}=	35	
	ϕ_{av}=	31	degree
	c_{av}=	0	t/m^2
	γ'=	1.00	t/m^3
15.00 m	γ=	1.75	t/m^3

Safe Bearing Capacity from Shear Failure

Design ϕ= 14 degree

Design C= 3.5 t/m^2

Q(safe)= $(2/3cN_cs_cd_ci_c+(\gamma*D)(N_{q-}1)s_qd_qi_q+0.5B\gamma N_\gamma s_\gamma d_\gamma i_{\gamma W}/FS$

FS=	2.5			w=	0.5
N_c=	8.10	N_q=	2.35	N_γ=	1.12
S_c=	1.30	S_q=	1.20	S_γ=	0.80

dc= 1+0.2*(D/B)*tan(45+φ/2)= 1.083

dq=dγ= 1+0.1*(D/B)*tan(45+φ/2) 1.042

ic=iq= (1-α/90)^2= 1.00 ig= (1-α/φ)^2= 1.00 α=0

Q_{safe-I}= 13.47 t/m^2

Design Bearing Capacity= 13.47 t/m^2

Settlement Calculation

Settlement for Layer - I

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= 0.00089 μ_g= 0.7

1st layer I = 0.86

δ (mm)= [2.303*(H/C)*log$_{10}$((p$_o$+Δp)/p$_o$)]*d$_f$*Rigidity Factor (0.8) for sand

C=1.5*(C_{kd}/p_o)= 229.09 C_{kd}/N= 30 t/m^2

p_o= 5.5 p = 13.47 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.80

δ_1 (mm)= 23.18

Settlement for Layer-II

δ (mm) = m_v*H*Δp*μ_g*d_f*Rigidity Factor (0.8) for clay

m_v= - μ_g= 0.7

2nd layer I = 0.312

δ (mm)= [2.303*(H/C)*$\log_{10}$(p_o+Δp)/p_o]*d_f*Rigidity Factor (0.8) for sand IS:8009 (Part I)

C=1.5*(C_{kd}/p_o)= 136.96 C_{kd}/N= 30 t/m^2

p_o= 11.5 p= 13.47 t/m^2

Rigidity factor= 0.8 Depth Factor, d_f= 0.67

δ_2 (mm)= 8.54

Total settlement =	**31.71**	**mm**
Safe bearing pressure for 75 mm settlement =	**31.86**	**t/m^2**
Allowable bearing capacity for 50 mm settlement =	**13.25**	**t/m^2**

Annexure- B
Pile Capacity Calculations

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78] | Ch. 44+758 | BH-A1

Pile Dia. (mm) =	1000	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.00	Scouring Depth (m) =	-

Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m^2)	Angle of Shear Resistance, ϕ (^{0})	Bulk/Submerged Density (t/m^3)	Eff. Over-burden Pressure at Pile Tip (t/m^2)	Bearing Capacity Factors: N_c	N_q	N_g	Ultimate Base Resistance, $P_{pu} = A_p*(c.N_c+q.N_q+0.5.\gamma.B.N_\gamma)$: A_p (m^2)	$c.N_c$	$q.N_q$	$0.5.\gamma.B.N_\gamma$	P_{pu} (t)	Ultimate Shaft Friction, P_{su} =(S($K_s.P_{di}$.Tand).A_{si} + a.c.A_s): Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m^2)	A_{si} (m^2)	($K_s.P_{di}$.Tand)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}	Total Ultimate Capacity, [P_u = (P_{pu} + P_{su})] (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression P_s =[(P_{pu} / FOS + P_{su} / FOS) - W_p] (t)	Safe Axial Capacity in uplift, P_u = [P_{su} / FOS) + W_p] (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
1	2.00	4.00	2.00	0.40	28	1.00	2.00	9.0	15.6	16.7	0.79	3.6	31.2	8.36	**33.9**	1.00	6.28	0.80	1.00	0.40	1.20	**7.5**					2.00		
2	4.00	6.00	2.00	0.40	28	1.00	4.00	9.0	15.6	16.7	0.79	3.6	62.4	8.36	**58.4**	3.00	6.28	2.39	1.00	0.40	2.79	**25.1**	83.5	5	29	12	4.00		
3	6.00	8.00	2.00	0.50	29	1.00	6.00	9.0	18.1	19.3	0.79	4.5	108.5	9.67	**96.3**	5.00	6.28	4.16	1.00	0.50	4.66	**54.3**	150.7	7	53	23	6.00		
4	8.00	10.00	2.00	0.00	31	1.00	8.00	9.0	24.0	26.0	0.79	0.0	192.0	13.00	**161.0**	7.00	6.28	6.31	1.00	0.00	6.31	**94.0**	255.0	9	93	36	8.00		
5	10.00	12.00	2.00	0.00	31	1.00	10.00	9.0	24.0	26.0	0.79	0.0	240.0	13.00	**198.7**	9.00	6.28	8.11	1.00	0.00	8.11	**144.9**	343.6	12	126	53	10.00		
6	12.00	14.00	2.00	0.00	32	1.00	12.00	9.0	30.0	30.0	0.79	0.0	360.0	15.00	**294.5**	11.00	6.28	10.31	1.00	0.00	10.31	**209.7**	504.2	14	188	74	12.00		
7	14.00	15.00	1.00	0.00	32	1.00	13.00	9.0	30.0	30.0	0.79	0.0	390.0	15.00	**318.1**	12.50	3.14	11.72	1.00	0.00	11.72	**246.5**	564.6	15	211	86	13.00		
8	15.00	16.00	1.00	0.00	32	1.00	14.00	9.0	30.0	30.0	0.79	0.0	420.0	15.00	**341.6**	13.50	3.14	12.65	1.00	0.00	12.65	**286.3**	627.9	16	235	98	14.00		
9	16.00	17.00	1.00	0.80	29	1.00	15.00	9.0	18.1	19.3	0.79	7.2	271.2	9.67	**226.2**	15.00	3.14	12.47	1.00	0.80	13.27	**328.0**	554.2	18	204	111	15.00		
10	17.00	18.00	1.00	15.70	9	1.00	15.00	9.0	2.3	1.1	0.79	141.3	34.4	0.53	**138.4**	15.00	3.14	3.56	0.36	5.65	9.22	**356.9**	495.3	19	179	121	16.00	170	120
11	18.00	19.00	1.00	15.70	9	1.00	15.00	9.0	2.3	1.1	0.79	141.3	34.4	0.53	**138.4**	15.00	3.14	3.56	0.36	5.65	9.22	**385.9**	524.3	20	190	130	17.00		
12	19.00	20.00	1.00	1.00	28	1.00	15.00	9.0	15.6	16.7	0.79	9.0	234.0	8.36	**138.4**	15.00	3.14	11.96	1.00	1.00	12.96	**426.6**	565.0	21	205	143	18.00	200	140
13	20.00	21.00	1.00	1.00	28	1.00	15.00	9.0	15.6	16.7	0.79	9.0	234.0	8.36	**138.4**	15.00	3.14	11.96	1.00	1.00	12.96	**467.3**	605.7	22	220	156	19.00	210	150
14	21.00	22.00	1.00	1.00	28	1.00	15.00	9.0	15.6	16.7	0.79	9.0	234.0	8.36	**138.4**	15.00	3.14	11.96	1.00	1.00	12.96	**508.1**	646.4	24	235	169	20.00		
15	22.00	23.00	1.00	15.70	9	1.00	15.00	9.0	2.3	1.1	0.79	141.3	34.4	0.53	**138.4**	15.00	3.14	3.56	0.36	5.65	9.22	**537.0**	675.4	25	245	178	21.00	240	170

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm^3)	Young's modulus of pile material E (kg/cm^2)	Moment of Inertia I (cm^4)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	100	0.24	320000	4908739	365.50	1.91	698	0	0.000	1.00	1.00	13850	13.00	14.62
Fixed	100	0.24	320000	4908739	365.50	2.19	799	0	0.000	1.00	1.00	37006	37.00	14.62

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78]	Ch. 44+758	BH-A1

Pile Dia. (mm) =	1200	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.30	Scouring Depth (m) =	-

Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m²)	Angle of Shear Resistance, φ (°)	Bulk/Submerged Density (t/m³)	Eff. Over-burden Pressure at Pile Tip (t/m²)	Bearing Capacity Factors: N_c	N_q	N_g	Ultimate Base Resistance, $P_{pu} = A_p*(c.N_c+q.N_q+0.5.\gamma.B.N_\gamma)$: A_p (m²)	c.N_c	q.N_q	0.5.γ.B.N_γ	P_{pu} (t)	Ultimate Shaft Friction, P_{su} =(S(K_s.P_{di}.Tand).A_{si} + a.c.A_s): Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m²)	A_{si} (m²)	(K_s.P_{di}.Tand)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}	Total Ultimate Capacity, [P_u = (P_{pu} + P_{su})] (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression P_s = [(P_{pu} / FOS + P_{su} / FOS) - W_p] (t)	Safe Axial Capacity in uplift, P_u = [P_{su} / FOS) + W_p] (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
1	2.30	4.00	1.70	0.40	28	1.00	1.70	9.0	15.6	16.7	1.13	3.6	26.5	10.03	**45.4**	0.85	6.41	0.68	1.00	0.40	1.08	**6.9**					1.70		
2	4.00	6.00	2.00	0.40	28	1.00	3.70	9.0	15.6	16.7	1.13	3.6	57.7	10.03	**80.7**	2.70	7.54	2.15	1.00	0.40	2.55	**26.2**	106.9	6	36	14	3.70		
3	6.00	8.00	2.00	0.50	29	1.00	5.70	9.0	18.1	19.3	1.13	4.5	103.1	11.60	**134.8**	4.70	7.54	3.91	1.00	0.50	4.41	**59.4**	194.2	10	68	27	5.70		
4	8.00	10.00	2.00	0.00	31	1.00	7.70	9.0	24.0	26.0	1.13	0.0	184.8	15.60	**226.6**	6.70	7.54	6.04	1.00	0.00	6.04	**104.9**	331.6	13	120	43	7.70		
5	10.00	12.00	2.00	0.00	31	1.00	9.70	9.0	24.0	26.0	1.13	0.0	232.8	15.60	**280.9**	8.70	7.54	7.84	1.00	0.00	7.84	**164.0**	445.0	16	162	63	9.70		
6	12.00	14.00	2.00	0.00	32	1.00	11.70	9.0	30.0	30.0	1.13	0.0	351.0	18.00	**417.3**	10.70	7.54	10.03	1.00	0.00	10.03	**239.7**	657.0	20	243	88	11.70		
7	14.00	15.00	1.00	0.00	32	1.00	12.70	9.0	30.0	30.0	1.13	0.0	381.0	18.00	**451.3**	12.20	3.77	11.44	1.00	0.00	11.44	**282.8**	734.0	22	272	102	12.70		
8	15.00	16.00	1.00	0.00	32	1.00	13.70	9.0	30.0	30.0	1.13	0.0	411.0	18.00	**485.2**	13.20	3.77	12.37	1.00	0.00	12.37	**329.4**	814.6	23	303	117	13.70		
9	16.00	17.00	1.00	0.80	29	1.00	14.70	9.0	18.1	19.3	1.13	7.2	265.8	11.60	**321.9**	14.20	3.77	11.81	1.00	0.80	12.61	**376.9**	698.8	25	255	133	14.70		
10	17.00	18.30	1.30	15.70	9	1.00	16.00	9.0	2.3	1.1	1.13	141.3	36.6	0.64	**202.0**	15.35	4.90	3.65	0.36	5.65	9.30	**422.5**	624.5	27	223	148	16.00	220	140
11	18.30	19.00	0.70	15.70	9	1.00	16.70	9.0	2.3	1.1	1.13	141.3	38.2	0.64	**203.8**	16.35	2.64	3.88	0.36	5.65	9.54	**447.7**	651.5	28	232	156	16.70		
12	19.00	19.30	0.30	1.00	28	1.00	17.00	9.0	15.6	16.7	1.13	9.0	265.2	10.03	**207.1**	16.85	1.13	13.44	1.00	1.00	14.44	**464.0**	671.2	29	240	161	17.00		
13	19.30	20.30	1.00	1.00	28	1.00	18.00	9.0	15.6	16.7	1.13	9.0	280.8	10.03	**207.1**	18.00	3.77	14.36	1.00	1.00	15.36	**521.9**	729.1	31	261	180	18.00	260	170
14	20.30	21.30	1.00	1.00	28	1.00	18.00	9.0	15.6	16.7	1.13	9.0	280.8	10.03	**207.1**	18.00	3.77	14.36	1.00	1.00	15.36	**579.8**	786.9	32	283	198	19.00	280	190
15	21.30	22.00	0.70	1.00	28	1.00	18.00	9.0	15.6	16.7	1.13	9.0	280.8	10.03	**207.1**	18.00	2.64	14.36	1.00	1.00	15.36	**620.3**	827.5	33	298	211	19.70		
16	22.00	23.30	1.30	15.70	9	1.00	18.00	9.0	2.3	1.1	1.13	141.3	41.2	0.64	**207.1**	18.00	4.90	4.28	0.36	5.65	9.93	**669.0**	876.1	36	315	227	21.00	310	220

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm³)	Young's modulus of pile material E (kg/cm²)	Moment of Inertia I (cm⁴)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	120	0.27	320000	10178761	413.45	1.91	790	0	0.000	1.20	1.20	23811	23.00	16.54
Fixed	120	0.27	320000	10178761	413.45	2.19	903	0	0.000	1.20	1.20	63618	63.00	16.54

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78]	Ch. 44+758	BH-P1

Pile Dia. (mm) =	1000	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.00	Scouring Depth (m) =	-

Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m^2)	Angle of Shear Resistance, ϕ (^{0})	Bulk/Submerged Density (t/m^3)	Eff. Over-burden Pressure at Pile Tip (t/m^2)	Bearing Capacity Factors			Ultimate Base Resistance, $P_{pu} = A_p*(c.N_c+q.N_q+0.5.\gamma.B.N_\gamma)$					Ultimate Shaft Friction, P_{su} =(S($K_s.P_{di}$.Tand).A_{si} + a.c.A_s)							Total Ultimate Capacity, [P_u = (P_{pu} + P_{su})] (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression P_s = [(P_{pu} / FOS + P_{su} / FOS) - W_p] (t)	Safe Axial Capacity in uplift, P_u = [P_{su} / FOS) + W_p] (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
								N_c	N_q	N_g	A_p (m^2)	c.N_c	q.N_q	0.5.γ.B.N_γ	P_{pu} (t)	Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m^2)	A_{si} (m^2)	(K_s.P_{di}.Tand)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}							
1	2.00	4.00	2.00	8.10	8	1.00	2.00	9.0	2.1	0.9	0.79	72.9	4.2	0.46	**60.9**	1.00	6.28	0.21	0.63	5.10	5.31	**33.4**					2.00		
2	4.00	6.00	2.00	8.10	8	1.00	4.00	9.0	2.1	0.9	0.79	72.9	8.4	0.46	**64.2**	3.00	6.28	0.63	0.63	5.10	5.74	**69.4**	133.7	5	49	25	4.00		
3	6.00	7.50	1.50	11.80	9	1.00	5.50	9.0	2.3	1.1	0.79	106.2	12.6	0.53	**93.7**	4.75	4.71	1.13	0.44	5.19	6.32	**99.2**	192.9	6	71	35	5.50		
4	7.50	9.00	1.50	0.00	31	1.00	7.00	9.0	24.0	26.0	0.79	0.0	168.0	13.00	**142.2**	6.25	4.71	5.63	1.00	0.00	5.63	**125.8**	267.9	8	99	44	7.00		
5	9.00	11.00	2.00	0.00	31	1.00	9.00	9.0	24.0	26.0	0.79	0.0	216.0	13.00	**179.9**	8.00	6.28	7.21	1.00	0.00	7.21	**171.1**	350.9	11	130	59	9.00		
6	11.00	13.00	2.00	0.00	31	1.00	11.00	9.0	24.0	26.0	0.79	0.0	264.0	13.00	**217.6**	10.00	6.28	9.01	1.00	0.00	9.01	**227.7**	445.2	13	165	78	11.00		
7	13.00	15.00	2.00	0.00	31	1.00	13.00	9.0	24.0	26.0	0.79	0.0	312.0	13.00	**255.3**	12.00	6.28	10.82	1.00	0.00	10.82	**295.6**	550.9	15	205	100	13.00		
8	15.00	16.00	1.00	0.00	32	1.00	14.00	9.0	30.0	30.0	0.79	0.0	420.0	15.00	**341.6**	13.50	3.14	12.65	1.00	0.00	12.65	**335.4**	677.0	16	254	112	14.00		
9	16.00	17.00	1.00	15.70	9	1.00	15.00	9.0	2.3	1.1	0.79	141.3	34.4	0.53	**138.4**	15.00	3.14	3.56	0.36	5.65	9.22	**364.3**	502.7	18	183	122	15.00	180	120
10	17.00	18.00	1.00	15.70	9	1.00	15.00	9.0	2.3	1.1	0.79	141.3	34.4	0.53	**138.4**	15.00	3.14	3.56	0.36	5.65	9.22	**393.3**	531.7	19	194	131	16.00		
11	18.00	19.00	1.00	0.90	28	1.00	15.00	9.0	15.6	16.7	0.79	8.1	234.0	8.36	**167.4**	15.00	3.14	11.96	1.00	0.90	12.86	**433.7**	601.1	20	220	144	17.00	220	140
12	19.00	20.00	1.00	0.90	28	1.00	15.00	9.0	15.6	16.7	0.79	8.1	234.0	8.36	**167.4**	15.00	3.14	11.96	1.00	0.90	12.86	**474.1**	641.5	21	235	157	18.00	230	150
13	20.00	21.00	1.00	0.90	28	1.00	15.00	9.0	15.6	16.7	0.79	8.1	234.0	8.36	**167.4**	15.00	3.14	11.96	1.00	0.90	12.86	**514.5**	681.9	22	250	169	19.00		
14	21.00	22.00	1.00	19.80	9	1.00	15.00	9.0	2.3	1.1	0.79	178.2	34.4	0.53	**167.4**	15.00	3.14	3.56	0.32	6.34	9.90	**545.6**	713.0	24	262	179	20.00	260	170

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm^3)	Young's modulus of pile material E (kg/cm^2)	Moment of Inertia I (cm^4)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	100	0.24	320000	4908739	365.50	1.91	698	0	0.000	1.00	1.00	13850	13.00	14.62
Fixed	100	0.24	320000	4908739	365.50	2.19	799	0	0.000	1.00	1.00	37006	37.00	14.62

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78]	Ch. 44+758	BH-P1

Pile Dia. (mm) =	1200	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.30	Scouring Depth (m) =	-

Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m^2)	Angle of Shear Resistance, ϕ (^{0})	Bulk/Submerged Density (t/m^3)	Eff. Over-burden Pressure at Pile Tip (t/m^2)	Bearing Capacity Factors: N_c	Bearing Capacity Factors: N_q	Bearing Capacity Factors: N_g	Ultimate Base Resistance, $P_{pu} = A_p*(c.N_c+q.N_q+0.5.\gamma.B.N_\gamma)$: A_p (m^2)	$c.N_c$	$q.N_q$	$0.5.\gamma.B.N_\gamma$	P_{pu} (t)	Ultimate Shaft Friction, P_{su} =(S($K_s.P_{di}$.Tand).A_{si} + a.c.A_s): Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m^2)	A_{si} (m^2)	($K_s.P_{di}$.Tand)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}	Total Ultimate Capacity, [P_u = (P_{pu} + P_{su})] (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression P_s = [(P_{pu}/ FOS + P_{su}/ FOS) - W_p] (t)	Safe Axial Capacity in uplift, P_u = [P_{su} / FOS) + W_p] (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
1	2.30	4.00	1.70	8.10	8	1.00	1.70	9.0	2.1	0.9	1.13	72.9	3.6	0.55	**87.1**	0.85	6.41	0.18	0.63	5.10	5.28	**33.9**					1.70		
2	4.00	6.00	2.00	8.10	8	1.00	3.70	9.0	2.1	0.9	1.13	72.9	7.8	0.55	**91.9**	2.70	7.54	0.57	0.63	5.10	5.67	**76.6**	168.5	6	61	28	3.70		
3	6.00	7.50	1.50	11.80	9	1.00	5.20	9.0	2.3	1.1	1.13	106.2	11.9	0.64	**134.3**	4.45	5.65	1.06	0.44	5.19	6.25	**112.0**	246.3	9	90	41	5.20		
4	7.50	9.00	1.50	0.00	31	1.00	6.70	9.0	24.0	26.0	1.13	0.0	160.8	15.60	**199.5**	5.95	5.65	5.36	1.00	0.00	5.36	**142.3**	341.8	11	125	52	6.70		
5	9.00	11.00	2.00	0.00	31	1.00	8.70	9.0	24.0	26.0	1.13	0.0	208.8	15.60	**253.8**	7.70	7.54	6.94	1.00	0.00	6.94	**194.6**	448.4	15	165	70	8.70		
6	11.00	13.00	2.00	0.00	31	1.00	10.70	9.0	24.0	26.0	1.13	0.0	256.8	15.60	**308.1**	9.70	7.54	8.74	1.00	0.00	8.74	**260.5**	568.6	18	209	93	10.70		
7	13.00	15.00	2.00	0.00	31	1.00	12.70	9.0	24.0	26.0	1.13	0.0	304.8	15.60	**362.4**	11.70	7.54	10.55	1.00	0.00	10.55	**340.0**	702.4	22	259	119	12.70		
8	15.00	16.00	1.00	0.00	32	1.00	13.70	9.0	30.0	30.0	1.13	0.0	411.0	18.00	**485.2**	13.20	3.77	12.37	1.00	0.00	12.37	**386.7**	871.9	23	326	134	13.70		
9	16.00	18.00	2.00	15.70	9	1.00	15.70	9.0	2.3	1.1	1.13	141.3	36.0	0.64	**201.2**	14.70	7.54	3.49	0.36	5.65	9.14	**455.6**	656.8	27	236	157	15.70		
10	18.00	19.30	1.30	0.90	28	1.00	17.00	9.0	15.6	16.7	1.13	8.1	265.2	10.03	**248.9**	16.35	4.90	13.04	1.00	0.90	13.94	**523.9**	772.8	29	280	179	17.00	280	170
11	19.30	20.30	1.00	0.90	28	1.00	18.00	9.0	15.6	16.7	1.13	8.1	280.8	10.03	**248.9**	18.00	3.77	14.36	1.00	0.90	15.26	**581.5**	830.3	31	302	197	18.00	300	190
12	20.30	21.00	0.70	0.90	28	1.00	18.00	9.0	15.6	16.7	1.13	8.1	280.8	10.03	**248.9**	18.00	2.64	14.36	1.00	0.90	15.26	**621.7**	870.6	32	317	209	18.70		
13	21.00	22.30	1.30	19.80	9	1.00	18.00	9.0	2.3	1.1	1.13	178.2	41.2	0.64	**248.9**	18.00	4.90	4.28	0.32	6.34	10.61	**673.7**	922.6	34	335	226	20.00	330	220
14	22.30	23.30	1.00	19.80	9	1.00	18.00	9.0	2.3	1.1	1.13	178.2	41.2	0.64	**248.9**	18.00	3.77	4.28	0.32	6.34	10.61	**713.7**	962.6	36	349	240	21.00	340	230

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm^3)	Young's modulus of pile material E (kg/cm^2)	Moment of Inertia I (cm^4)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	120	0.27	320000	10178761	413.45	1.91	790	0	0.000	1.20	1.20	23811	23.00	16.54
Fixed	120	0.27	320000	10178761	413.45	2.19	903	0	0.000	1.20	1.20	63618	63.00	16.54

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78]	Ch. 44+758	BH-A2

Pile Dia. (mm) =	1000	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.00	Scouring Depth (m) =	-

Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m^2)	Angle of Shear Resistance, ϕ (^{0})	Bulk/Submerged Density (t/m^3)	Eff. Over-burden Pressure at Pile Tip (t/m^2)	Bearing Capacity Factors: N_c	N_q	N_g	Ultimate Base Resistance, $P_{pu} = A_p$*($c.N_c$+$q.N_q$+$0.5.\gamma.B.N_\gamma$): A_p (m^2)	$c.N_c$	$q.N_q$	$0.5.\gamma.B.N_\gamma$	P_{pu} (t)	Ultimate Shaft Friction, P_{su} =(S(K_s.P_{di}.Tand).A_{si} + a.c.A_s): Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m^2)	A_{si} (m^2)	(K_s.P_{di}.Tand)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}	Total Ultimate Capacity, [P_u = (P_{pu} + P_{su})] (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression P_s = [(P_{pu}/ FOS + P_{su}/ FOS) - W_p] (t)	Safe Axial Capacity in uplift, P_u = [P_{su} / FOS) + W_p] (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
1	2.00	4.50	2.50	7.60	6	1.00	2.50	9.0	1.8	0.6	0.79	68.4	4.4	0.30	**57.4**	1.25	7.85	0.20	0.66	5.02	5.21	**40.9**					2.50		
2	4.50	6.00	1.50	0.40	28	1.00	4.00	9.0	15.6	16.7	0.79	3.6	62.4	8.36	**58.4**	3.25	4.71	2.59	1.00	0.40	2.99	**55.0**	113.4	5	41	20	4.00		
3	6.00	7.00	1.00	0.40	28	1.00	5.00	9.0	15.6	16.7	0.79	3.6	78.0	8.36	**70.7**	4.50	3.14	3.59	1.00	0.40	3.99	**67.6**	138.2	6	49	25	5.00		
4	7.00	8.00	1.00	11.50	8	1.00	6.00	9.0	2.1	0.9	0.79	103.5	12.7	0.46	**91.6**	5.50	3.14	1.16	0.44	5.06	6.22	**87.1**	178.7	7	64	32	6.00		
5	8.00	10.00	2.00	0.00	31	1.00	8.00	9.0	24.0	26.0	0.79	0.0	192.0	13.00	**161.0**	7.00	6.28	6.31	1.00	0.00	6.31	**126.8**	287.8	9	106	46	8.00		
6	10.00	11.00	1.00	0.00	31	1.00	9.00	9.0	24.0	26.0	0.79	0.0	216.0	13.00	**179.9**	8.50	3.14	7.66	1.00	0.00	7.66	**150.8**	330.7	11	122	54	9.00		
7	11.00	13.00	2.00	0.00	32	1.00	11.00	9.0	30.0	30.0	0.79	0.0	330.0	15.00	**271.0**	10.00	6.28	9.37	1.00	0.00	9.37	**209.7**	480.7	13	179	73	11.00		
8	13.00	16.00	3.00	0.00	32	1.00	14.00	9.0	30.0	30.0	0.79	0.0	420.0	15.00	**341.6**	12.00	9.42	11.25	1.00	0.00	11.25	**315.7**	657.4	16	246	107	14.00		
9	16.00	17.00	1.00	0.80	28	1.00	15.00	9.0	15.6	16.7	0.79	7.2	234.0	8.36	**146.1**	14.50	3.14	11.56	1.00	0.80	12.36	**354.6**	500.7	18	183	119	15.00	180	110
10	17.00	18.00	1.00	0.80	28	1.00	15.00	9.0	15.6	16.7	0.79	7.2	234.0	8.36	**146.1**	14.50	3.14	11.56	1.00	0.80	12.36	**393.4**	539.6	19	197	131	16.00		
11	18.00	19.00	1.00	16.80	9	1.00	15.00	9.0	2.3	1.1	0.79	151.2	34.4	0.53	**146.1**	14.50	3.14	3.44	0.35	5.88	9.32	**422.7**	568.9	20	208	141	17.00	200	140
12	19.00	20.00	1.00	16.80	9	1.00	15.00	9.0	2.3	1.1	0.79	151.2	34.4	0.53	**146.1**	14.50	3.14	3.44	0.35	5.88	9.32	**452.0**	598.2	21	218	150	18.00	210	150
13	20.00	21.00	1.00	16.80	9	1.00	15.00	9.0	2.3	1.1	0.79	151.2	34.4	0.53	**146.1**	14.50	3.14	3.44	0.35	5.88	9.32	**481.3**	627.4	22	229	160	19.00	220	150

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm^3)	Young's modulus of pile material E (kg/cm^2)	Moment of Inertia I (cm^4)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	100	0.28	320000	4908739	353.64	1.91	675	0	0.000	1.00	1.00	15292	15.00	14.15
Fixed	100	0.28	320000	4908739	353.64	2.19	773	0	0.000	1.00	1.00	40857	40.00	14.15

Calculation of Vertical Load Bearing Capacity of "Concrete Pile" [As per IS : 2911(Part 1) and IRC : 78]	Ch. 44+758	BH-A2

Pile Dia. (mm) =	1200	K_s =	1.5	Design Water Level (m) =	0.00	FOS for Shaft Friction =	2.5	FOS for End Bearing =	2.5	Cut - off Level (m) =	2.30	Scouring Depth (m) =	-

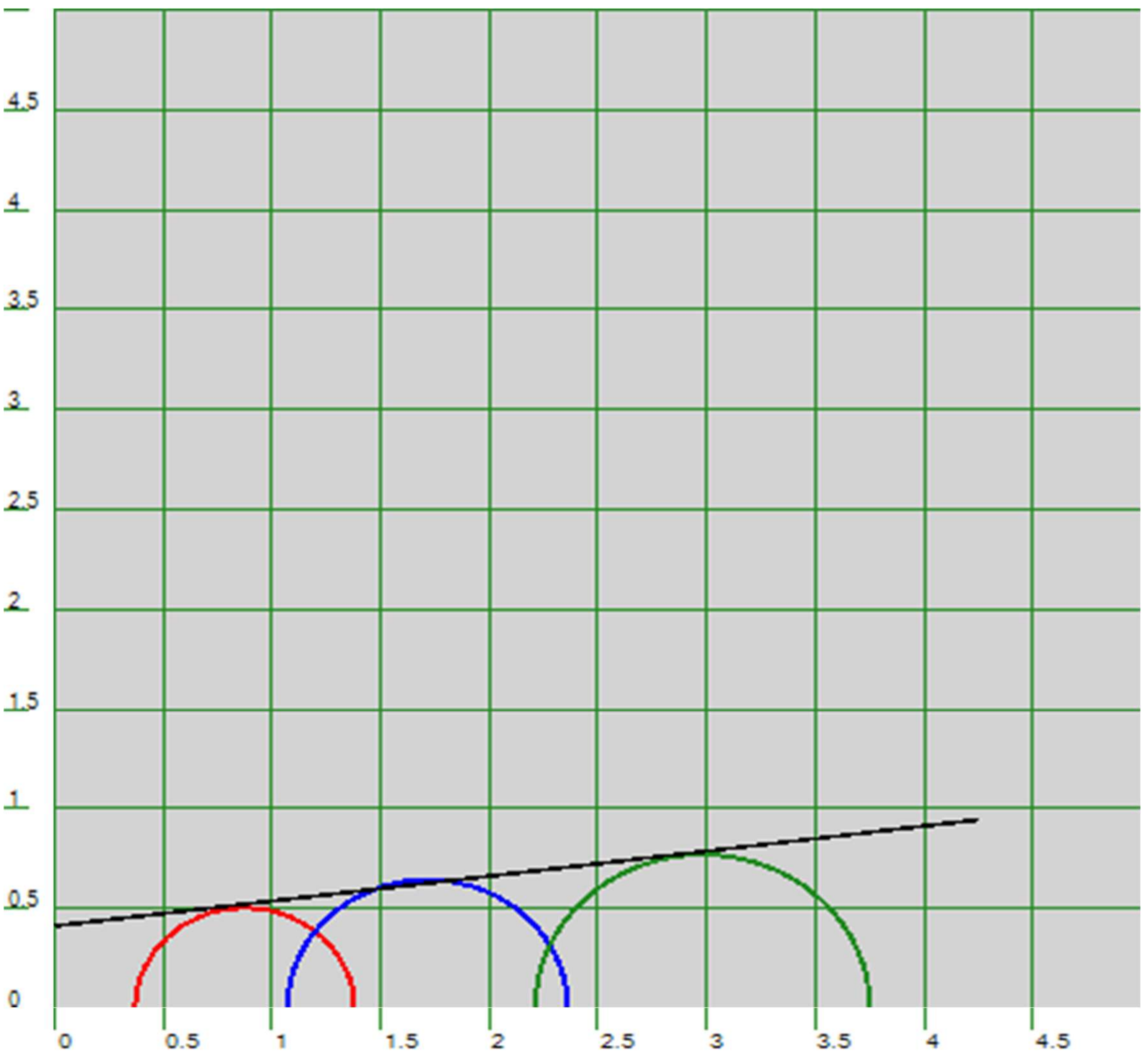
Layer No	Depth of Layer Top	Depth of Layer Bottom	Thickness of Layer (m)	Cohesion, c (t/m^2)	Angle of Shear Resistance, ϕ (^{0})	Bulk/Submerged Density (t/m^3)	Eff. Over-burden Pressure at Pile Tip (t/m^2)	Bearing Capacity Factors			Ultimate Base Resistance, $P_{pu} = A_p*(c.N_c+q.N_q+0.5.\gamma.B.N_\gamma)$					Ultimate Shaft Friction, $P_{su} = (S(K_s.P_{di}.Tan\delta).A_{si} + a.c.A_s)$							Total Ultimate Capacity, $[P_u = (P_{pu} + P_{su})]$ (t)	Self Weight of Pile, W_p (t)	Safe Capacity in Compression $P_s = [(P_{pu}/FOS + P_{su}/FOS) - W_p]$ (t)	Safe Axial Capacity in uplift, $P_u = [P_{su}/FOS) + W_p]$ (t)	Pile Length below COL (m)	Recom. Vertical Capapcity (Comp)	Recom. Vertical Capapcity (Uplift)
								N_c	N_q	N_g	A_p (m^2)	$c.N_c$	$q.N_q$	$0.5.\gamma.B.N_\gamma$	P_{pu} (t)	Eff. Over-burden Pressure at c.g of the layer 'P_{di}' (t/m^2)	A_{si} (m^2)	($K_s.P_{di}.Tan\delta$)	Adhesion Factor, a	a.c	Max. Frictional Resistance	P_{su}							
1	2.30	4.50	2.20	7.60	6	1.00	2.20	9.0	1.8	0.6	1.13	68.4	3.9	0.36	**82.1**	1.10	8.29	0.17	0.66	5.02	5.19	**43.0**					2.20		
2	4.50	6.00	1.50	0.40	28	1.00	3.70	9.0	15.6	16.7	1.13	3.6	57.7	10.03	**80.7**	2.95	5.65	2.35	1.00	0.40	2.75	**58.6**	139.3	6	49	23	3.70		
3	6.00	7.00	1.00	0.40	28	1.00	4.70	9.0	15.6	16.7	1.13	3.6	73.3	10.03	**98.3**	4.20	3.77	3.35	1.00	0.40	3.75	**72.7**	171.1	8	60	29	4.70		
4	7.00	8.00	1.00	11.50	8	1.00	5.70	9.0	2.1	0.9	1.13	103.5	12.0	0.55	**131.3**	5.20	3.77	1.10	0.44	5.06	6.16	**96.0**	227.2	10	81	37	5.70		
5	8.00	10.00	2.00	0.00	31	1.00	7.70	9.0	24.0	26.0	1.13	0.0	184.8	15.60	**226.6**	6.70	7.54	6.04	1.00	0.00	6.04	**141.5**	368.1	13	134	53	7.70		
6	10.00	11.00	1.00	0.00	31	1.00	8.70	9.0	24.0	26.0	1.13	0.0	208.8	15.60	**253.8**	8.20	3.77	7.39	1.00	0.00	7.39	**169.3**	423.1	15	154	63	8.70		
7	11.00	13.00	2.00	0.00	32	1.00	10.70	9.0	30.0	30.0	1.13	0.0	321.0	18.00	**383.4**	9.70	7.54	9.09	1.00	0.00	9.09	**237.9**	621.3	18	230	86	10.70		
8	13.00	16.00	3.00	0.00	32	1.00	13.70	9.0	30.0	30.0	1.13	0.0	411.0	18.00	**485.2**	12.20	11.31	11.44	1.00	0.00	11.44	**367.2**	852.4	23	318	128	13.70		
9	16.00	17.00	1.00	0.80	28	1.00	14.70	9.0	15.6	16.7	1.13	7.2	229.3	10.03	**278.8**	14.20	3.77	11.33	1.00	0.80	12.13	**412.9**	691.8	25	252	143	14.70		
10	17.00	18.00	1.00	0.80	28	1.00	15.70	9.0	15.6	16.7	1.13	7.2	244.9	10.03	**296.5**	15.20	3.77	12.12	1.00	0.80	12.92	**461.7**	758.1	27	277	159	15.70		
11	18.00	19.30	1.30	16.80	9	1.00	17.00	9.0	2.3	1.1	1.13	151.2	38.9	0.64	**215.8**	16.35	4.90	3.88	0.35	5.88	9.76	**509.5**	725.3	29	261	174	17.00	260	170
12	19.30	20.30	1.00	16.80	9	1.00	18.00	9.0	2.3	1.1	1.13	151.2	41.2	0.64	**218.3**	18.00	3.77	4.28	0.35	5.88	10.16	**547.8**	766.1	31	276	187	18.00	270	180
13	20.30	21.30	1.00	16.80	9	1.00	18.00	9.0	2.3	1.1	1.13	151.2	41.2	0.64	**218.3**	18.00	3.77	4.28	0.35	5.88	10.16	**586.1**	804.4	32	290	200	19.00	280	190
14	21.30	22.30	1.00	18.90	10	1.00	18.00	9.0	2.5	1.2	1.13	170.1	44.5	0.73	**243.5**	18.00	3.77	4.76	0.33	6.24	11.00	**627.5**	871.0	34	314	213	20.00	310	210

Estimation of Lateral Load Carrying Capacity of "Concrete Pile" [As per IS : 2911(Part 1/Sec 2) and IRC : 78]

End Condition	Diameter of pile in cm	Modulus of subgrade reaction (kg/cm^3)	Young's modulus of pile material E (kg/cm^2)	Moment of Inertia I (cm^4)	Stiffness Factor T (cm)	L_f/T	Depth to Point of fixity L_f (cm)	Cantilever length above ground L1 (cm)	L1/T	Permissible deflection Y (cm)	Permissible deflection at COL Y, cm	Calculated Lateral Load in kg	Recommended Lateral Load, T	Minimum Pile length required, cm
Free	120	0.28	320000	10178761	409.17	1.91	782	0	0.000	1.20	1.20	24566	24.00	16.37
Fixed	120	0.28	320000	10178761	409.17	2.19	894	0	0.000	1.20	1.20	65635	65.00	16.37

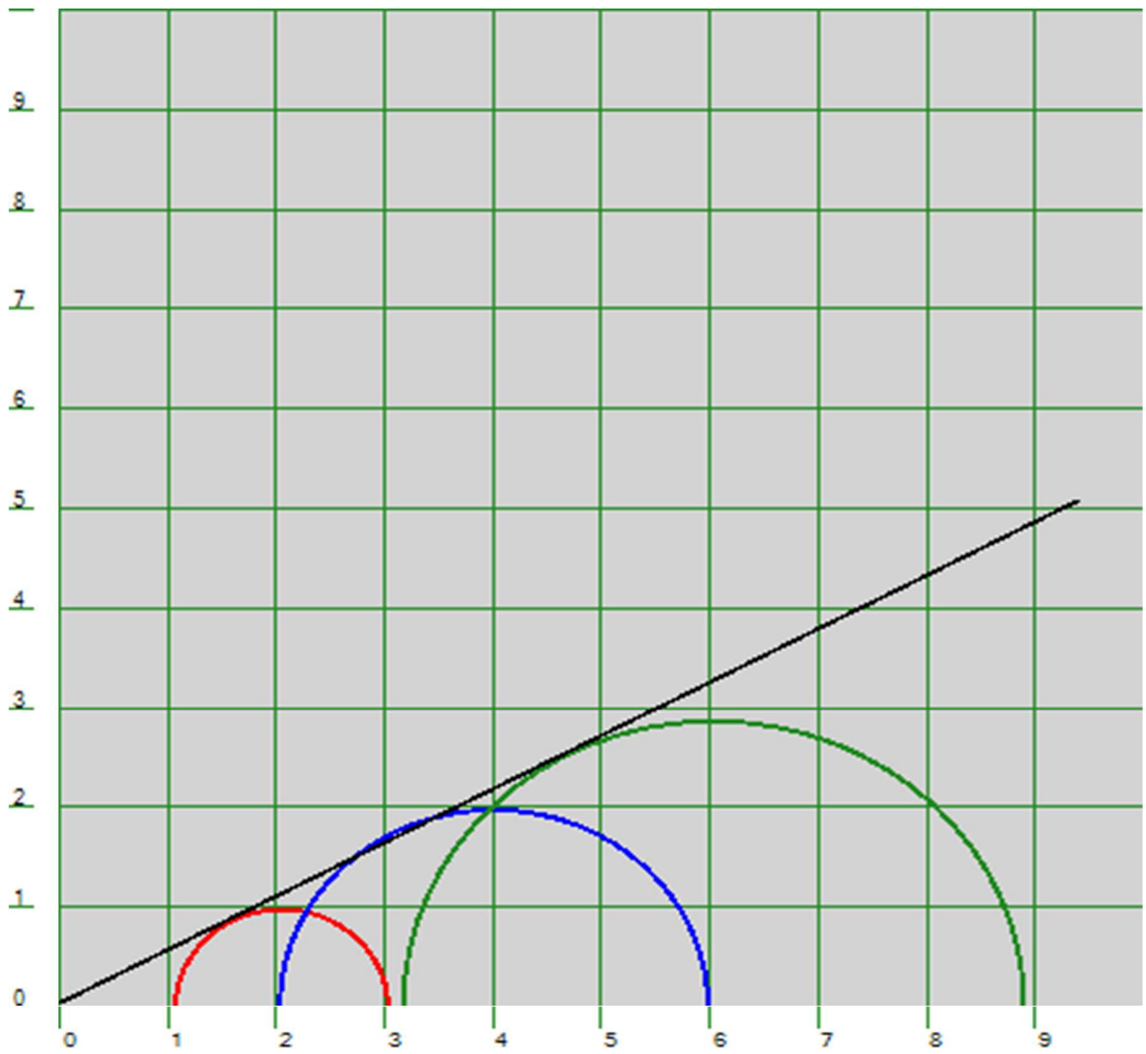
Annexure- C
Test Results

Test Type	**=**	**CU Test**
Chainage / BH No.	=	44+758 / BH-A1
Sampling Depth	=	1 m
Soil Type	=	Silty Clay



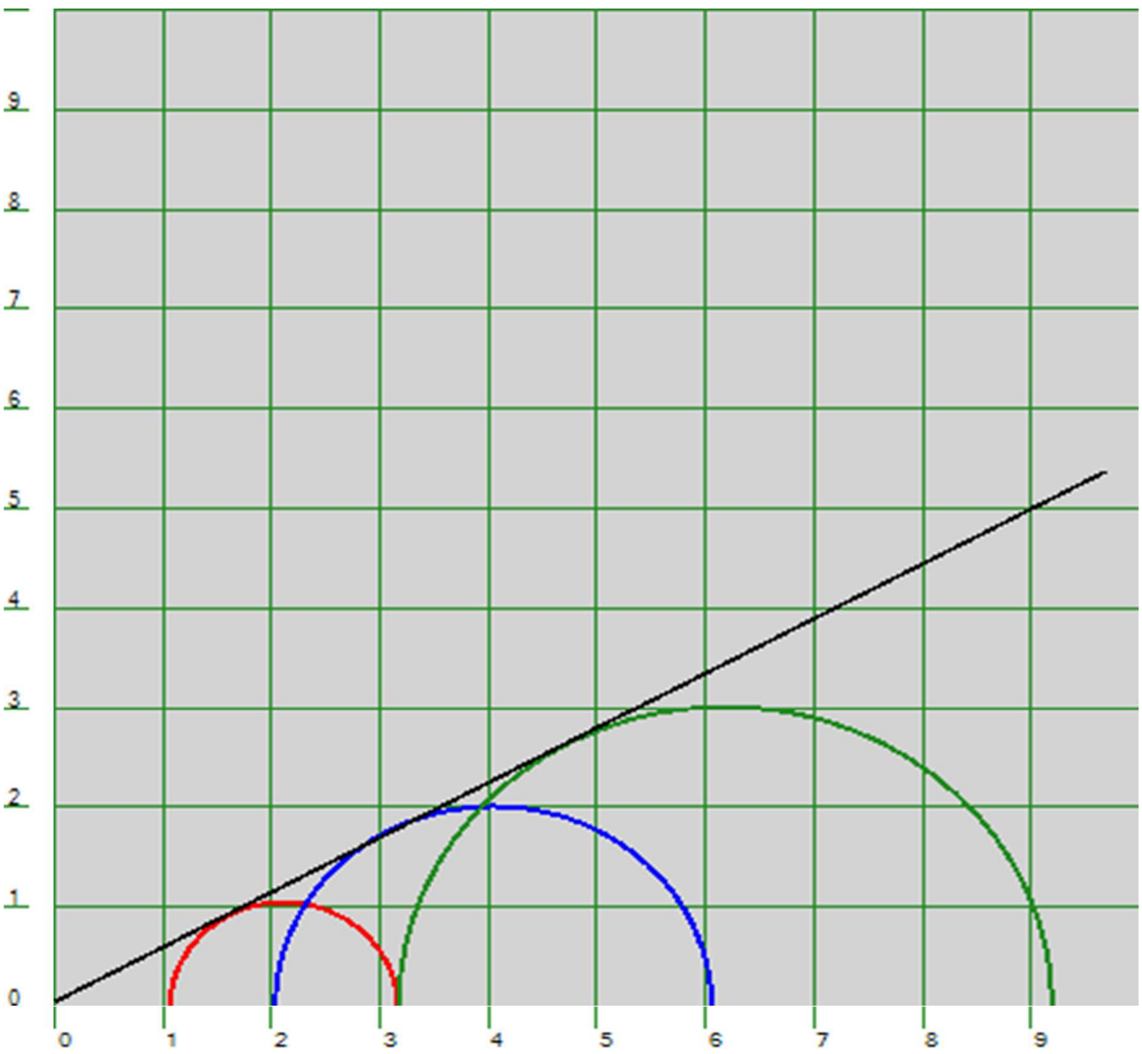
Mohr-Coulomb Plot [c=0.41 kg/sq.cm Phi=7.1 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CD Test**
Chainage / BH No.	=	44+758 / BH-A1
Sampling Depth	=	4 m
Soil Type	=	Sandy Silt



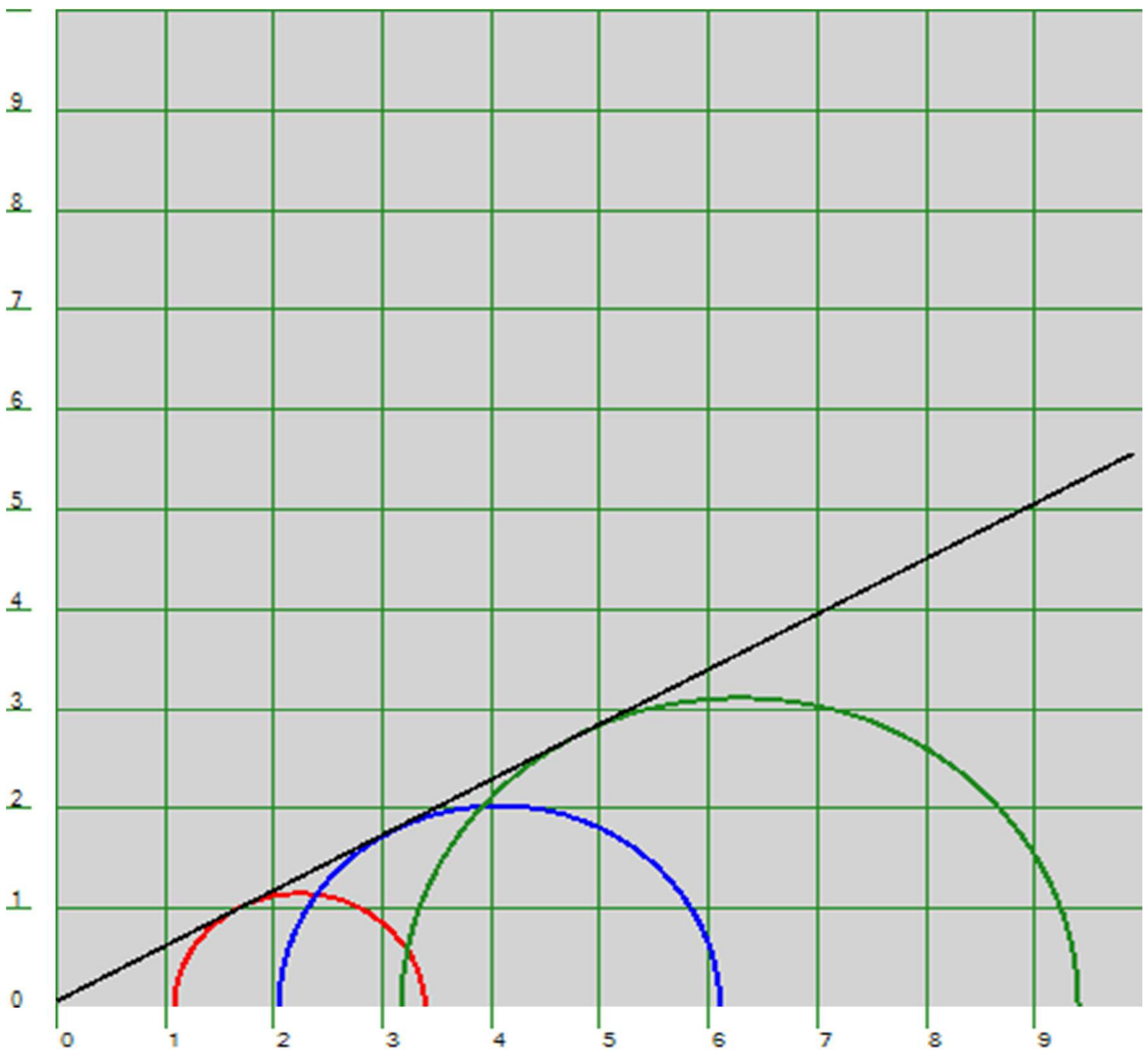
Mohr-Coulomb Plot [c=0.04 kg/sq.cm Phi=28.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CD Test
Chainage / BH No.	=	44+758 / BH-A1
Sampling Depth	=	7 m
Soil Type	=	Sandy Silt



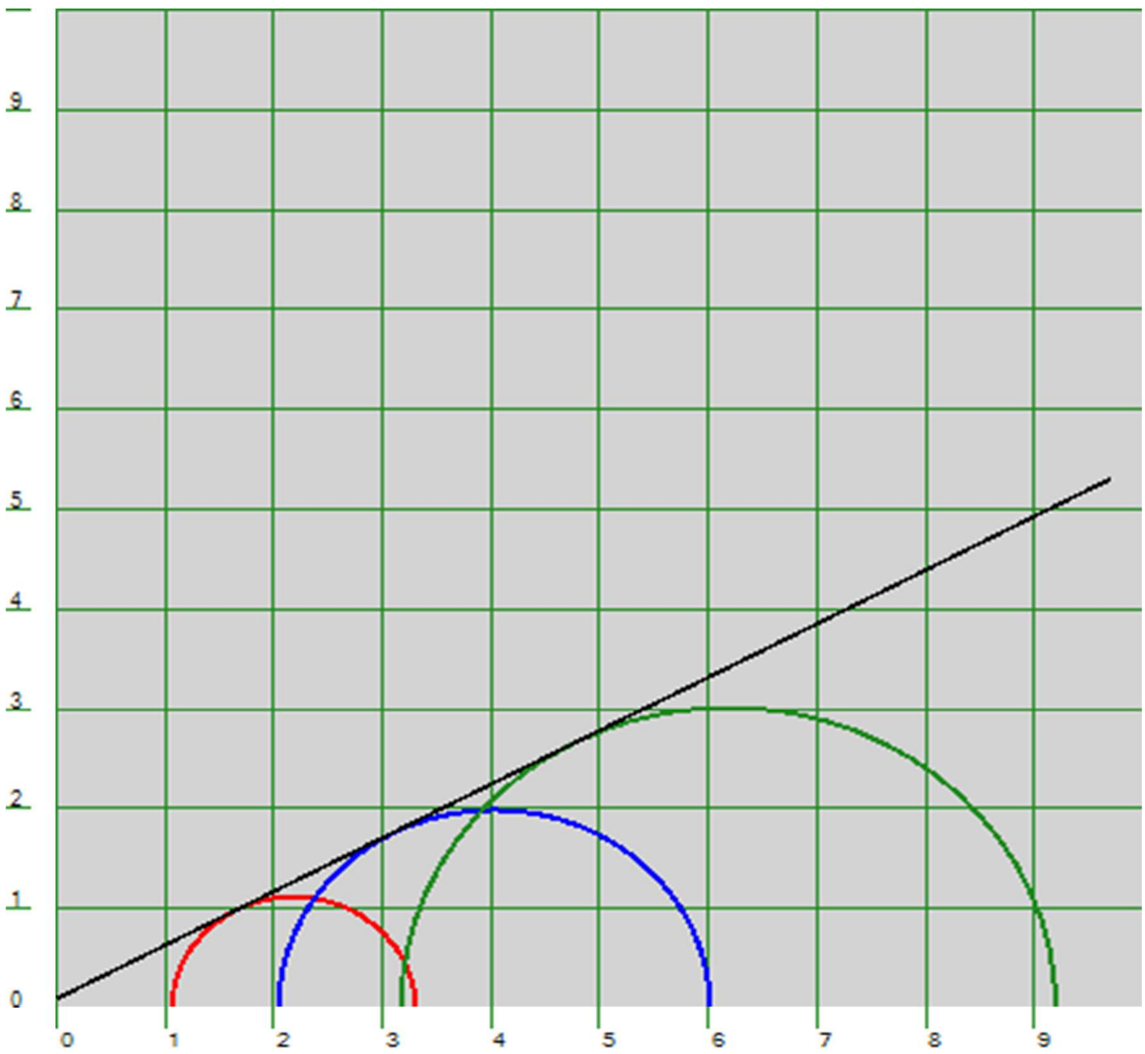
Mohr-Coulomb Plot [c=0.05 kg/sq.cm Phi=28.8 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CD Test**
Chainage / BH No.	=	44+758 / BH-A1
Sampling Depth	=	16 m
Soil Type	=	Sandy Silt



Mohr-Coulomb Plot [c=0.08 kg/sq.cm Phi=29 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CD Test**
Chainage / BH No.	=	44+758 / BH-A1
Sampling Depth	=	19 m
Soil Type	=	Sandy Silt

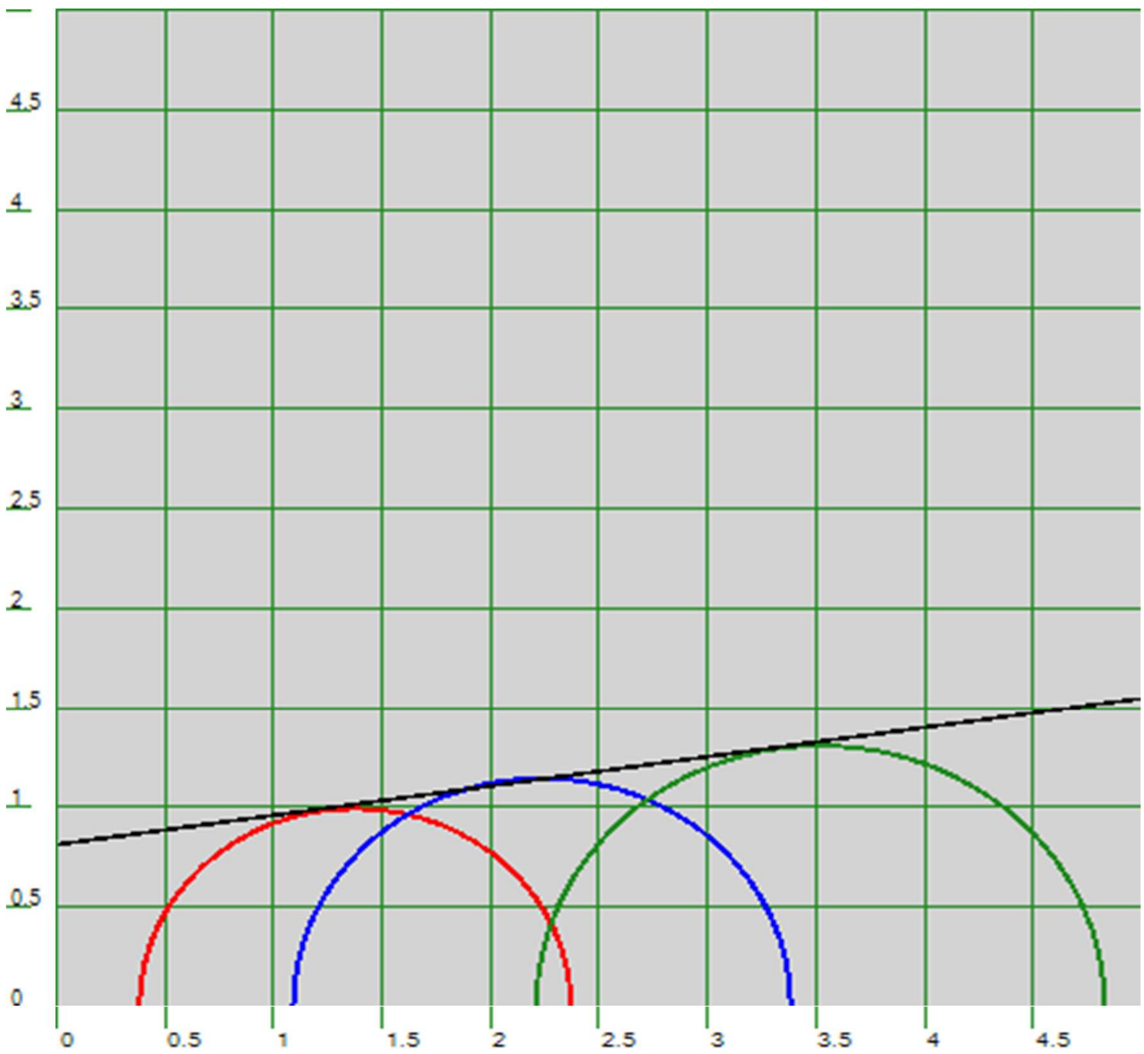


Mohr-Coulomb Plot [c=0.10 kg/sq.cm Phi=28.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	1 m
Soil Type	=	Silty Clay

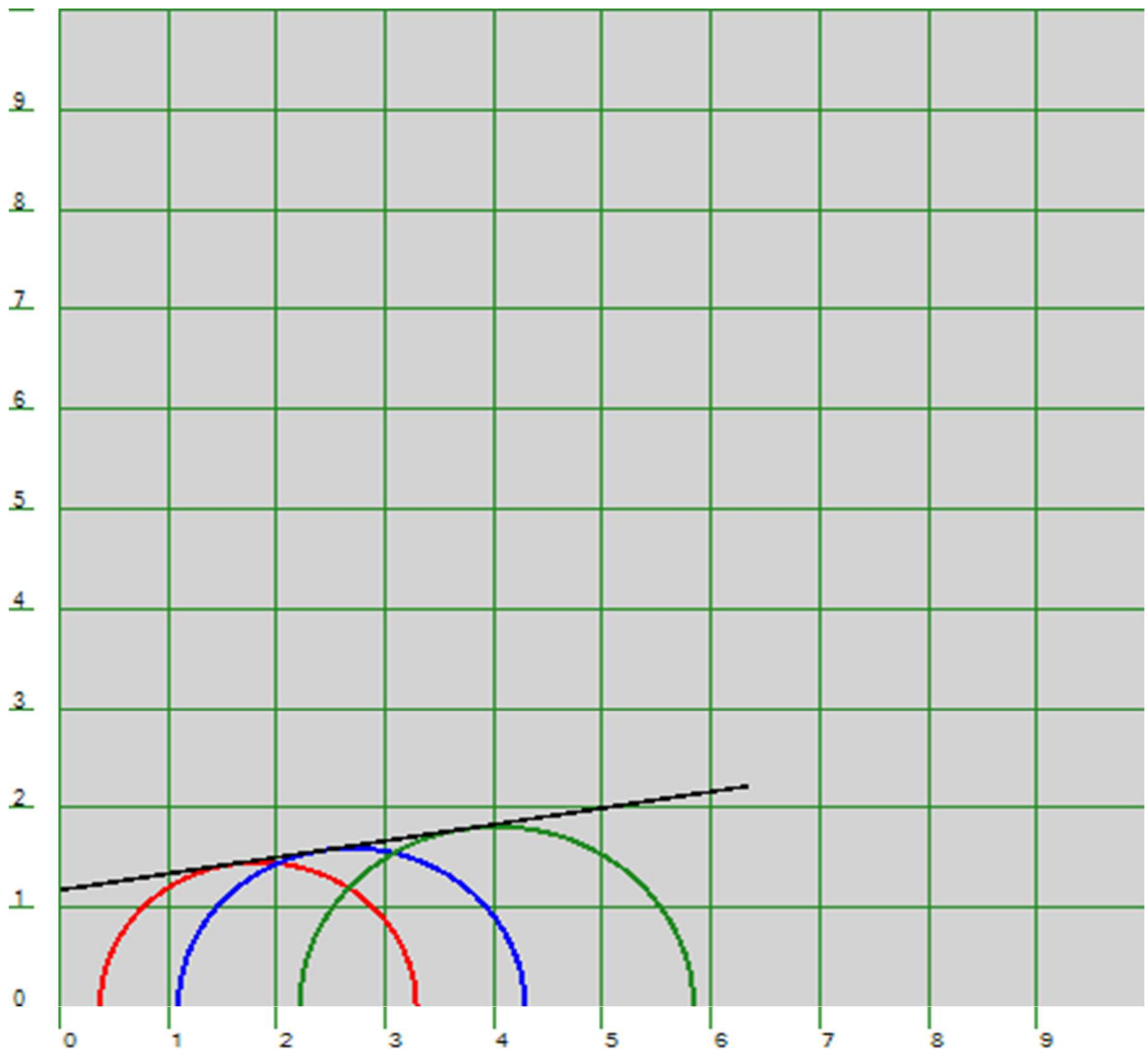
Mohr-Coulomb Plot [c=0.64 kg/sq.cm Phi=7.9 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	4 m
Soil Type	=	Silty Clay



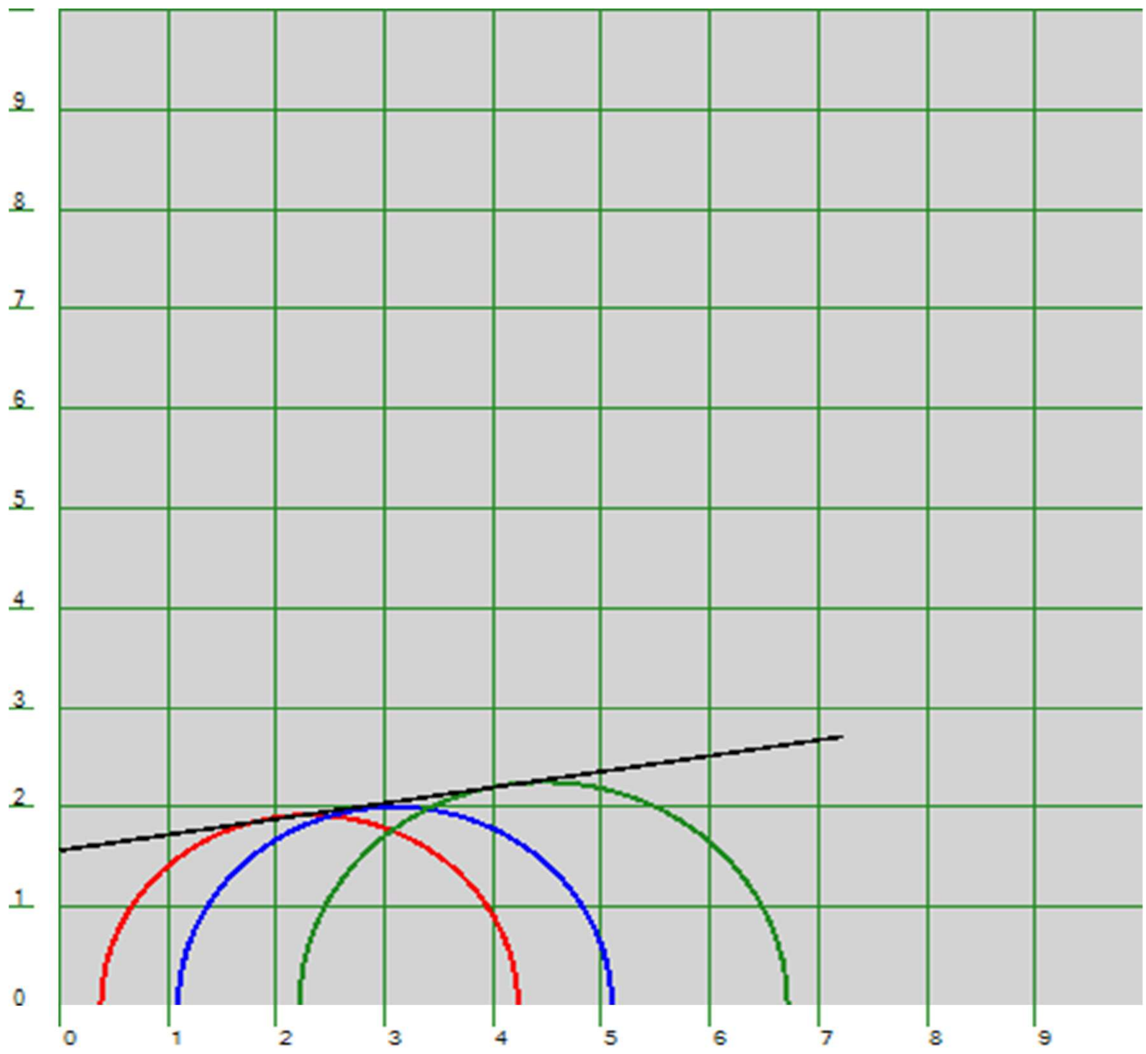
Mohr-Coulomb Plot [c=0.81 kg/sq.cm Phi=8.3 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	7 m
Soil Type	=	Silty Clay



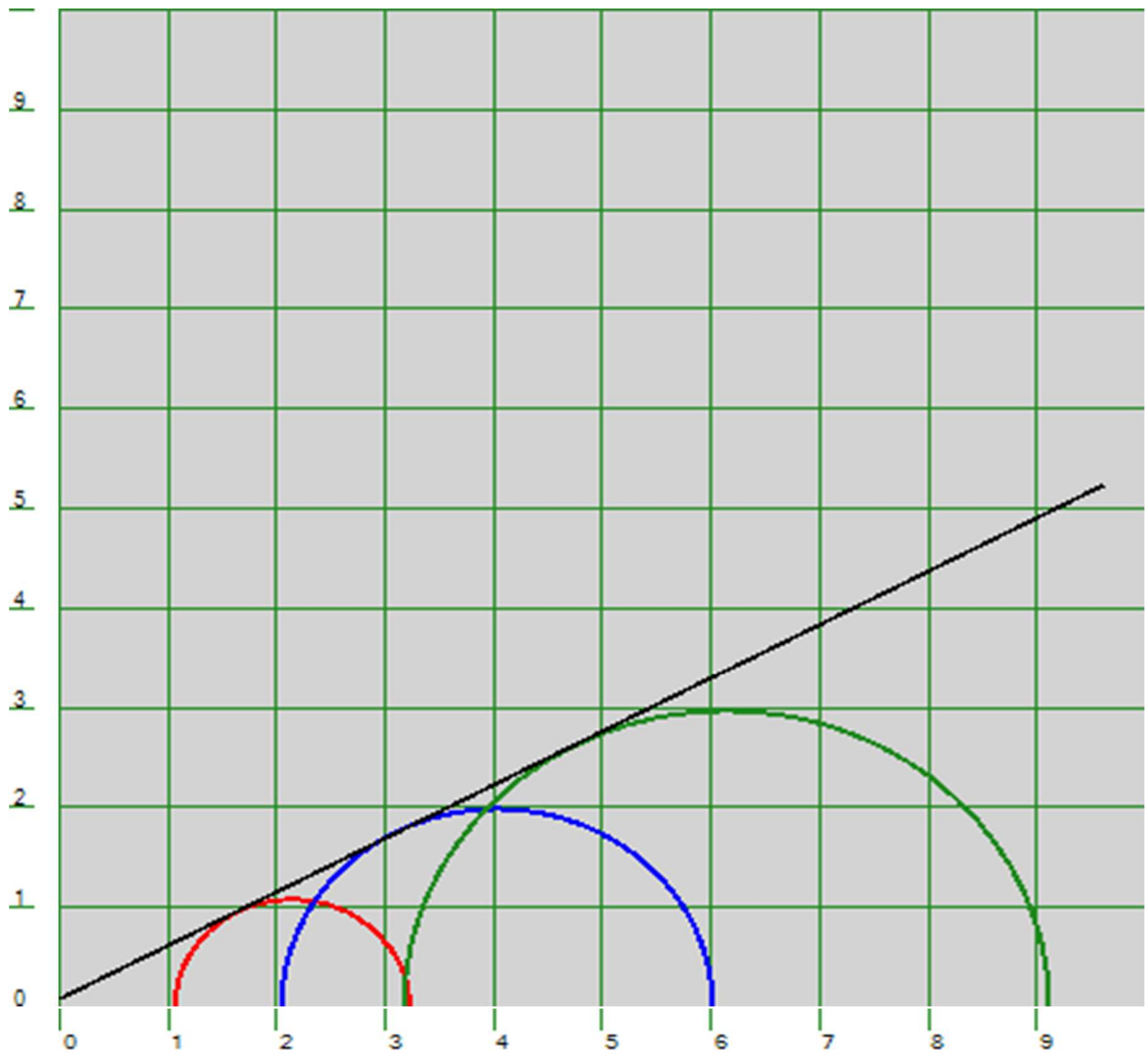
Mohr-Coulomb Plot [c=1.18 kg/sq.cm Phi=9.3 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CU Test**
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	16 m
Soil Type	=	Silty Clay



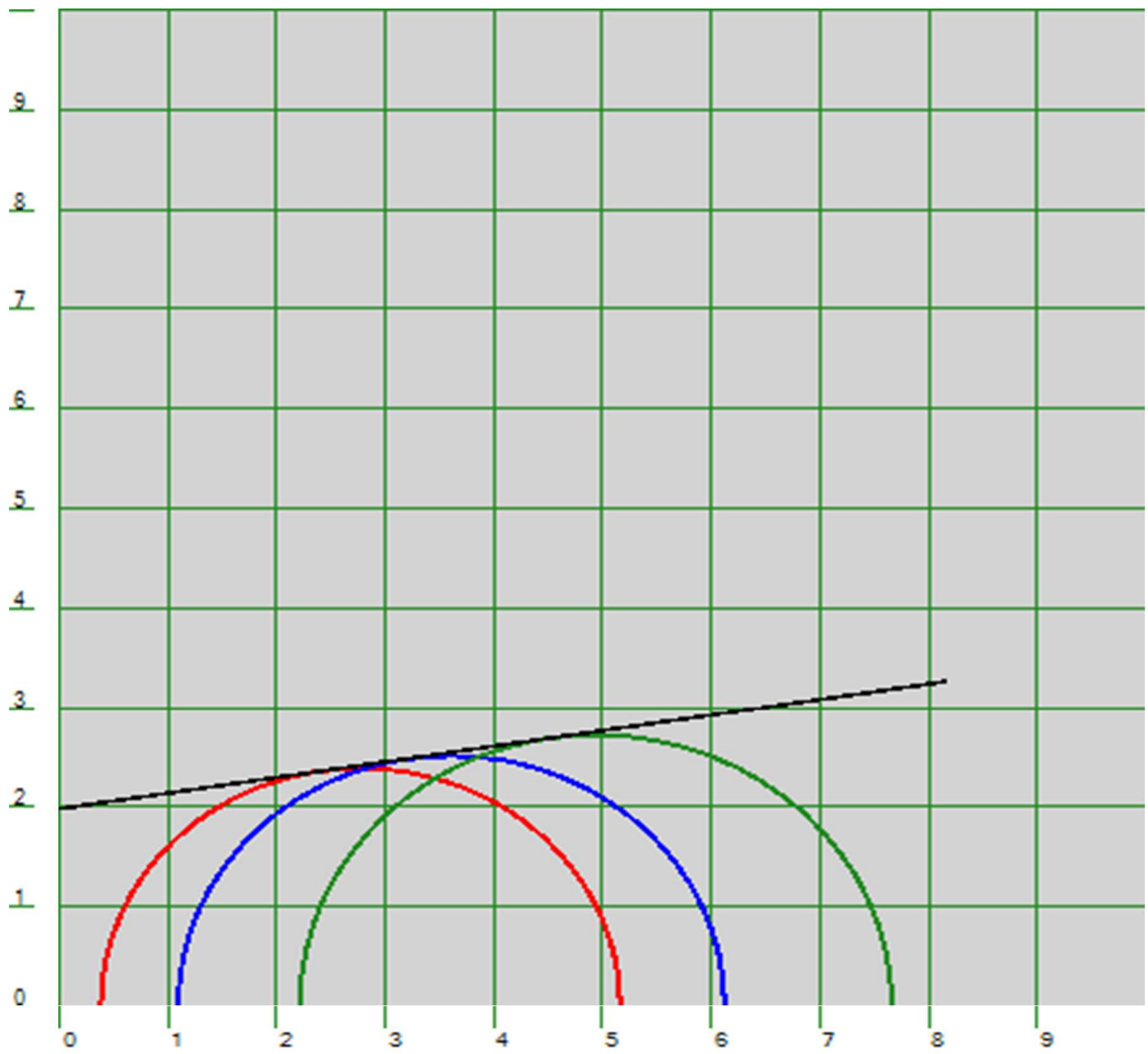
Mohr-Coulomb Plot [c=1.57 kg/sq.cm Phi=9.0 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CD Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	19 m
Soil Type	=	Sandy Silt



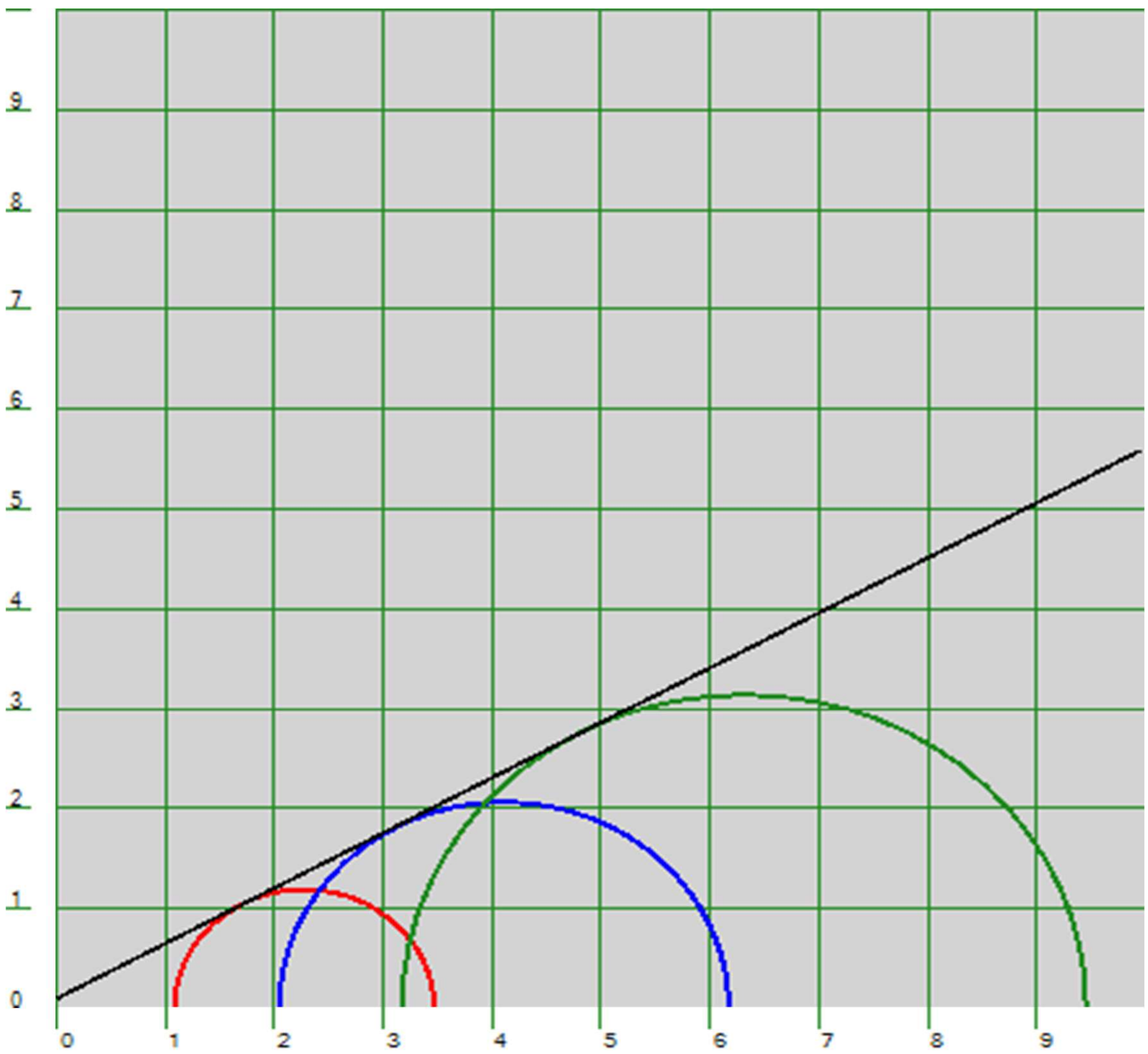
Mohr-Coulomb Plot [c=0.09 kg/sq.cm Phi=28.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	22 m
Soil Type	=	Silty Clay



Mohr-Coulomb Plot [c=1.98 kg/sq.cm Phi=8.9 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CD Test
Chainage / BH No.	=	44+758 / BH-P1
Sampling Depth	=	25 m
Soil Type	=	Sandy Silt

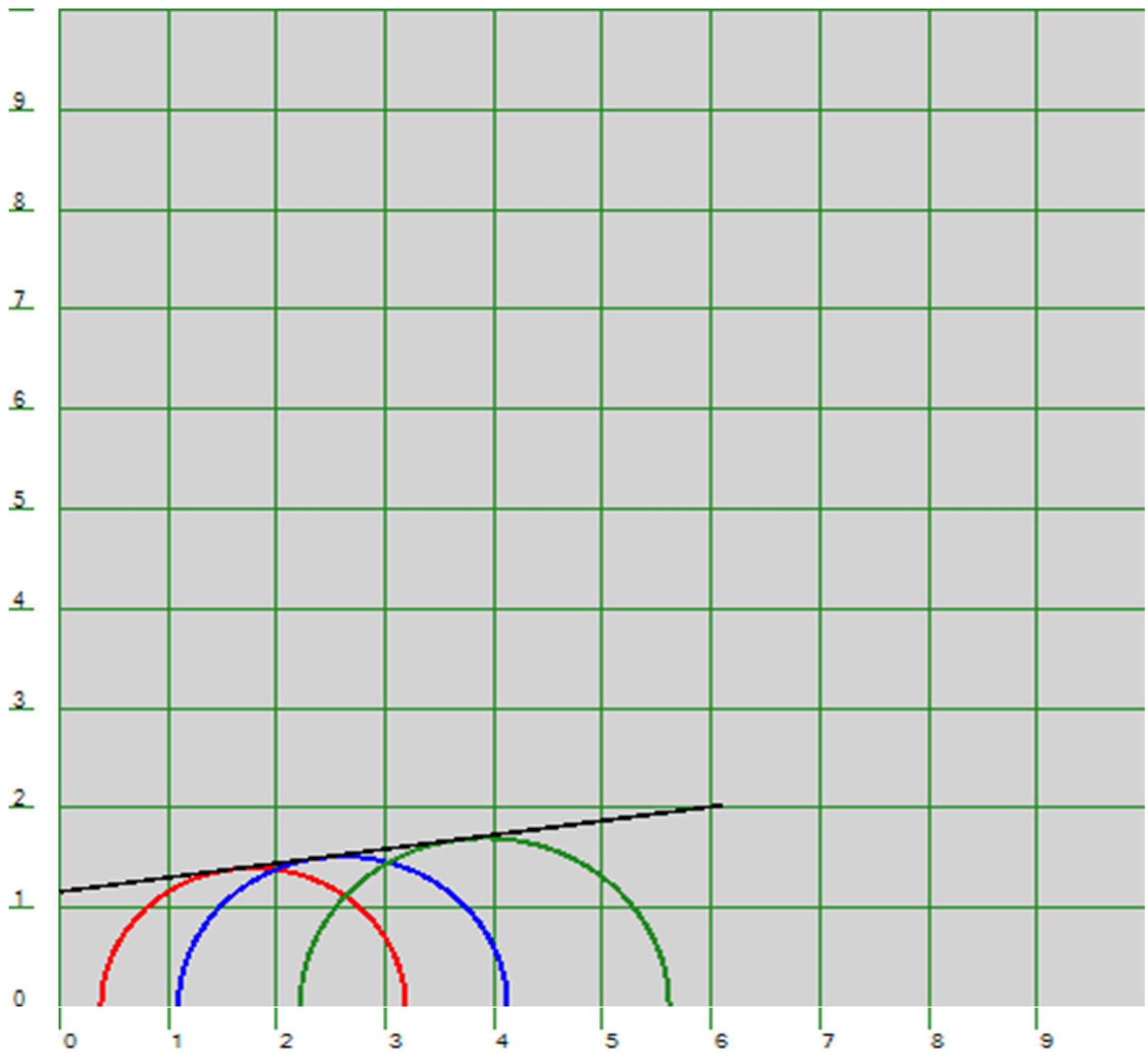


Mohr-Coulomb Plot [c=0.11 kg/sq.cm Phi=28.8 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CU Test**
Chainage / BH No.	=	44+758 / BH-A2
Sampling Depth	=	4 m
Soil Type	=	Silty Clay

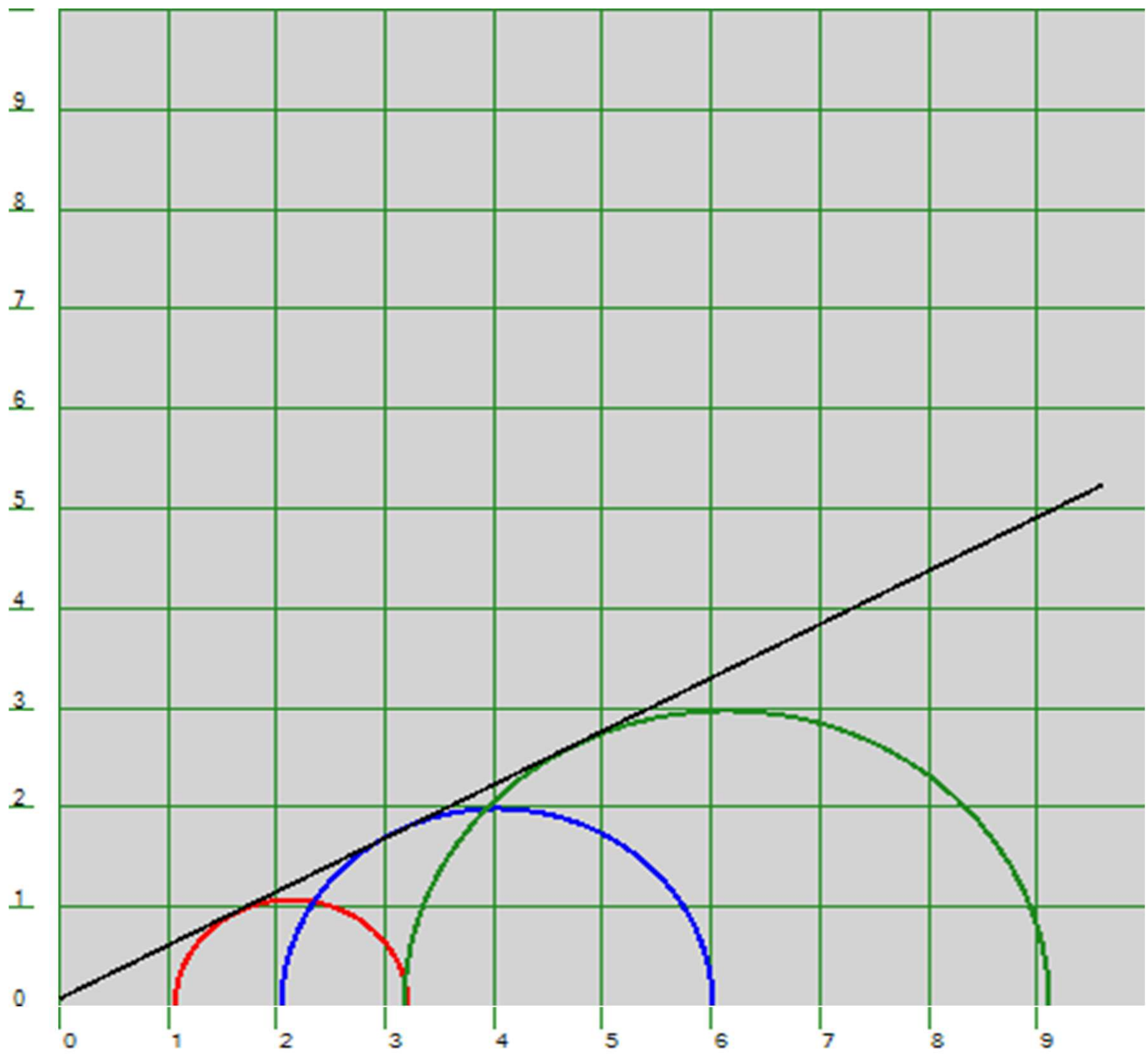
Mohr-Coulomb Plot [c=0.76 kg/sq.cm Phi=5.9 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-A2
Sampling Depth	=	7 m
Soil Type	=	Silty Clay



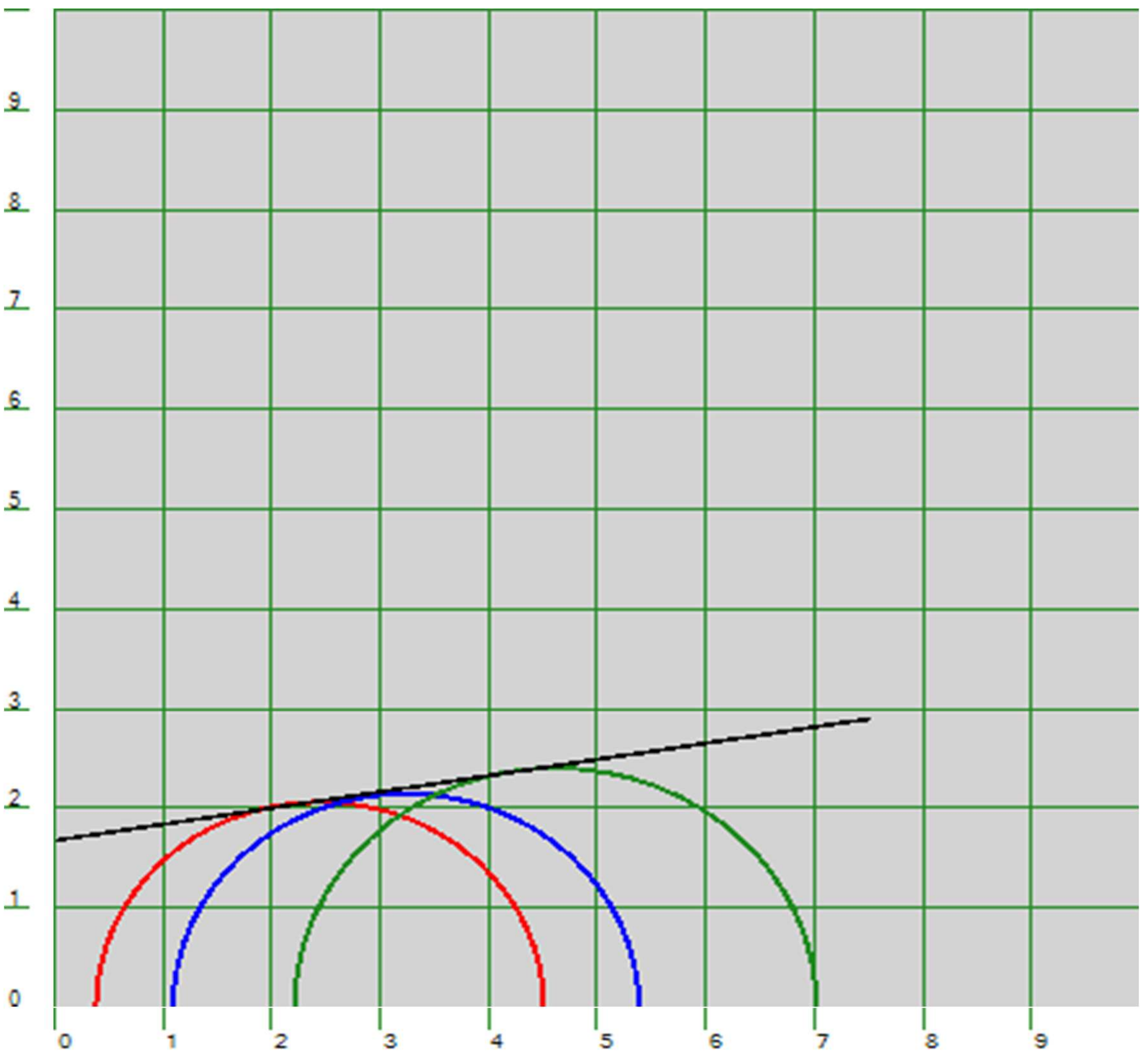
Mohr-Coulomb Plot [c=1.16 kg/sq.cm Phi=8.1 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CD Test**
Chainage / BH No.	=	44+758 / BH-A2
Sampling Depth	=	16 m
Soil Type	=	Sandy Silt



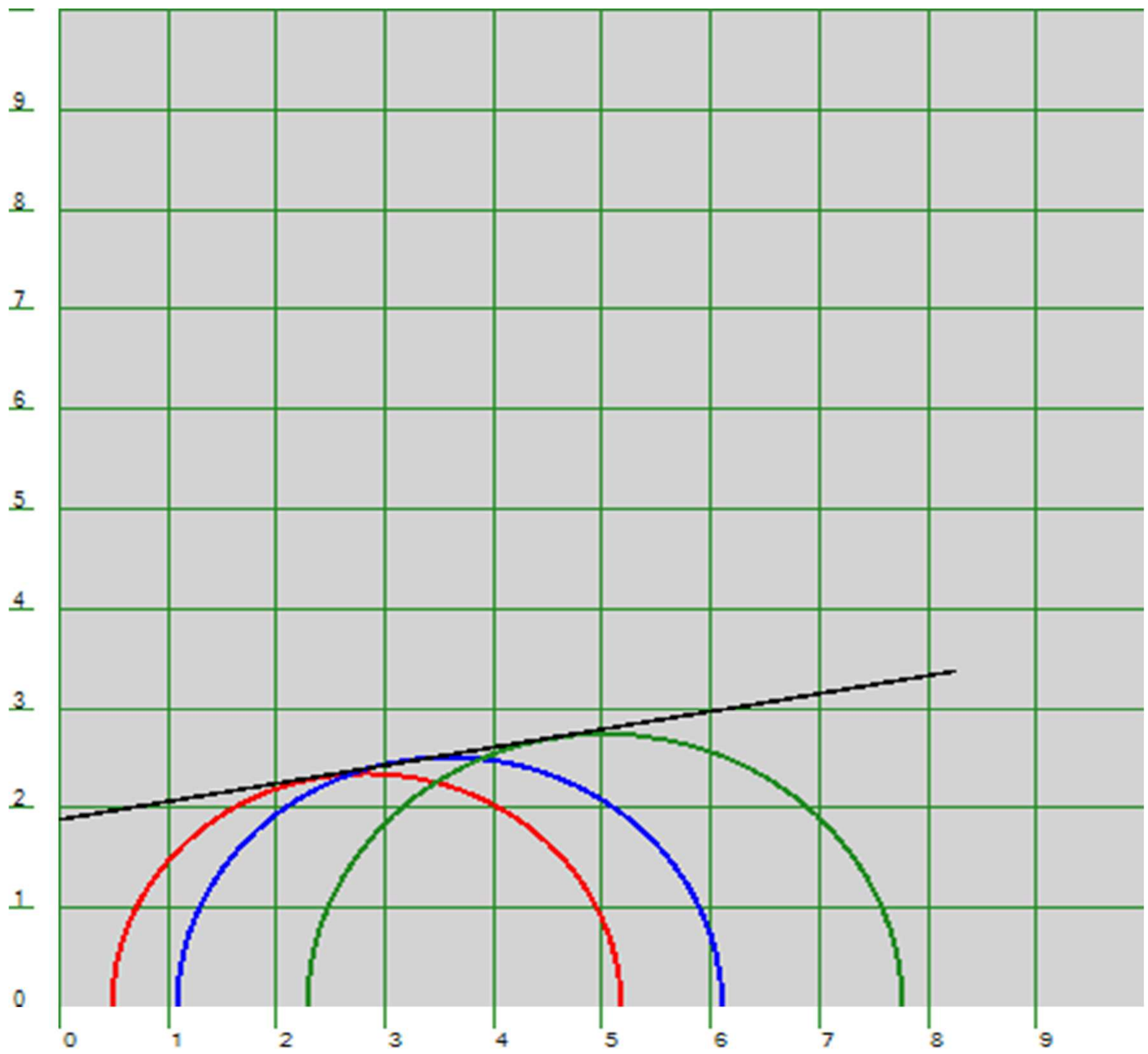
Mohr-Coulomb Plot [c=0.08 kg/sq.cm Phi=28.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / A2
Sampling Depth	=	19 m
Soil Type	=	Silty Clay



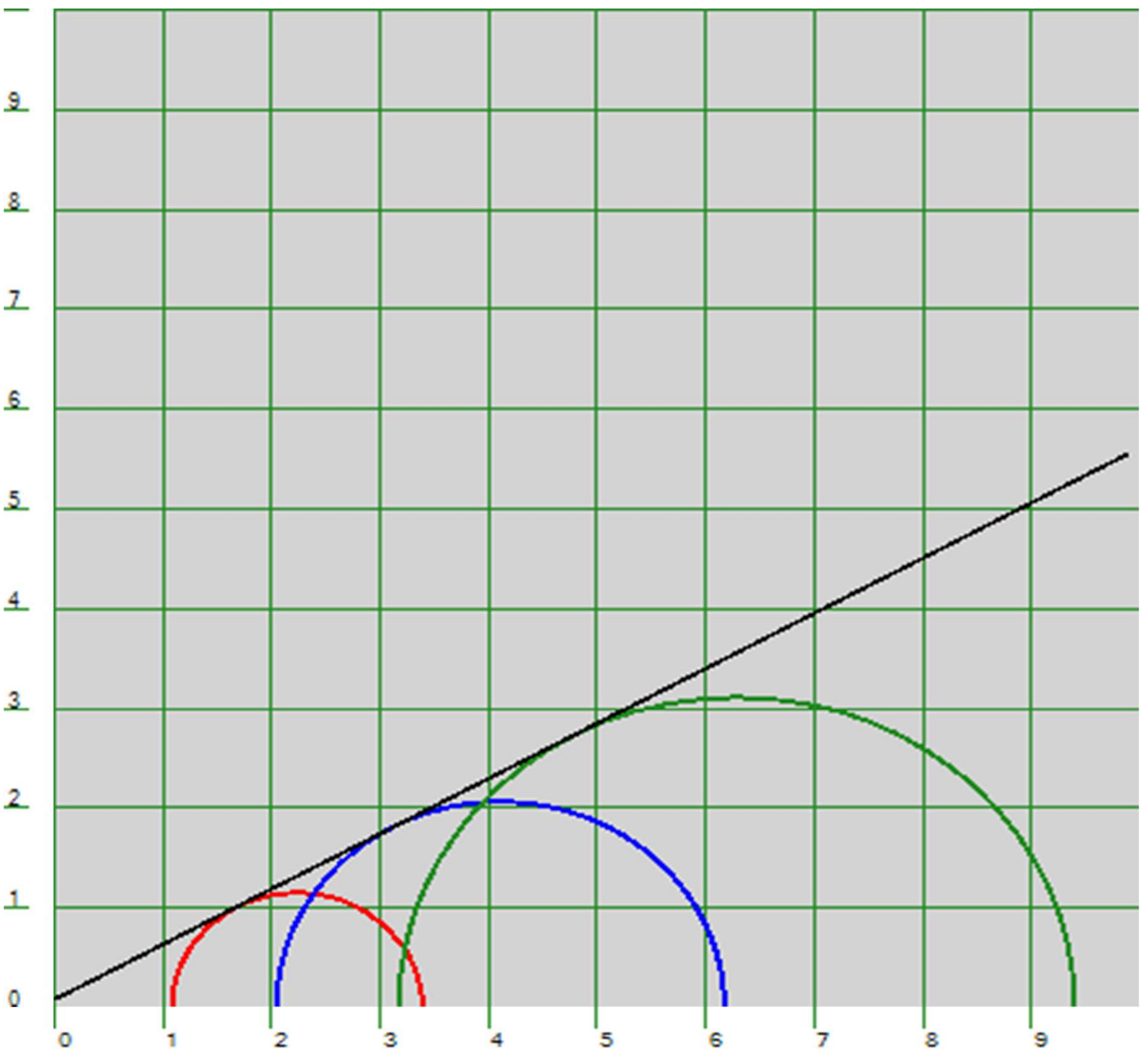
Mohr-Coulomb Plot [c=1.68 kg/sq.cm Phi=9.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	=	CU Test
Chainage / BH No.	=	44+758 / BH-A2
Sampling Depth	=	22 m
Soil Type	=	Silty Clay



Mohr-Coulomb Plot [c=1.89 kg/sq.cm Phi=10.2 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

Test Type	**=**	**CD Test**
Chainage / BH No.	=	44+758 / BH-A2
Sampling Depth	=	25 m
Soil Type	=	Sandy Silt



Mohr-Coulomb Plot [c=0.09 kg/sq.cm Phi=28.9 deg]
ShearStress(kg/Sq.cm) vs NormalStress(kg/Sq.cm)

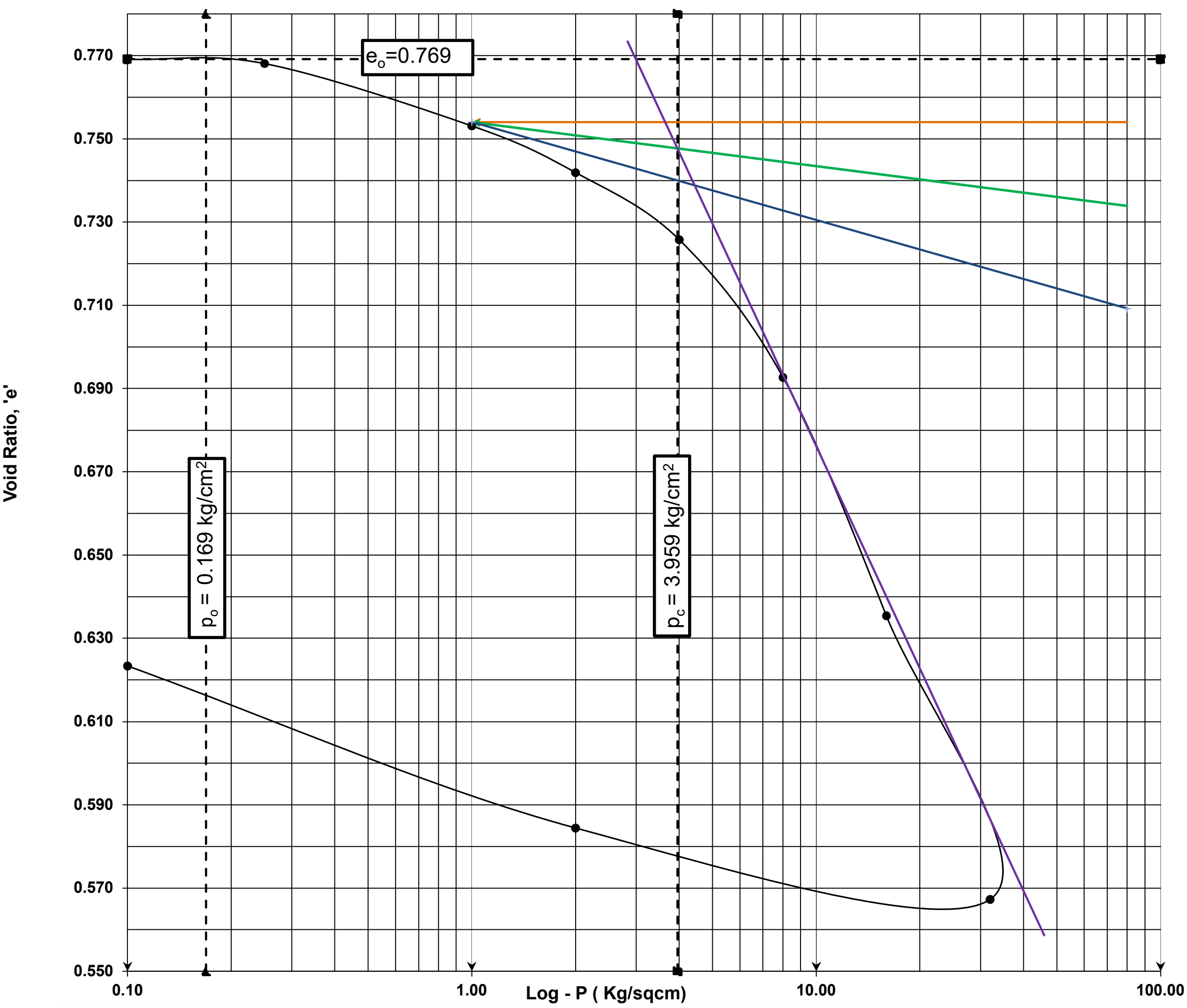
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+758
Borehole No : A1
Depth (m) : 1.0
Least count of dial gauge (mm) = 0.002

C_r = 0.046
C_c = 0.226
H_s (mm) = 11.31

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
11.53	21.0	0.769

Pressure Increment (Kg/cm²)	Void Ratio	Coefficient of Volume compressibility m_v (cm²/kg)
0.25 - 1.0	0.753	**0.0094**
1.0 - 2.0	0.742	0.0064
2.0 - 4.0	0.726	0.0046
4.0 - 8.0	0.693	0.0048
8.0-16.0	0.635	0.0042



Applied Pressure (Kg/cm²)	Final Dial Reading	Compression (mm)	Specimen Height (mm)	e = H/H_s - 1	d_e	d	m_v (cm²/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm²/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	2034	0	20	0.769								
0.25	2028	0.012	19.988	0.768	0.001	0.250	0.0024	0				
1	1944	0.169	19.819	0.753	0.015	0.900	0.0094	2306	1.99095	0.00036	0.342	
2	1880	0.127	19.692	0.742	0.011	1.000	0.0064	5302	1.97555	0.00016	0.100	
4	1789	0.182	19.510	0.726	0.016	2.000	0.0046	7194	1.9601	0.00011	0.052	0.226
8	1602	0.374	19.136	0.693	0.033	4.000	0.0048	9375	1.9323	8.4E-05	0.040	
16	1278	0.648	18.488	0.635	0.057	8.000	0.0042	2306	1.8812	0.00033	0.138	
32	893	0.770	17.718	0.567	0.068	16.000	0.0026	866	1.8103	0.0008	0.209	
2	990	0.194	17.912	0.584								
0.1	1210	0.440	18.352	0.623								

EXPLORE

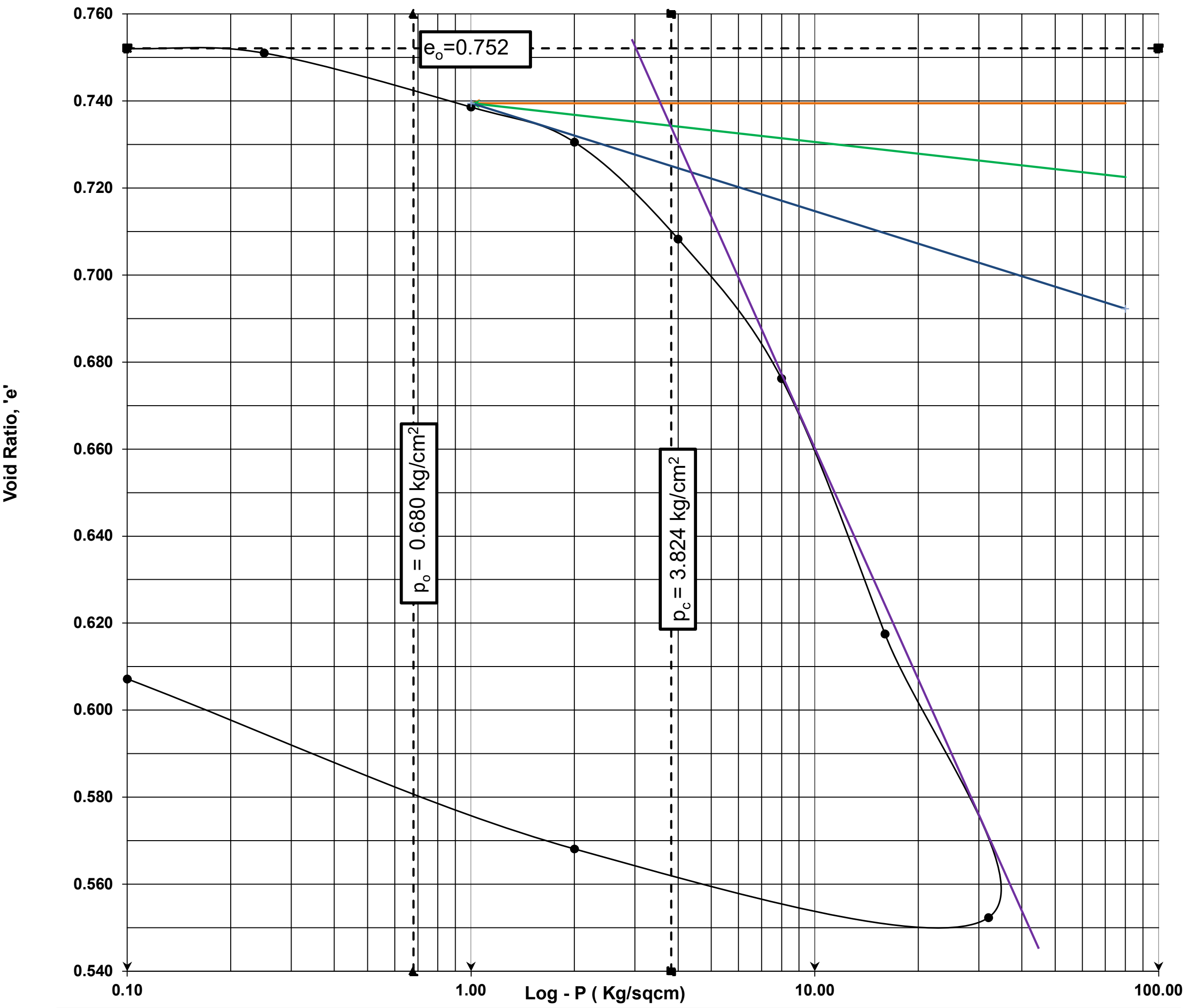
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+758
Borehole No : P1
Depth (m) : 4.0
Least count of dial gauge (mm) = **0.002**

C_r = **0.038**
C_c = **0.217**
H_s (mm) = **11.42**

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
11.81	21.5	0.752

Pressure Increment (Kg/cm^2)	Void Ratio	Coefficient of Volume compressibility m_v (cm^2/kg)
0.25 - 1.0	0.739	**0.0079**
1.0 - 2.0	0.731	0.0046
2.0 - 4.0	0.708	0.0064
4.0 - 8.0	0.676	0.0047
8.0-16.0	0.618	0.0044



Applied Pressure (Kg/cm^2)	Final Dial Reading	Compression (mm)	Specimen Height (mm)	e = H/H$_s$ - 1	d_e	d	m_v (cm^2/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm^2/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	2122	0	20	0.752								
0.25	2116	0.012	19.988	0.751	0.001	0.250	0.0024	0				
1	2045	0.142	19.846	0.739	0.012	0.900	0.0079	960	1.9923	0.00088	0.692	
2	1999	0.092	19.754	0.731	0.008	1.000	0.0046	1685	1.98	0.00049	0.229	
4	1872	0.254	19.500	0.708	0.022	2.000	0.0064	2940	1.9627	0.00028	0.179	0.217
8	1689	0.366	19.134	0.676	0.032	4.000	0.0047	1382	1.9317	0.00057	0.269	
16	1354	0.670	18.464	0.618	0.059	8.000	0.0044	866	1.8799	0.00086	0.379	
32	982	0.744	17.720	0.552	0.065	16.000	0.0025	614	1.8092	0.00113	0.284	
2	1072	0.180	17.900	0.568								
0.1	1295	0.446	18.346	0.607								

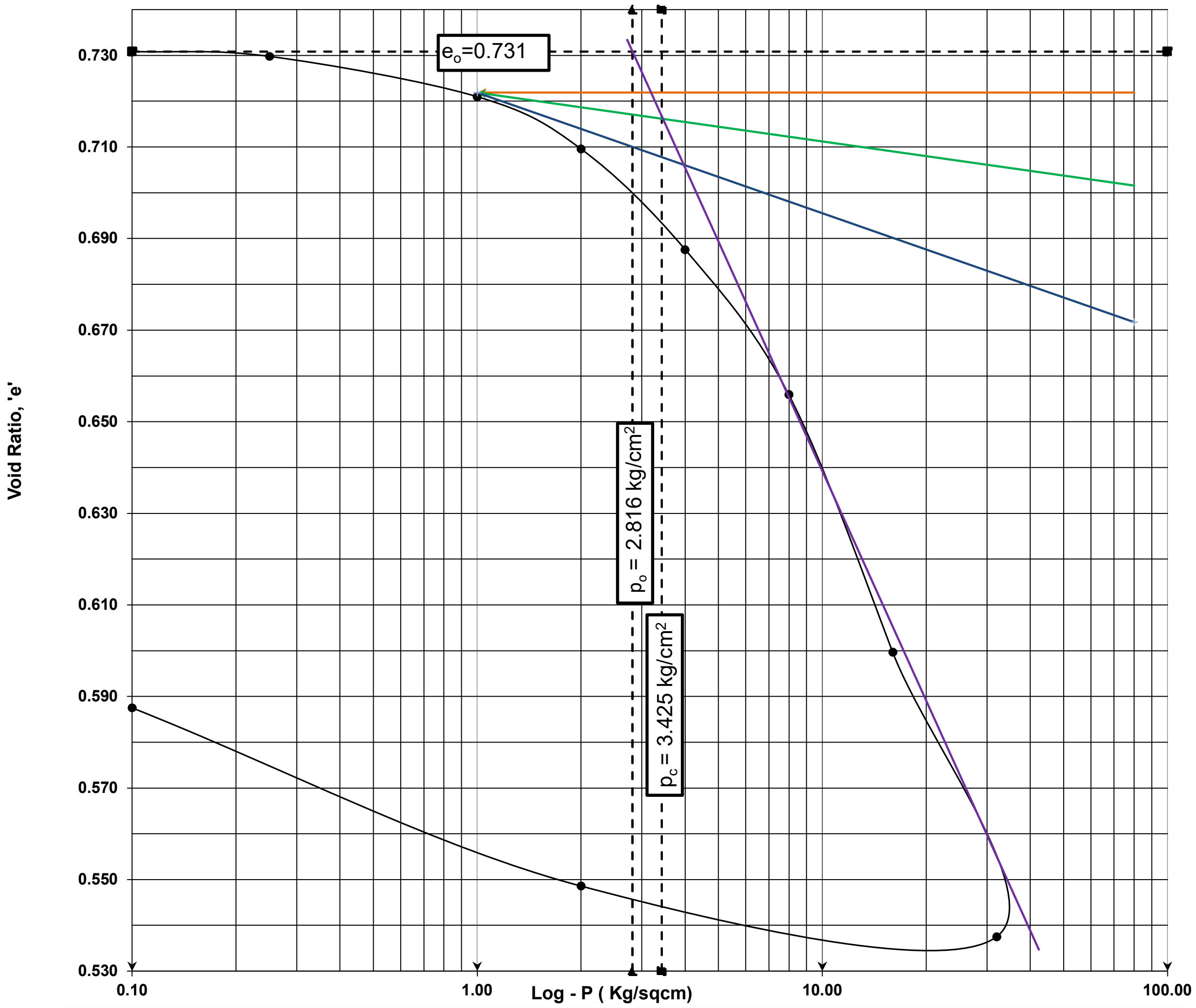
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+758
Borehole No : P1
Depth (m) : 16.0
Least count of dial gauge (mm) = **0.002**

C_r = **0.042**
C_c = **0.206**
H_s (mm) = **11.56**

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
13.69	24.9	0.731

Pressure Increment (Kg/cm^2)	Void Ratio	Coefficient of Volume compressibility m_v (cm^2/kg)
0.25 - 1.0	0.721	0.0057
1.0 - 2.0	0.710	0.0066
2.0 - 4.0	0.688	**0.0064**
4.0 - 8.0	0.656	0.0047
8.0-16.0	0.600	0.0042



Applied Pressure (Kg/cm^2)	Final Dial Reading	Compression (mm)	Specimen Height (mm)	e = H/H$_s$ - 1	d_e	d	m_v (cm^2/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm^2/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	2079	0	20	0.731								
0.25	2073	0.012	19.988	0.730	0.001	0.250	0.0024	0				
1	2022	0.102	19.886	0.721	0.009	0.900	0.0057	960	1.9943	0.00088	0.498	
2	1956	0.132	19.754	0.710	0.011	1.000	0.0066	2940	1.982	0.00028	0.188	
4	1829	0.254	19.500	0.688	0.022	2.000	0.0064	960	1.9627	0.00085	0.547	0.206
8	1646	0.366	19.134	0.656	0.032	4.000	0.0047	540	1.9317	0.00146	0.687	
16	1321	0.650	18.484	0.600	0.056	8.000	0.0042	1500	1.8809	0.0005	0.212	
32	962	0.718	17.766	0.538	0.062	16.000	0.0024	653	1.8125	0.00107	0.259	
2	1026	0.128	17.894	0.549								
0.1	1251	0.450	18.344	0.588								

EXPLORE

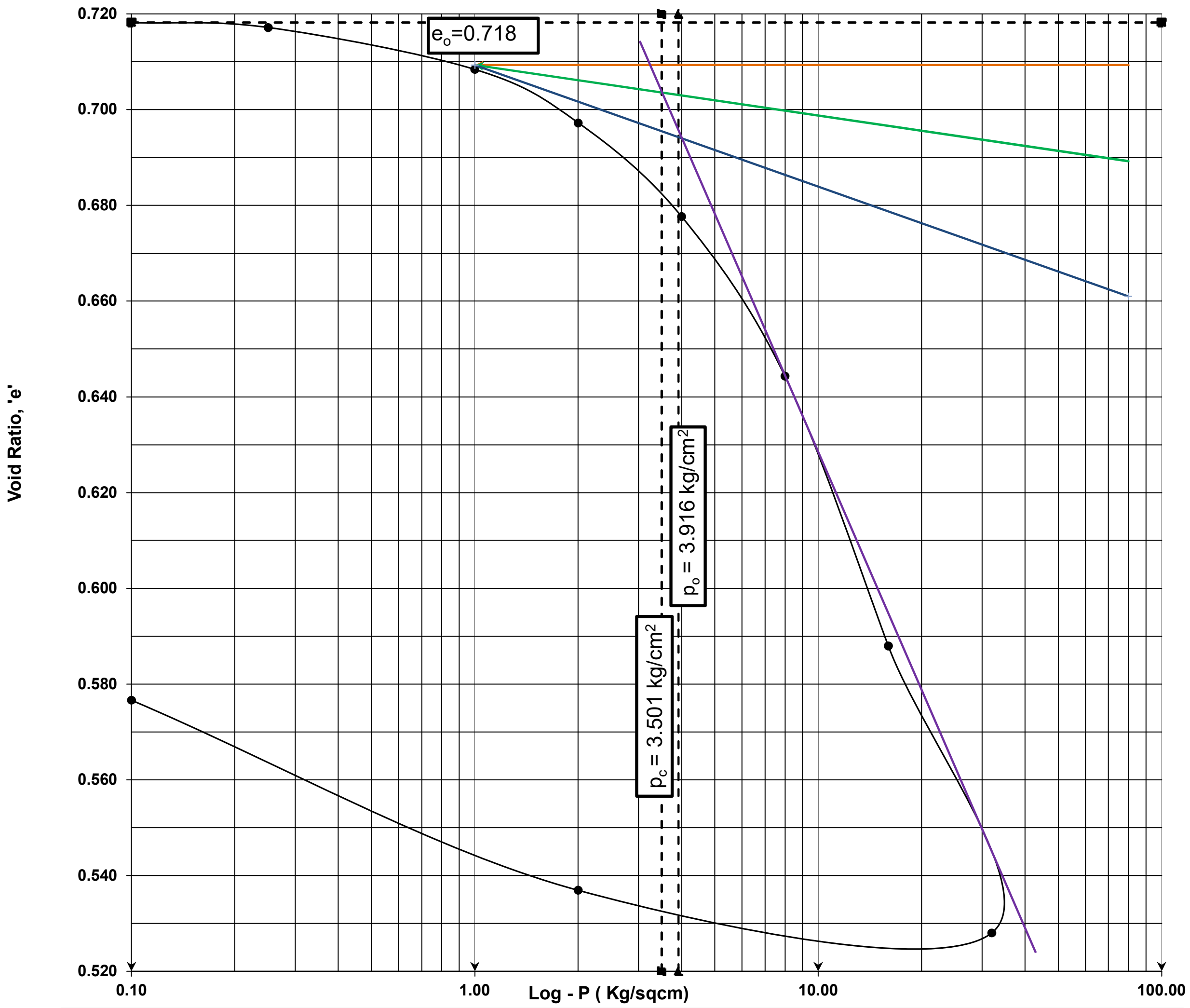
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+758
Borehole No : P1
Depth (m) : 22.0
Least count of dial gauge (mm) = **0.002**

C_r = 0.042
C_c = 0.199
H_s (mm) = 11.64

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
14.51	26.4	0.718

Pressure Increment (Kg/cm^2)	Void Ratio	Coefficient of Volume compressibility m_v (cm^2/kg)
0.25 - 1.0	0.708	0.0057
1.0 - 2.0	0.697	0.0065
2.0 - 4.0	0.678	**0.0058**
4.0 - 8.0	0.644	0.0050
8.0-16.0	0.588	0.0043



Applied Pressure (Kg/cm^2)	Final Dial Reading	Compression (mm)	Specimen Height (mm)	e = H/H_s - 1	d_e	d	m_v (cm^2/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm^2/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	2060	0	20	0.718								
0.25	2054	0.012	19.988	0.717	0.001	0.250	0.0024	0				
1	2003	0.102	19.886	0.708	0.009	0.900	0.0057	1500	1.9943	0.00056	0.319	
2	1938	0.130	19.756	0.697	0.011	1.000	0.0065	1058	1.9821	0.00079	0.514	
4	1824	0.228	19.528	0.678	0.020	2.000	0.0058	913	1.9642	0.00090	0.517	0.199
8	1630	0.388	19.140	0.644	0.033	4.000	0.0050	317	1.9334	0.0025	1.240	
16	1302	0.656	18.484	0.588	0.056	8.000	0.0043	1382	1.8812	0.00054	0.233	
32	953	0.698	17.786	0.528	0.060	16.000	0.0024	1815	1.8135	0.00038	0.091	
2	1005	0.104	17.890	0.537								
0.1	1236	0.462	18.352	0.577								

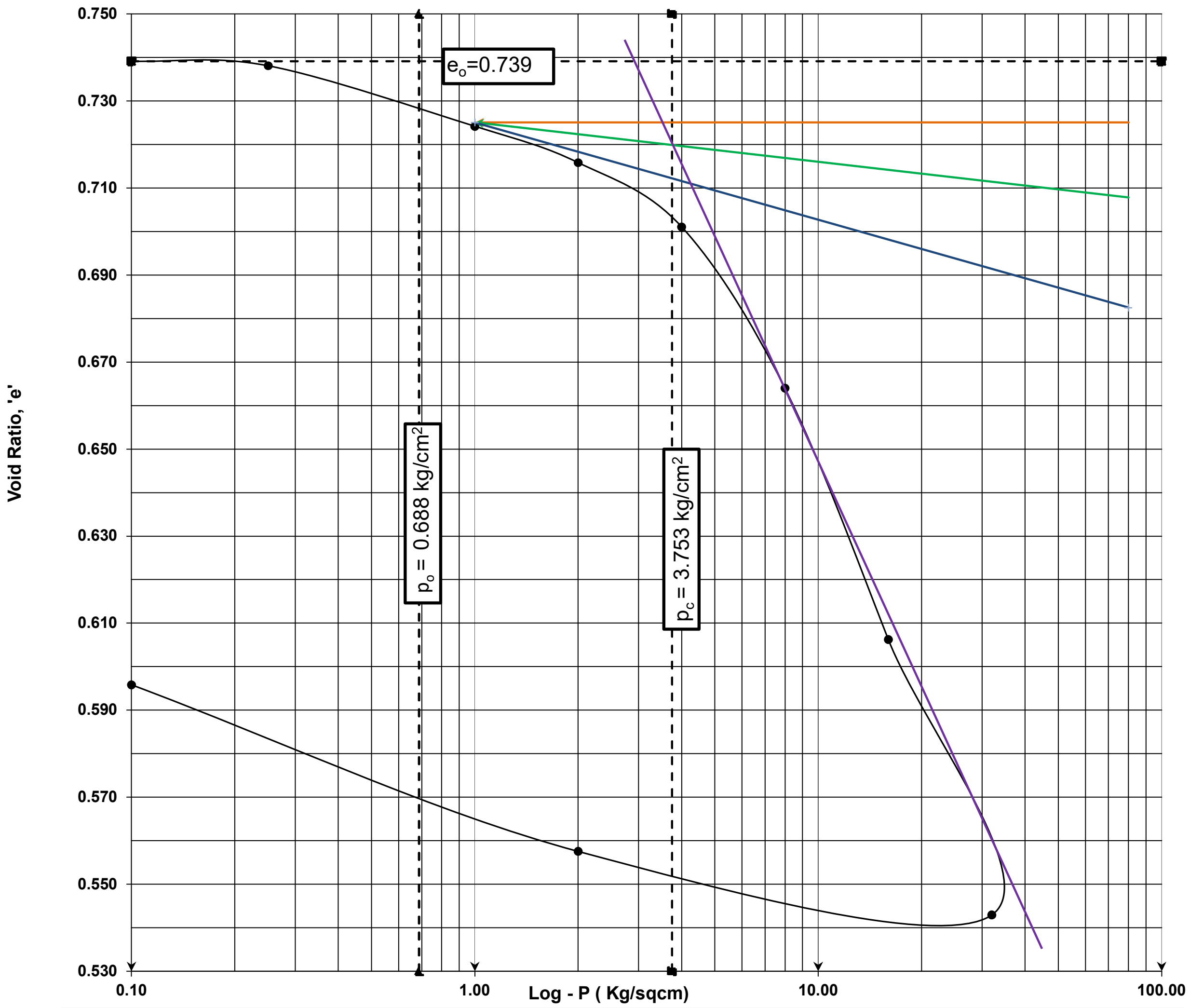
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+758
Borehole No : A2
Depth (m) : 4.0
Least count of dial gauge (mm) = **0.002**

C_r = **0.042**
C_c = **0.210**
H_s (mm) = **11.50**

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
11.62	21.1	0.739

Pressure Increment (Kg/cm^2)	Void Ratio	Coefficient of Volume compressibility m_v (cm^2/kg)
0.25 - 1.0	0.724	**0.0089**
1.0 - 2.0	0.716	0.0048
2.0 - 4.0	0.701	0.0043
4.0 - 8.0	0.664	0.0054
8.0-16.0	0.606	0.0043



Applied Pressure (Kg/cm^2)	Final Dial Reading	Compres sion (mm)	Specimen Height (mm)	e = H/H$_s$ - 1	d_e	d	m_v (cm^2/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm^2/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	2133	0	20	0.739								
0.25	2127	0.012	19.988	0.738	0.001	0.250	0.0024	0				
1	2047	0.160	19.828	0.724	0.014	0.900	0.0089	1382	1.9914	0.00061	0.541	
2	1999	0.096	19.732	0.716	0.008	1.000	0.0048	2381	1.978	0.00035	0.169	
4	1914	0.170	19.562	0.701	0.015	2.000	0.0043	7194	1.9647	0.00011	0.049	0.210
8	1701	0.426	19.136	0.664	0.037	4.000	0.0054	9375	1.9349	8.5E-05	0.046	
16	1369	0.664	18.472	0.606	0.058	8.000	0.0043	2306	1.8804	0.00033	0.141	
32	1005	0.728	17.744	0.543	0.063	16.000	0.0025	866	1.8108	0.0008	0.198	
2	1089	0.168	17.912	0.558								
0.1	1309	0.440	18.352	0.596								

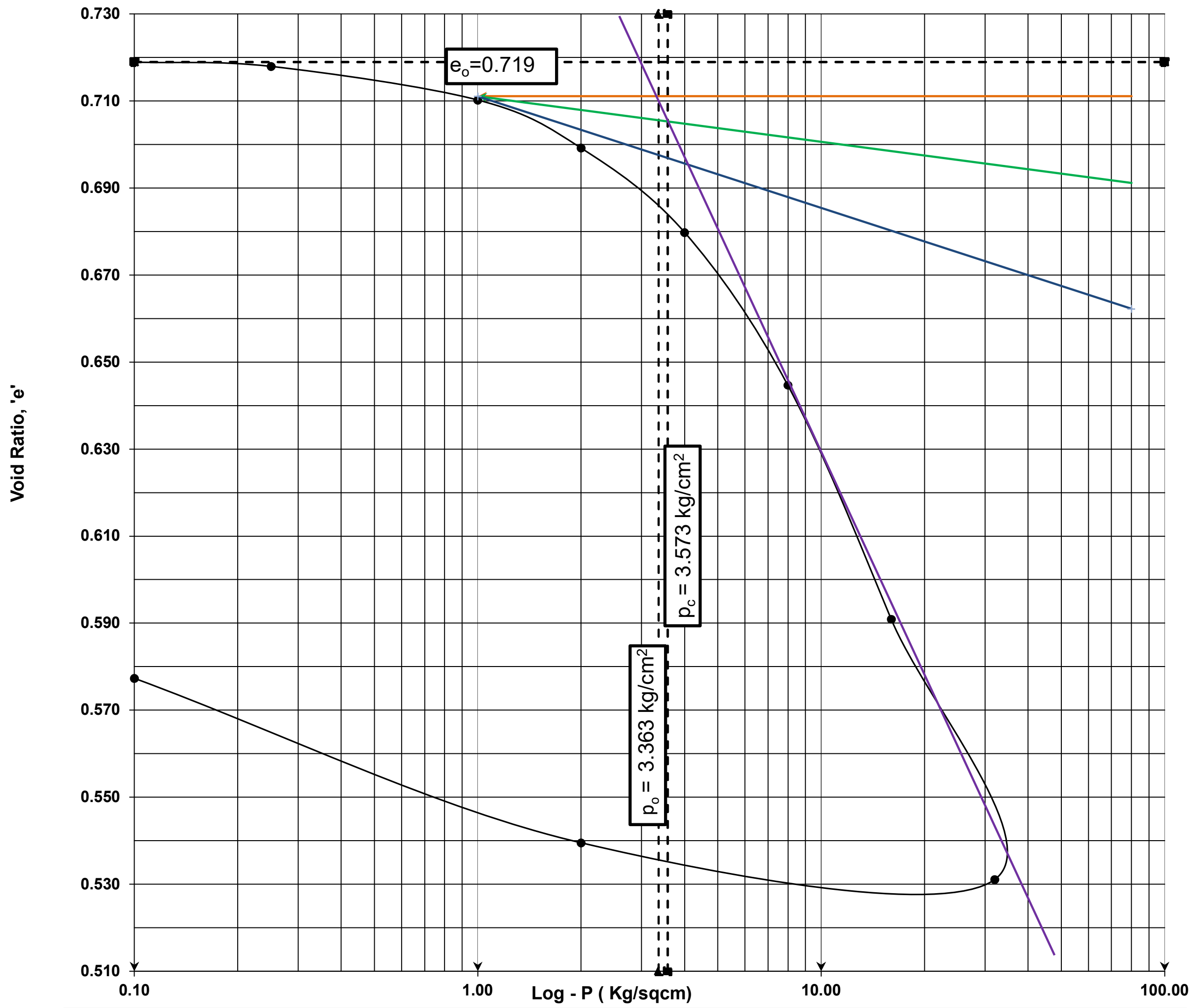
CONSOLIDATION CURVE (e - logp)

Reference No : 25065139
Chainage No : 44+578
Borehole No : A2
Depth (m) : 19.0
Least count of dial gauge (mm) = **0.002**

C_r = **0.048**
C_c = **0.199**
H_s (mm) = **11.64**

MOISTURE CONTENT		INITIAL
INITIAL	FINAL	VOID RATIO
13.93	25.4	0.719

Pressure Increment (Kg/cm^2)	Void Ratio	Coefficient of Volume compressibility m_v (cm^2/kg)
0.25 - 1.0	0.710	0.0050
1.0 - 2.0	0.699	0.0064
2.0 - 4.0	0.680	**0.0057**
4.0 - 8.0	0.645	0.0052
8.0-16.0	0.591	0.0041



Applied Pressure (Kg/cm^2)	Final Dial Reading	Compression (mm)	Specimen Height (mm)	e = H/H$_s$ - 1	d_e	d	m_v (cm^2/kg)	t_{90} (Sec)	H_{av} (cm)	C_v (cm^2/sec)	Permeability cm/sec (10^{-5})	C_c
0.1	1997	0	20	0.719								
0.25	1991	0.012	19.988	0.718	0.001	0.250	0.0024	0				
1	1946	0.090	19.898	0.710	0.008	0.900	0.0050	470	1.9949	0.00179	0.897	
2	1882	0.128	19.770	0.699	0.011	1.000	0.0064	3650	1.9834	0.00023	0.147	
4	1769	0.226	19.544	0.680	0.019	2.000	0.0057	8354	1.9657	0.00010	0.056	0.199
8	1565	0.408	19.136	0.645	0.035	4.000	0.0052	3110	1.934	0.00025	0.133	
16	1252	0.626	18.510	0.591	0.054	8.000	0.0041	2306	1.8823	0.00033	0.133	
32	904	0.696	17.814	0.531	0.060	16.000	0.0024	866	1.8162	0.00081	0.190	
2	953	0.098	17.912	0.539								
0.1	1173	0.440	18.352	0.577								

ANNEXURE-D
LIST OF REFERRED IS CODES

Field Investigation

1. IS: 1498-1970 Classification and identification of soils for general engineering purposes (first revision) Amendment 2

2. IS: 1892-1979 Code of practice for sub surface investigations for foundations

3. IS: 2131-1981 Method of standard penetration tests for soils

4. IS: 2132-1986 Code of practice for thin walled tube sampling of soils

Laboratory tests

1. IS: 2720-1983 (Part 1) Methods of tests for soils: Preparation of dry soil samples for various tests (second revision)

2. IS: 2720-1973 (Part-2) Methods of test for soils: Determination of water content (second revision) Amendment 1

3. IS: 2720-1980 (Part-3/Sec 1) Method of test for soil: Determination of specific gravity: Fine grained soils

4. IS: 2720-1980 (Part-3/Sec 2) Method of test for soil: Determination of specific gravity: Fine, medium & coarse grained soils. (First revision)

5. IS: 2720-1985 (Part-4) Methods of test for soils: Grain size analysis (Second revision)

6. IS: 2720-1985 (Part-5) Methods of test for soils: Determination of liquid and plastic limit (Second revision)

Foundation construction

1. IS: 1904-1986 Code of practice for design and construction of foundation in soils: General requirements (Third revision)

2. IS: 6403-1981 Code of practice for determination of bearing capacity of shallow foundations

3. IS: 8009-1976 (Part-1) Code of practice for calculation of settlements of foundations: Shallow foundations subjected to symmetrical static vertical loads (Amendment 2)

4. IS: 2911-1979 (Part-1-Sec.2) Code of practice for design and construction of Pile Foundation (Bored Cast in situ Pile)

5. IS: 2911-1985 (Part-4) Code of practice for design and construction of Pile Foundation (Load Test on Piles)

6. Design Aids in Soil Mechanics and Foundation Engineering (5th reprint) by S.R. Kaniraj

7. IS: 1893 (Part-1): 2002 Criteria for Earthquake Resistant Design of Structures

8. IRC:78-2014 Standard Specifications and Code of Practice for Road Bridges – Section: VII Foundations and Substructure